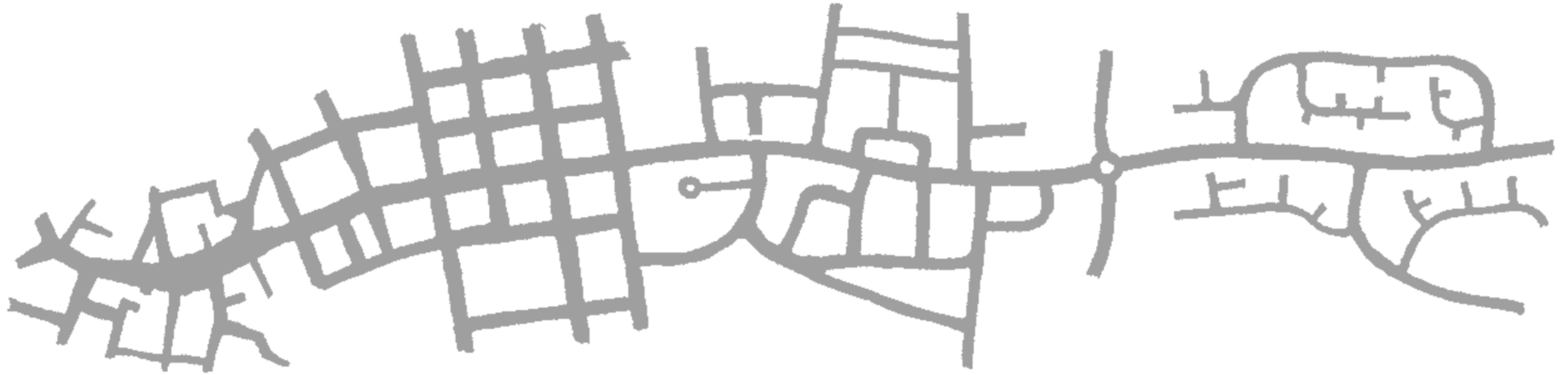




Instructor: Anastassia Vybornova

Lecture 10: Spatial Networks

Geospatial Data Science @ ITU, April 8th, 2026

Geospatial Data Science - pt II

(post-Easter)

Project proposals due **today!**

April 8: Spatial Networks

April 15: Human Mobility

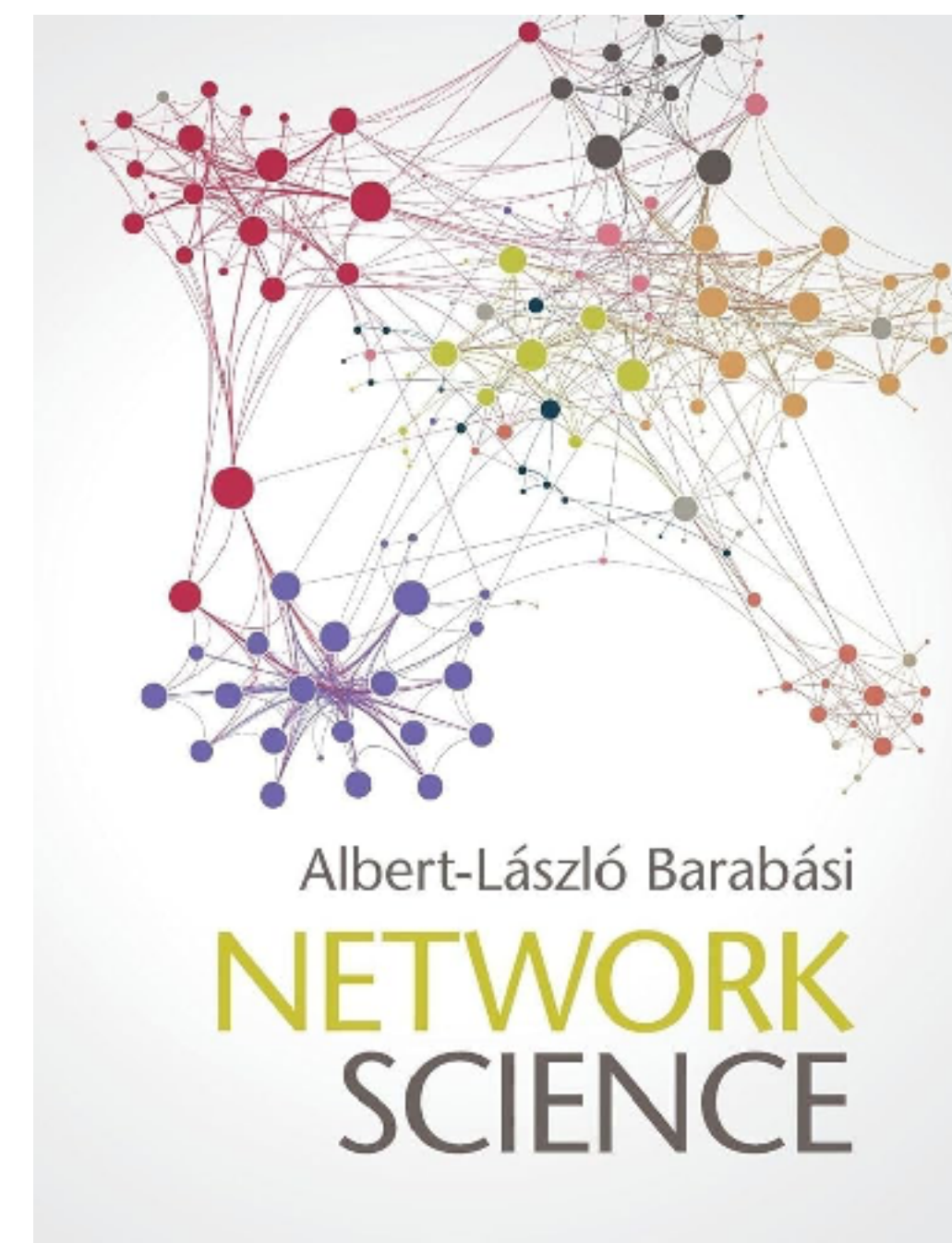
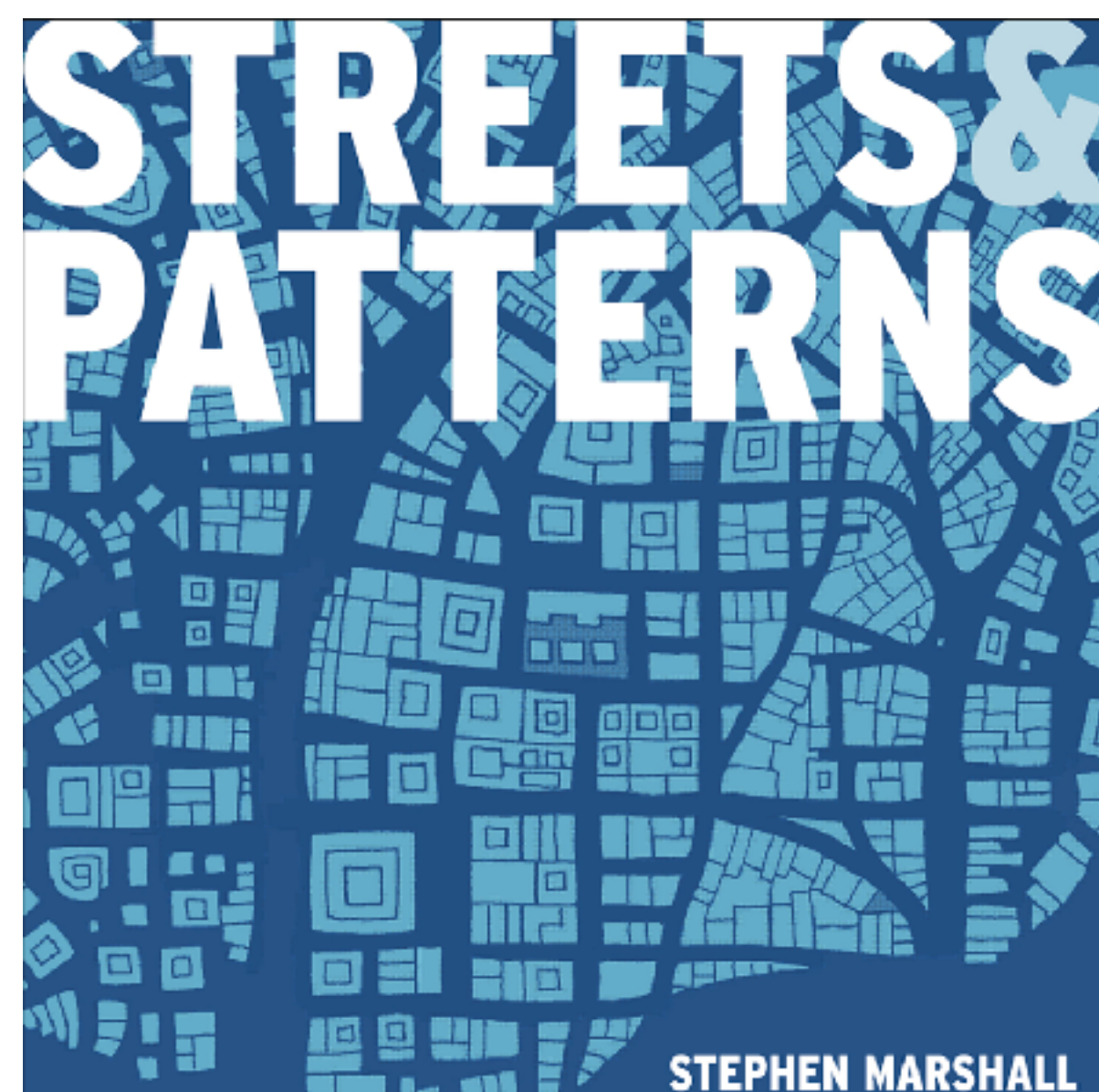
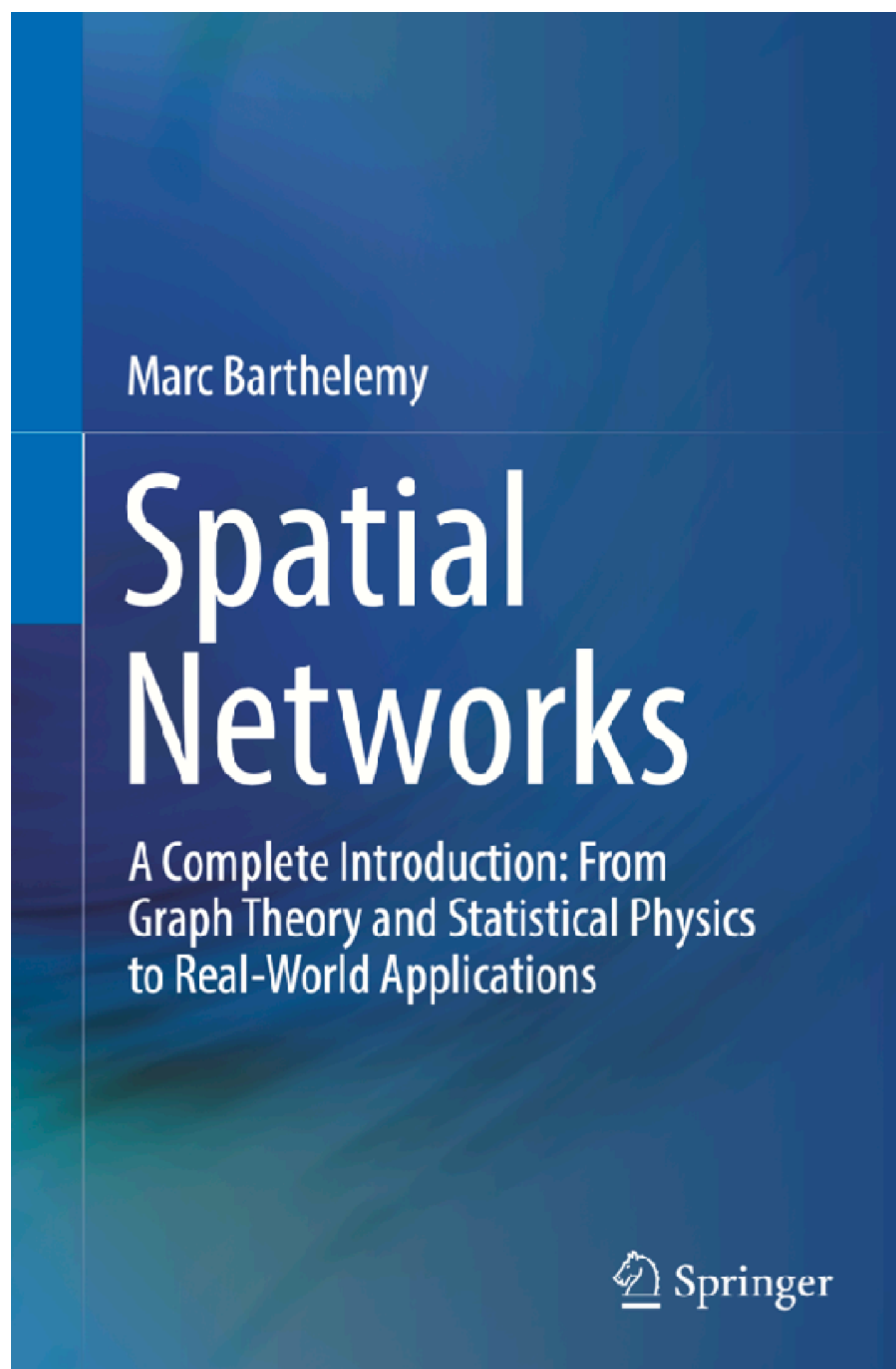
April 22: Raster Data, EO & GIS Guest lecture with Imke Sievers

April 29: Human Geography Guest lecture with Sofia Kazmi Høgsbro

Mandatory Readings: for April 15, 22, and 29

Python notebooks: only in exercises

Today's ~~mandatory~~ strongly suggested reading



Today's lecture in 3 sentences

A spatial network has nodes in metric space and is often (but not necessarily!) planar.

We can model "optimal" networks with different triangulation algorithms.

Street network degree distributions are boring, but street network patterns are interesting.

Today's lecture in 3 sentences

Bridges of Königsberg: Graph theory & network science 101

Spatial networks: Distance

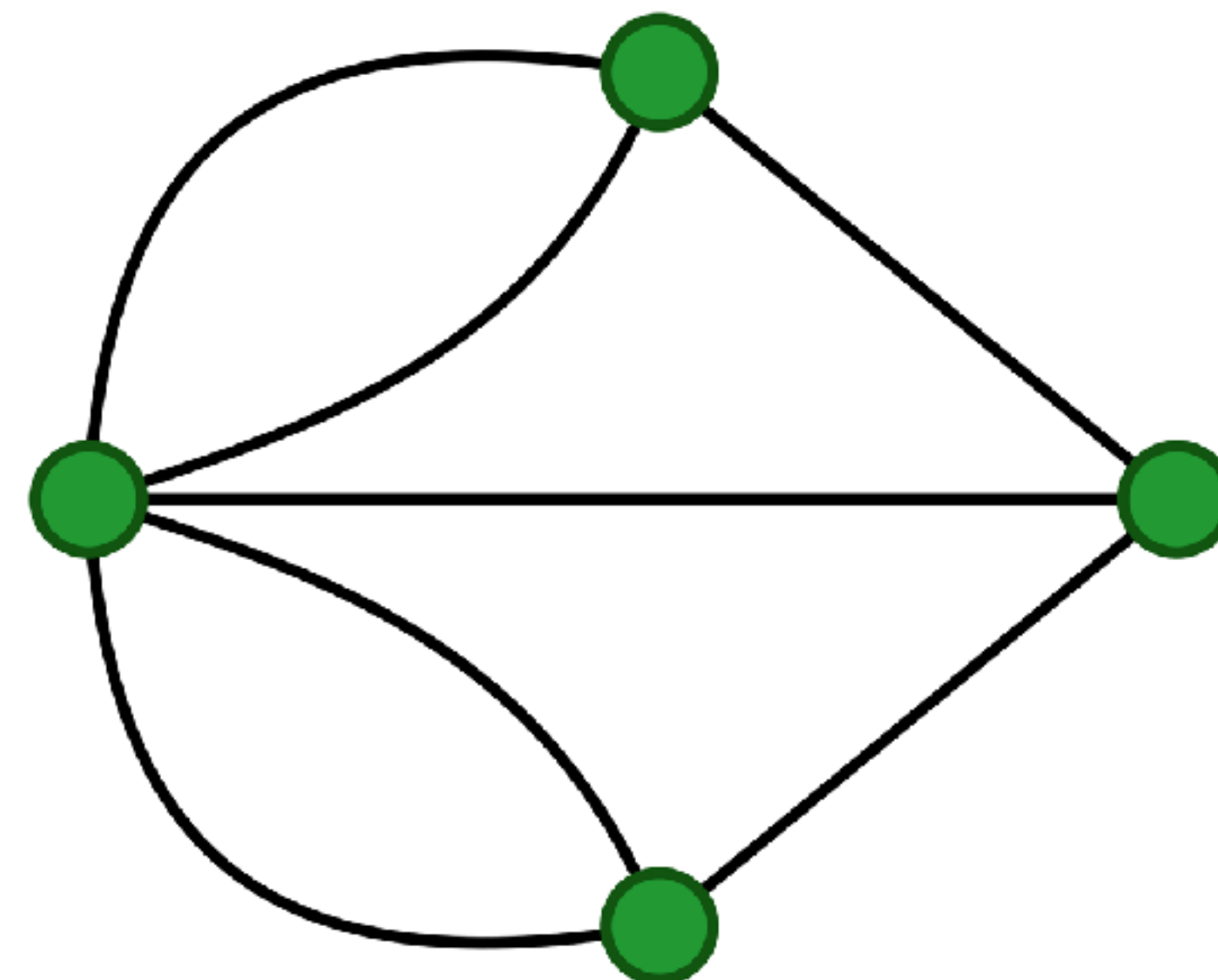
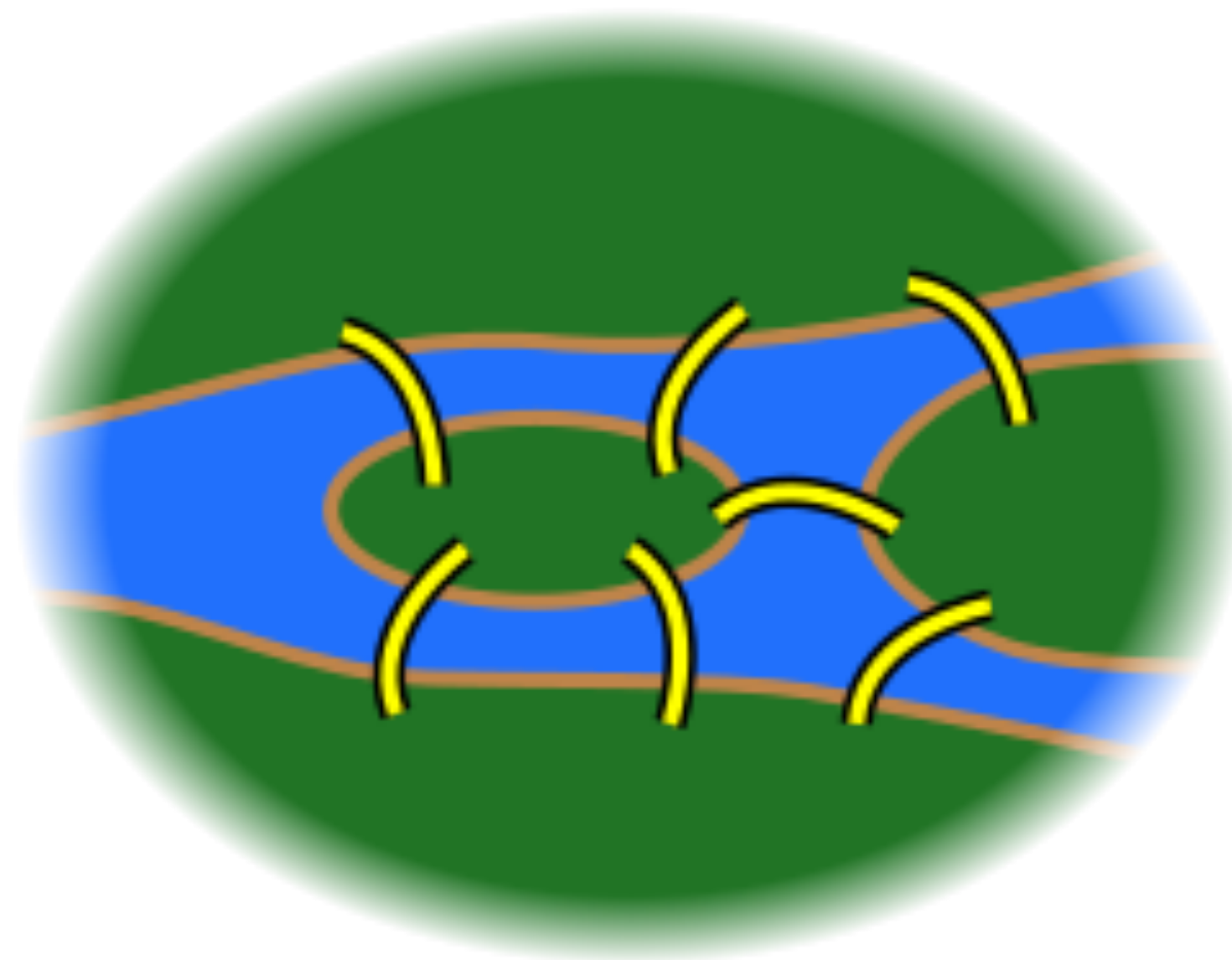
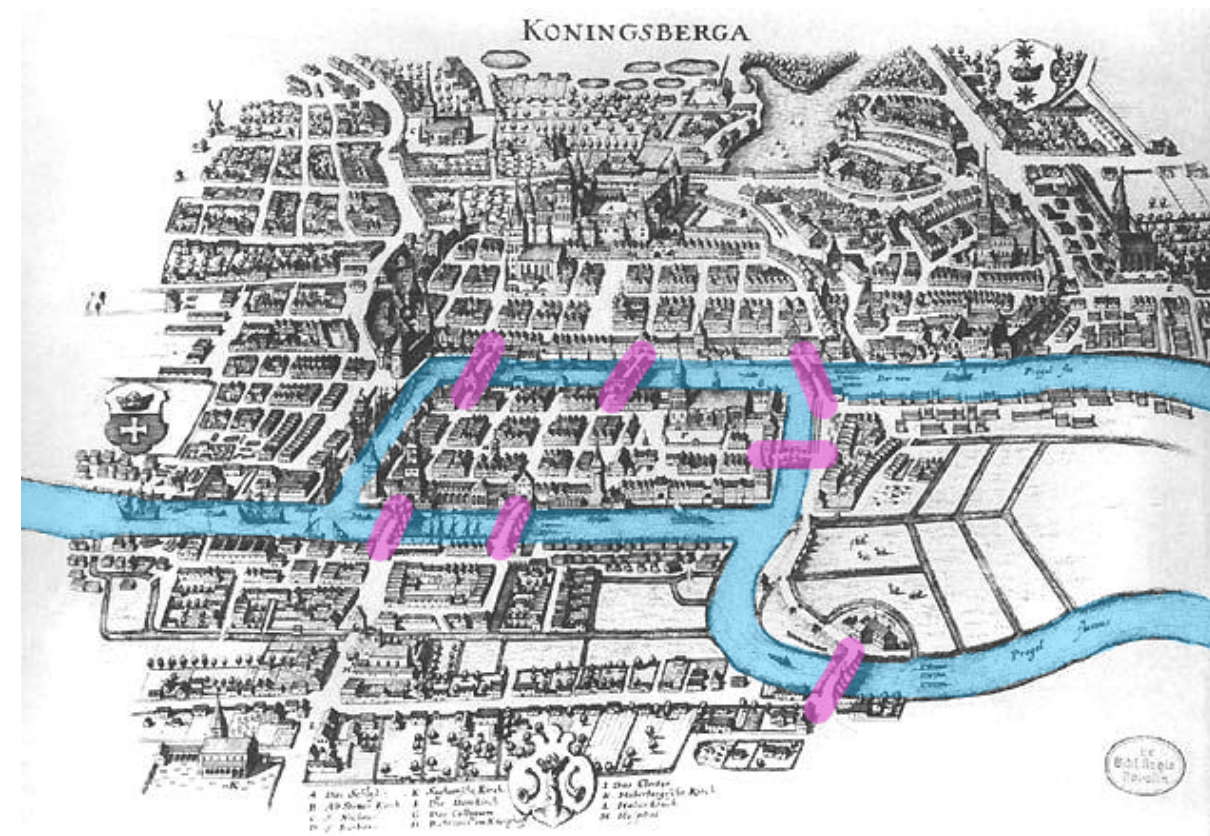
Spatial networks: Planarity

Spatial networks: Degree distributions

Triangulations & “optimal” graphs

Delaunay triangulation, minimum weight triangulation, Voronoi tessellation, dual graph (of a planar graph), minimum spanning tree (MST) & Kruskal's algorithm, ...

Bridges of Königsberg: Graph theory & network science 101

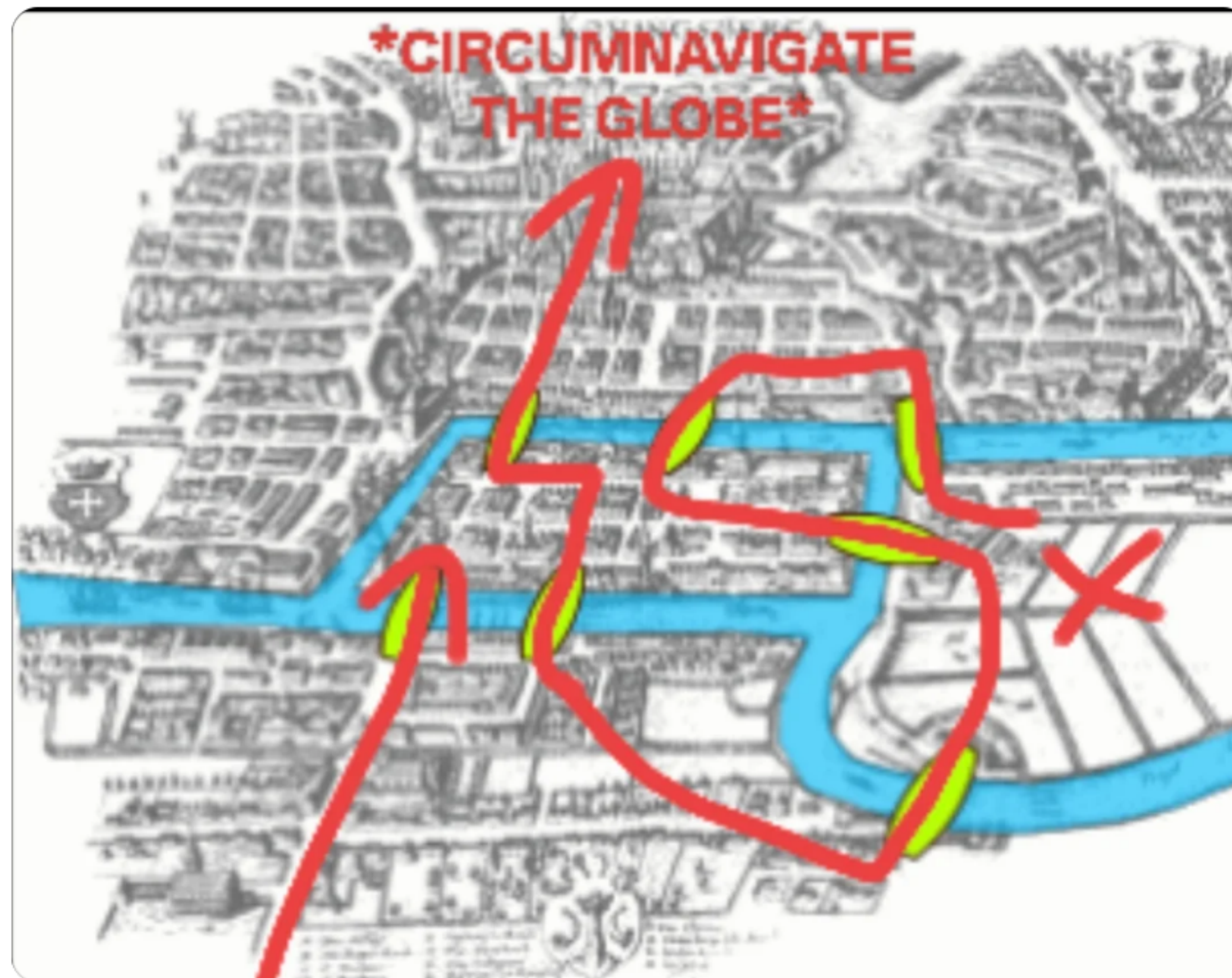




r/mathmemes • 9mo ago
totallyordinaryyy

...

Euler in shambles, solution found to the seven bridges of Königsberg

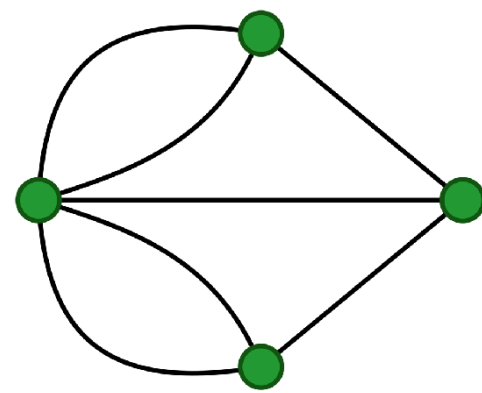


↑ 10K ↓ · 106

https://www.reddit.com/r/mathmemes/comments/1m46l0m/euler_in_shambles_solution_found_to_the_seven/

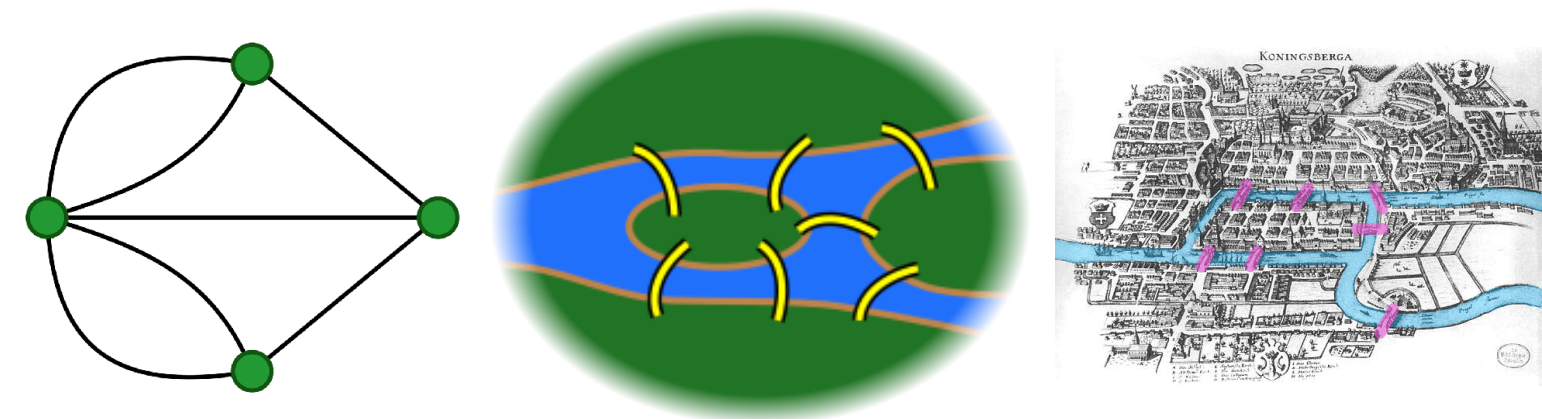
Disambiguation / terminology

graph theory

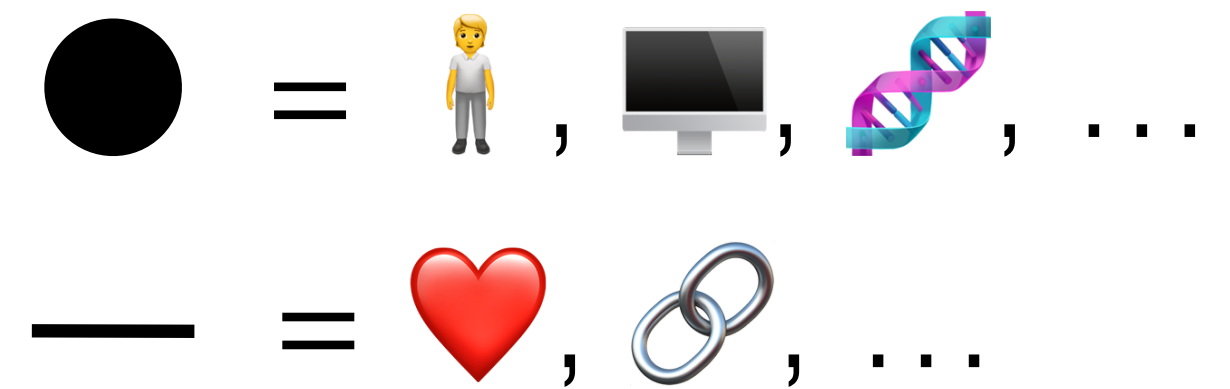
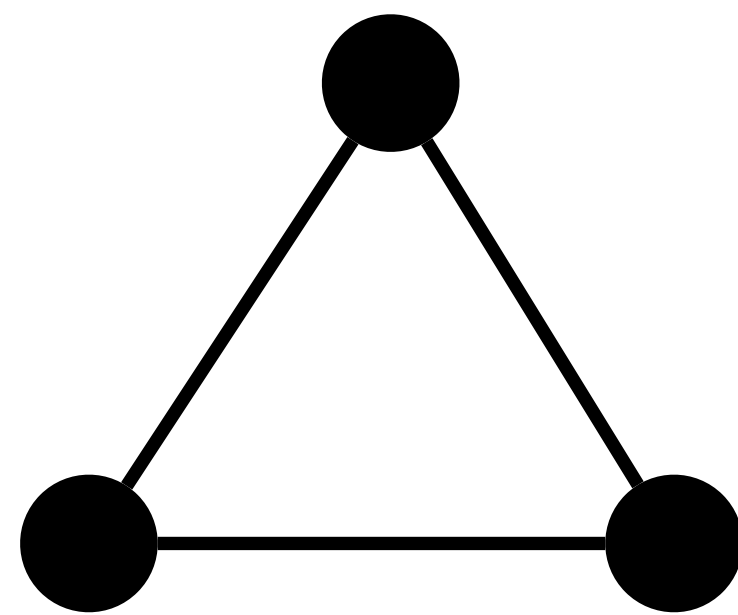
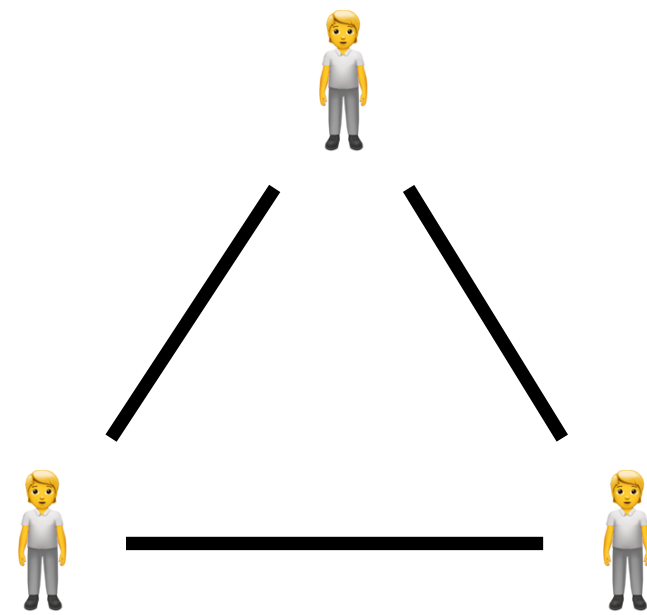


vs.

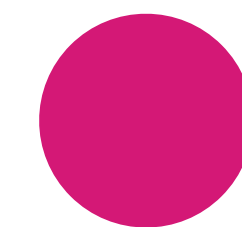
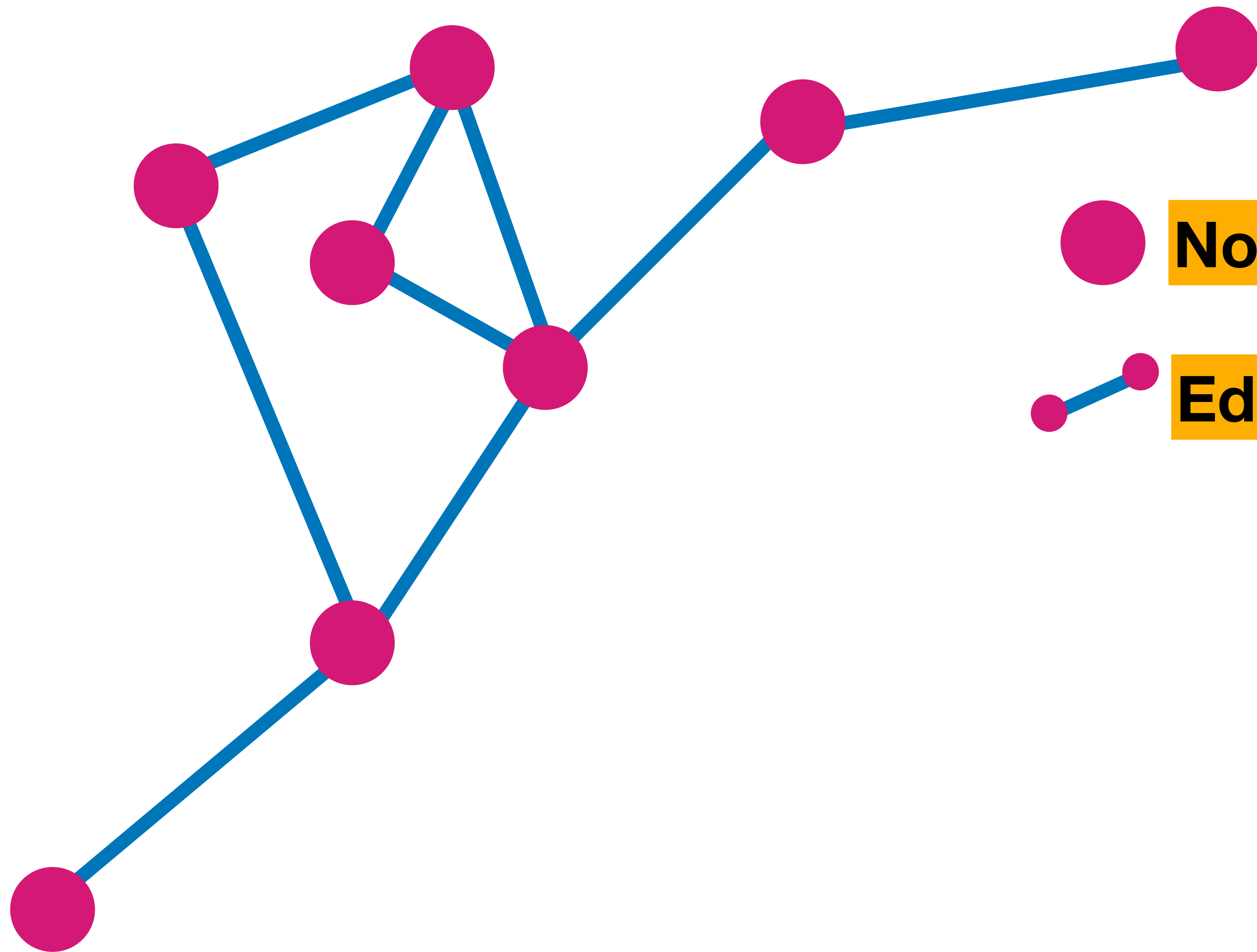
network science



What is the difference between a graph and a network ?

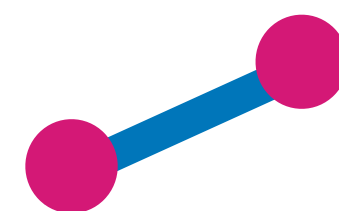


$$\text{Network} = \text{Graph} + \text{Meaning}$$



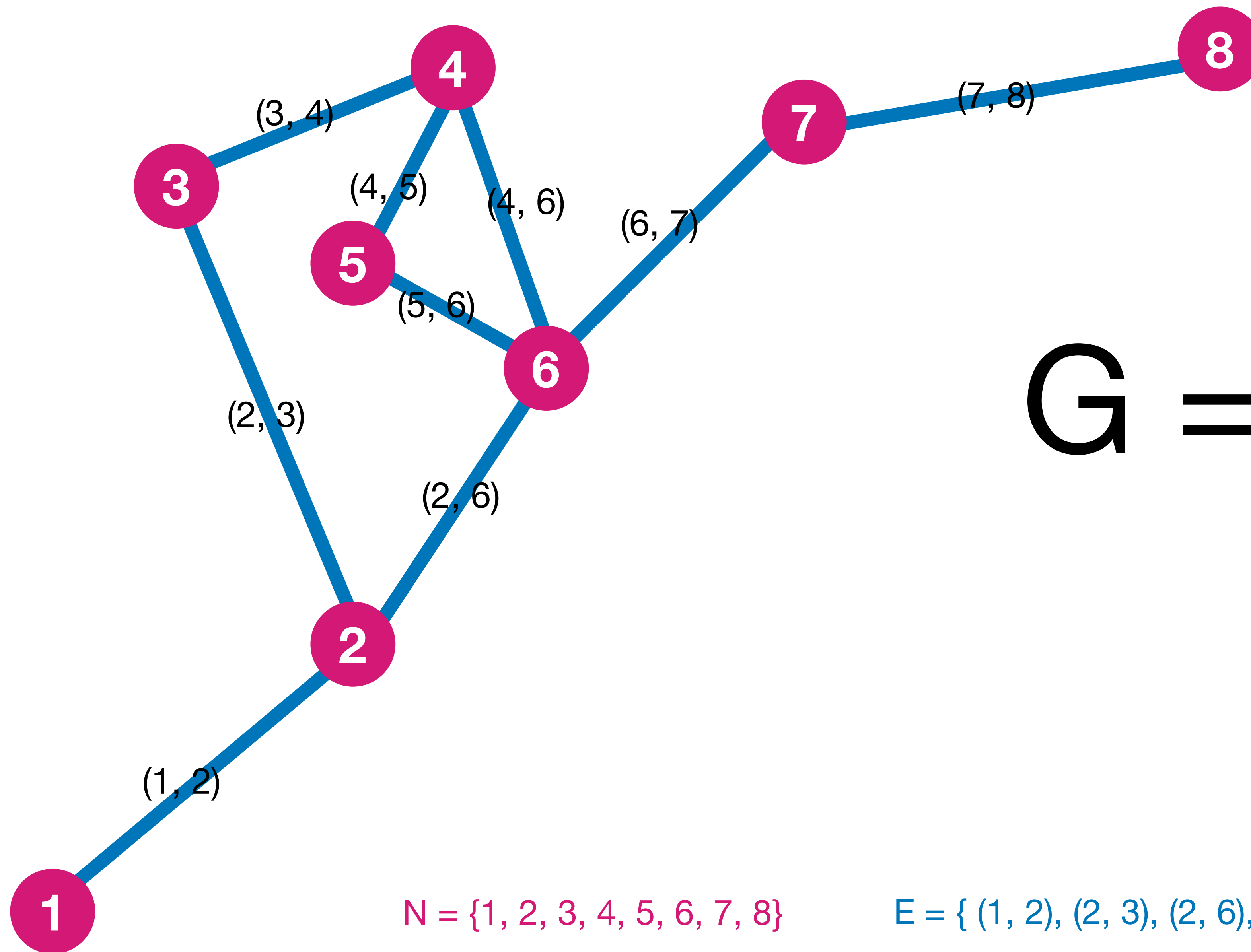
Node

aka “vertex”, “actor”



Edge

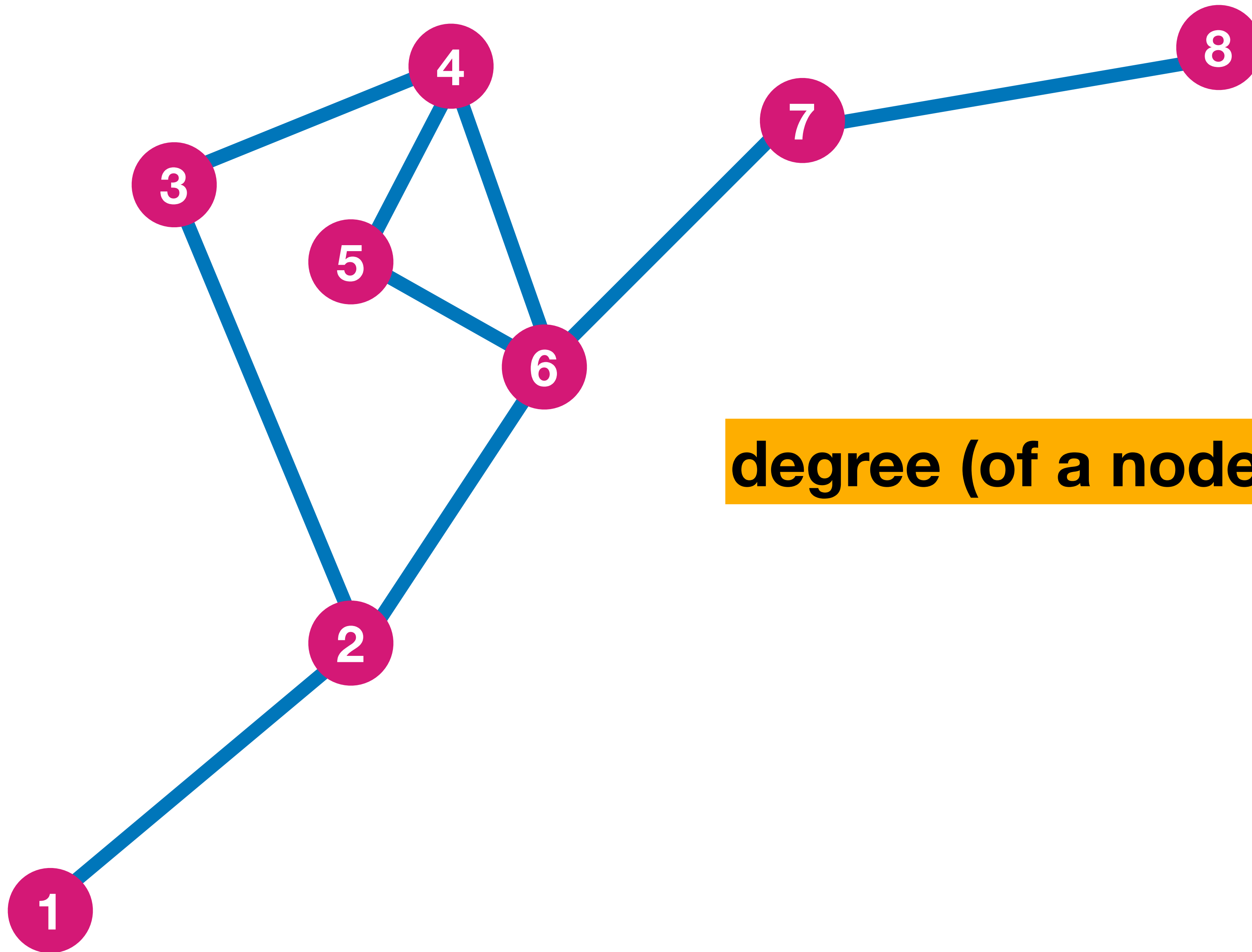
aka “link”, “arc”



$$G = (N, E)$$

$$N = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

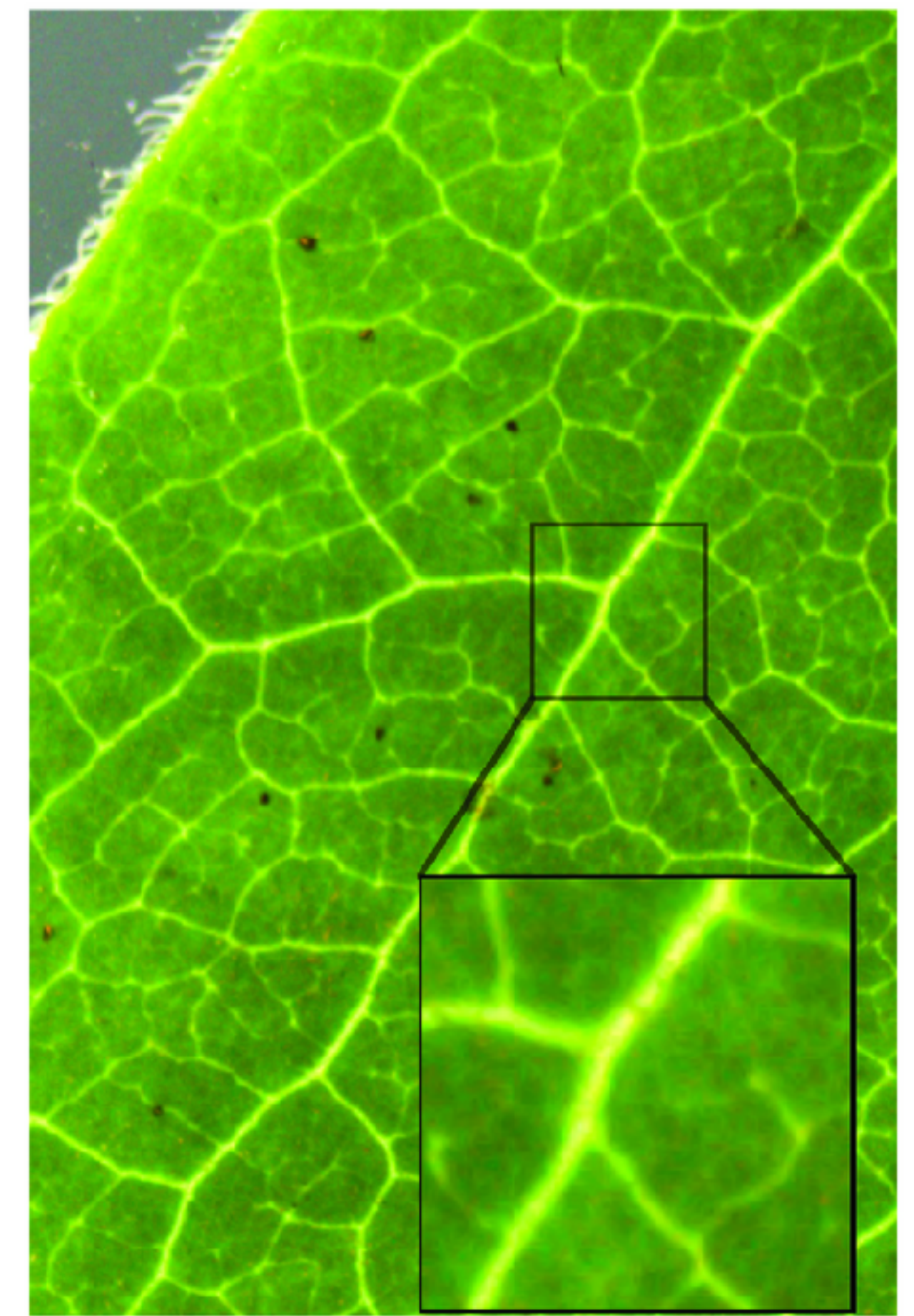
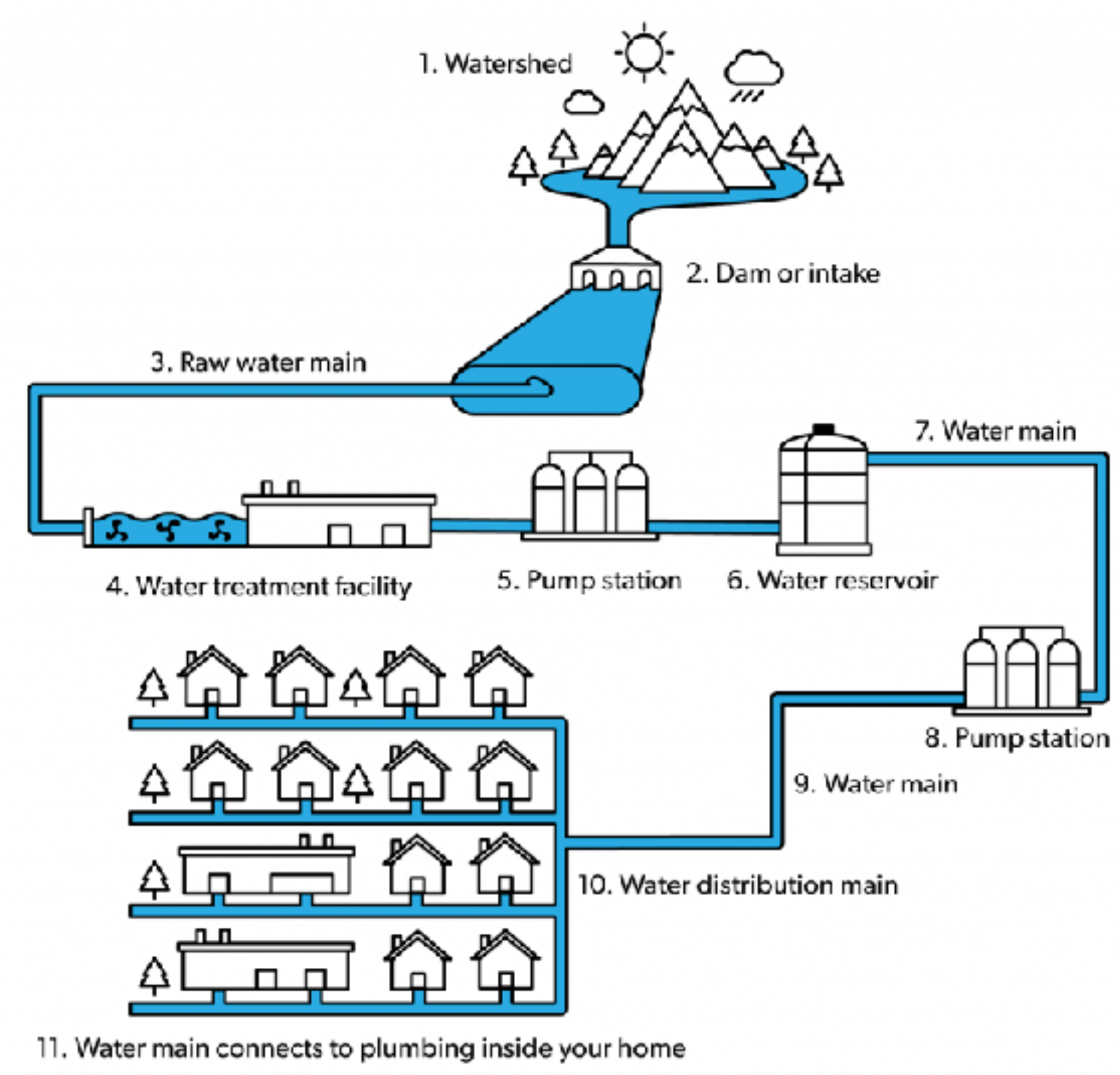
$$E = \{ (1, 2), (2, 3), (2, 6), (3, 4), (4, 5), (4, 6), (5, 6), (6, 7), (7, 8) \}$$

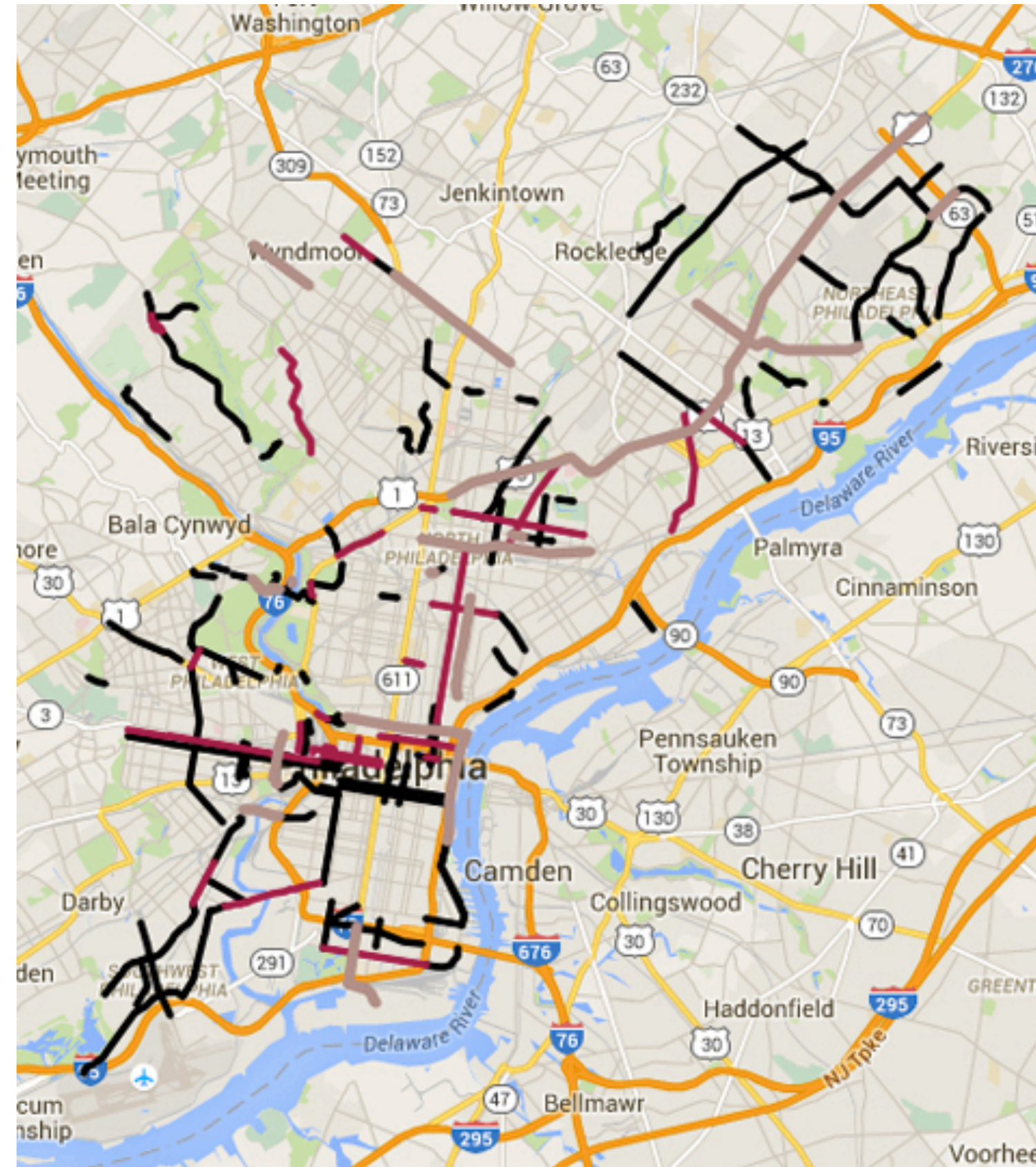
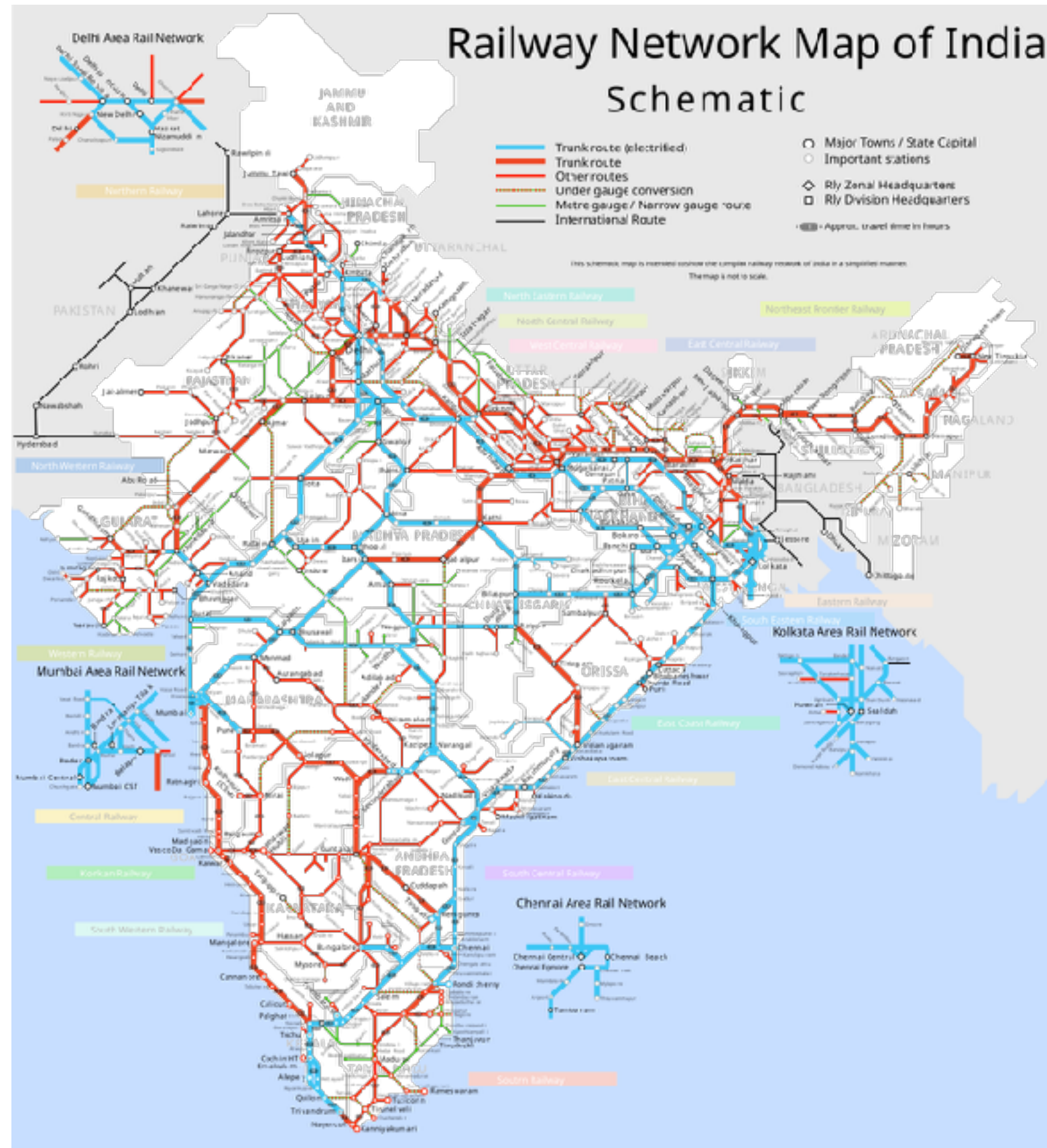


degree (of a node)

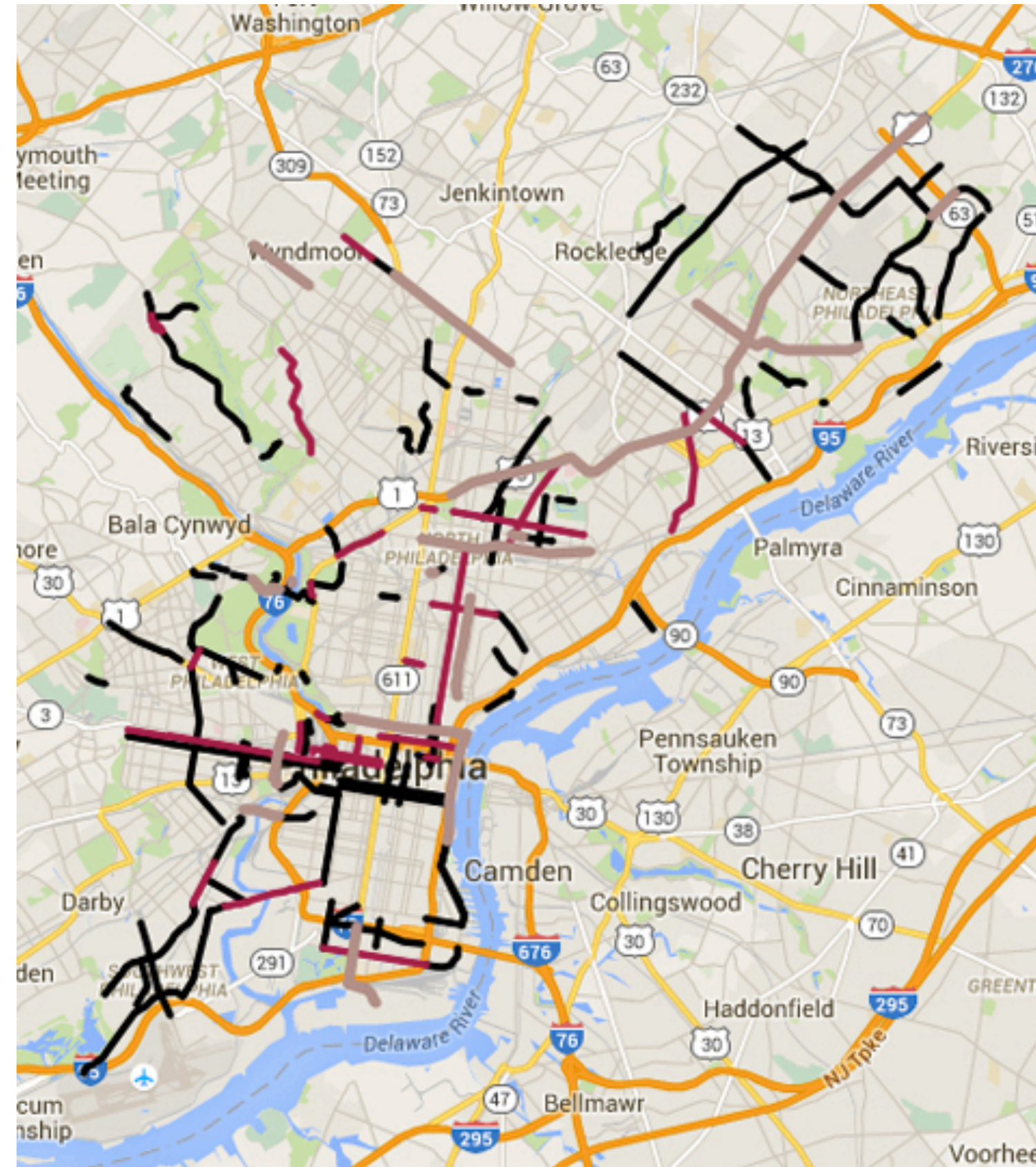
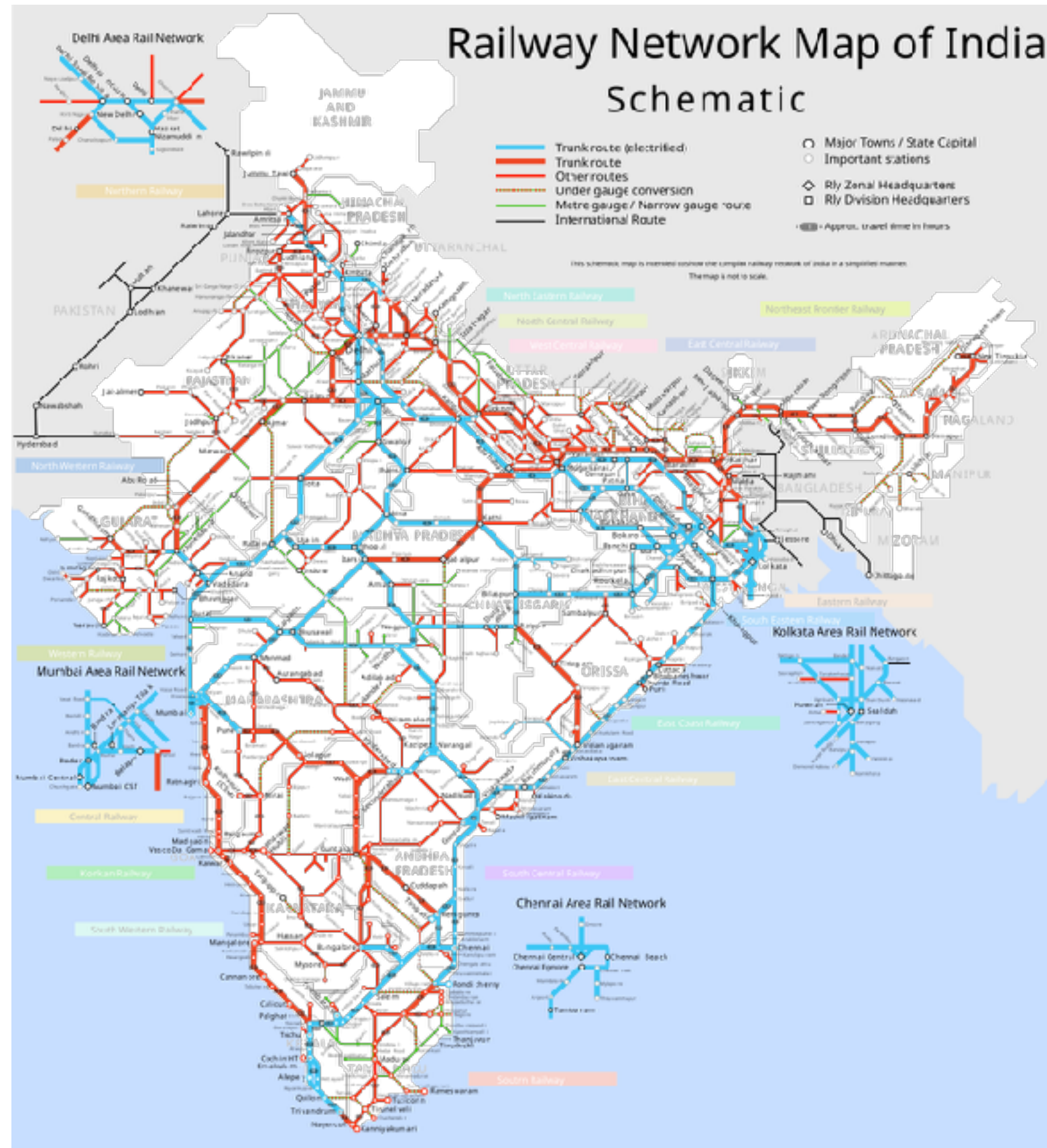
The number of edges
incident on the node

Examples of spatial networks



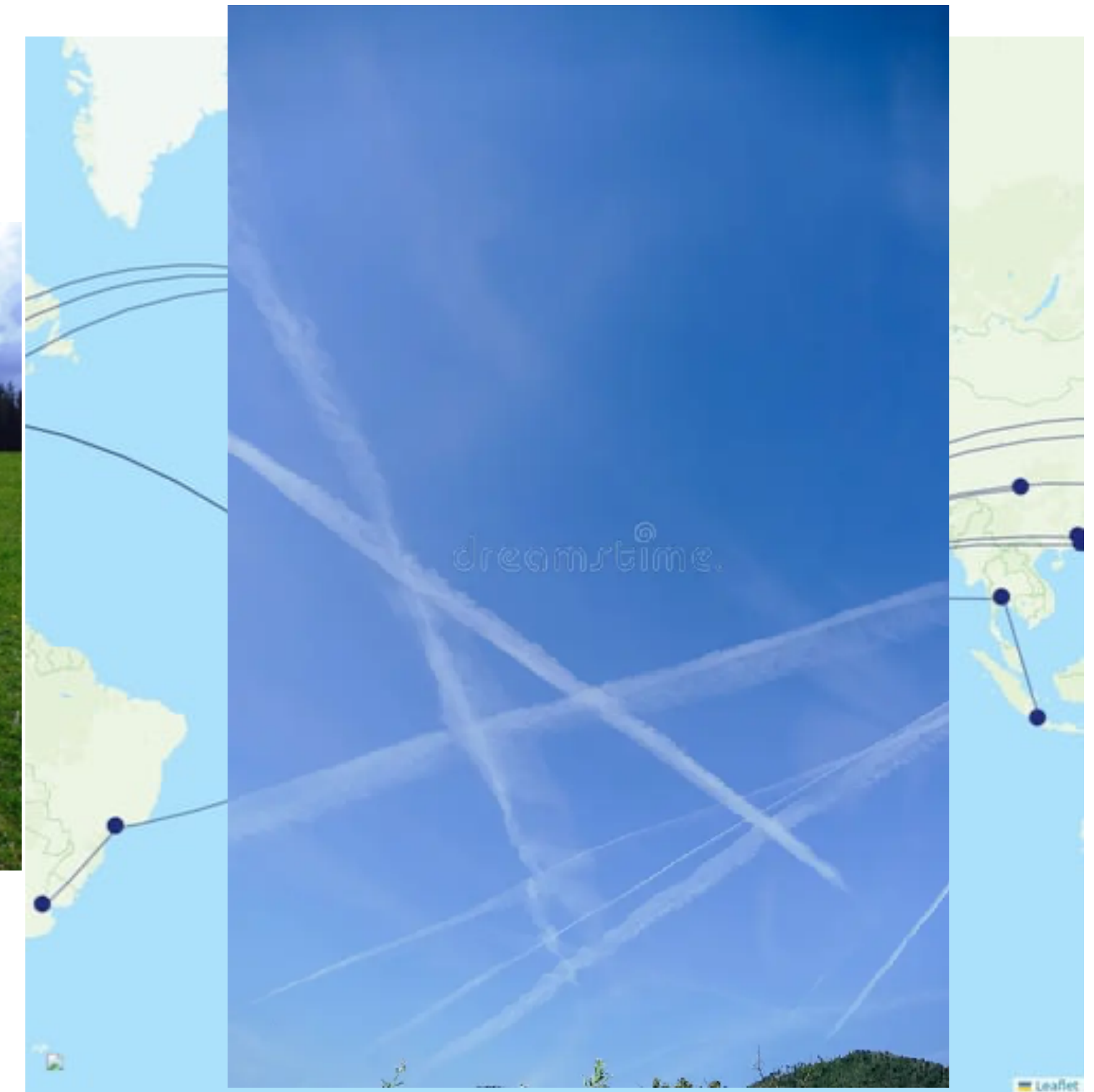
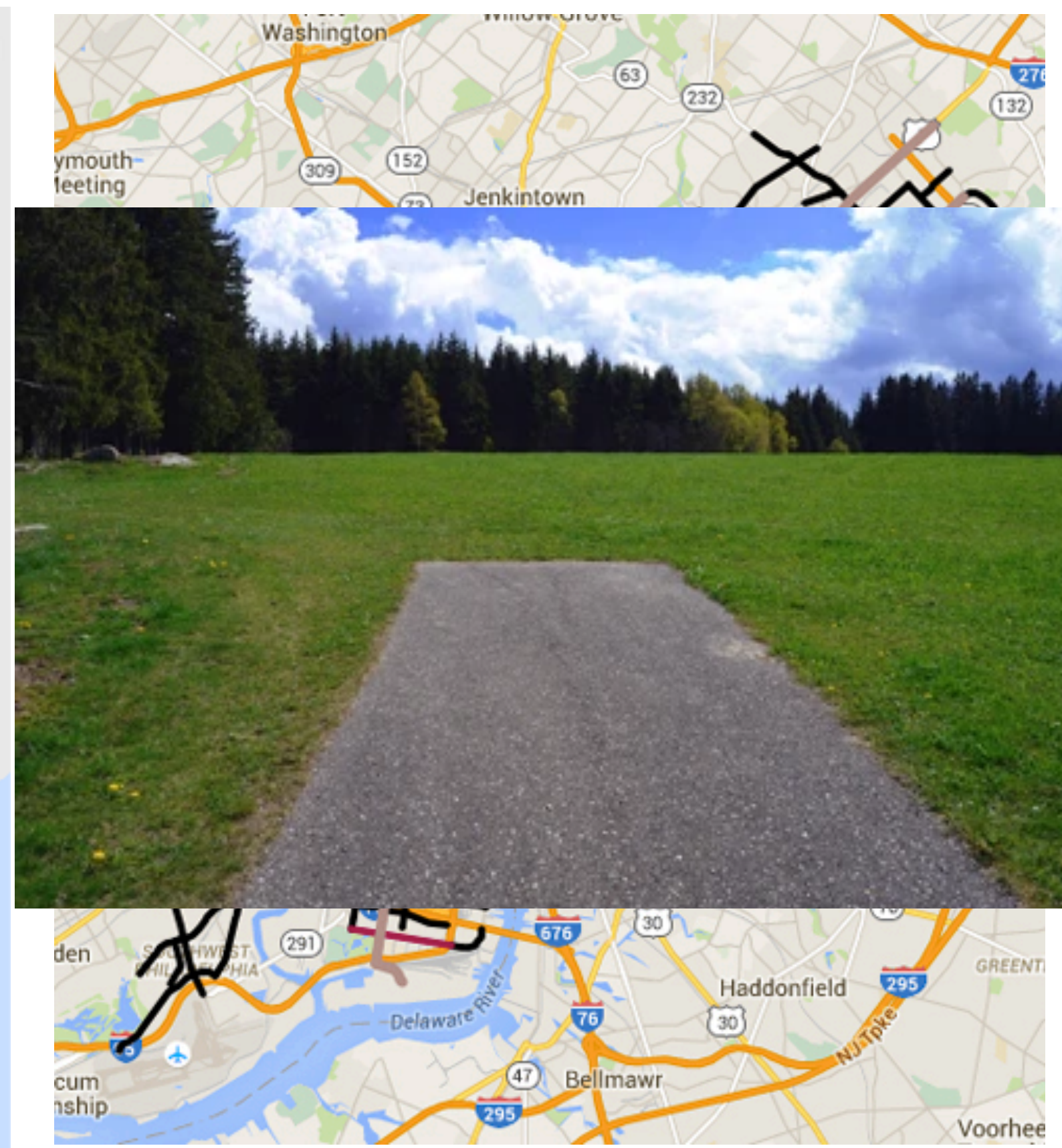
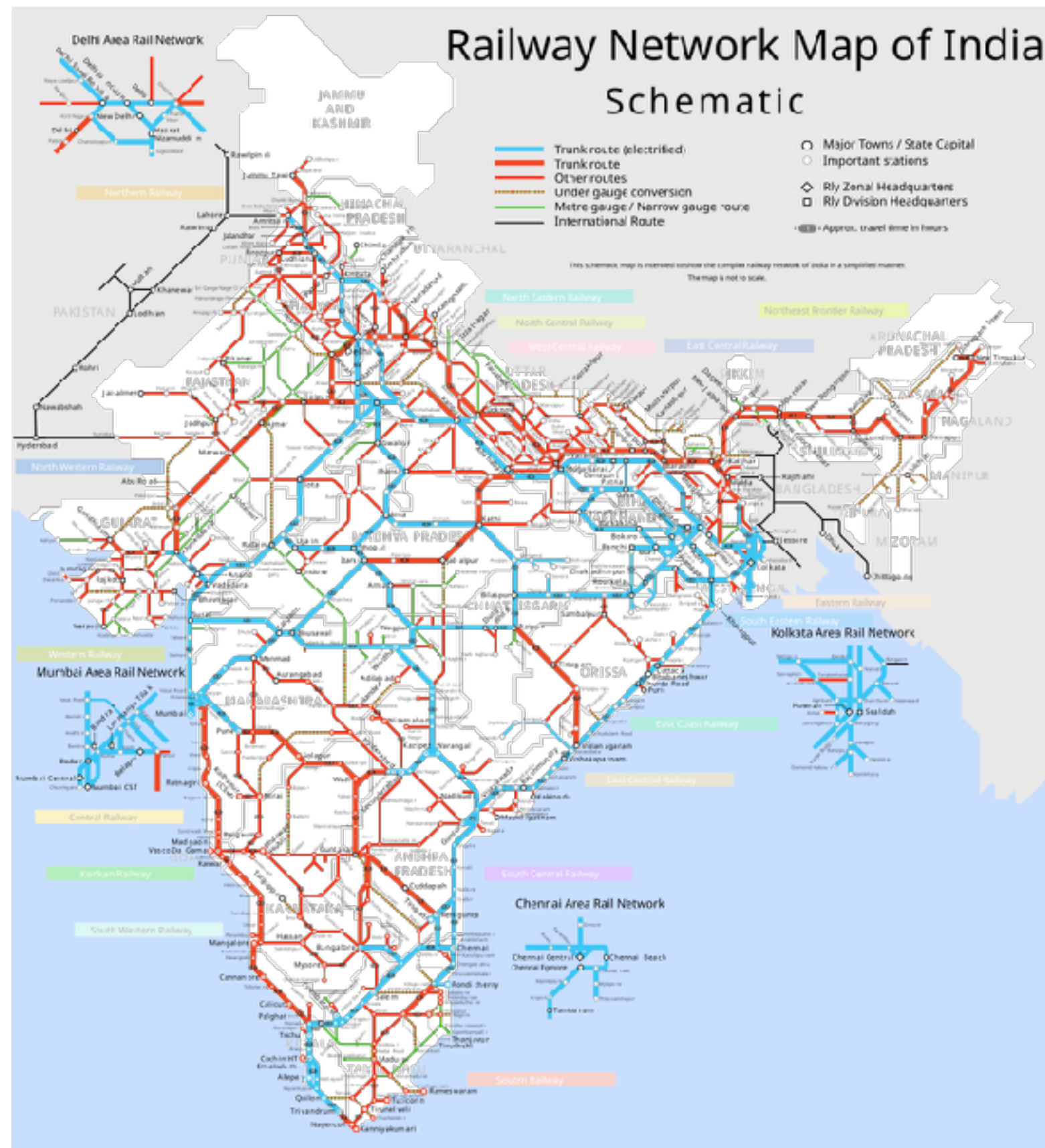


https://www.researchgate.net/figure/Illustration-of-a-power-grid-system_fig1_329873334
https://commons.wikimedia.org/wiki/File:Railway_network_map_of_India_-_Schematic.svg
<https://www.airlineinformation.com/ET>
<https://www.phillymag.com/be-well-philly/2016/02/09/protected-bike-lanes-map/>
<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013292>



+ car, bus, metro, ...

https://www.researchgate.net/figure/Illustration-of-a-power-grid-system_fig1_329873334
https://commons.wikimedia.org/wiki/File:Railway_network_map_of_India_-_Schematic.svg
<https://www.airlineinformation.com/ET>
<https://www.phillymag.com/be-well-philly/2016/02/09/protected-bike-lanes-map/>
<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013292>



+ car, bus, metro, ...

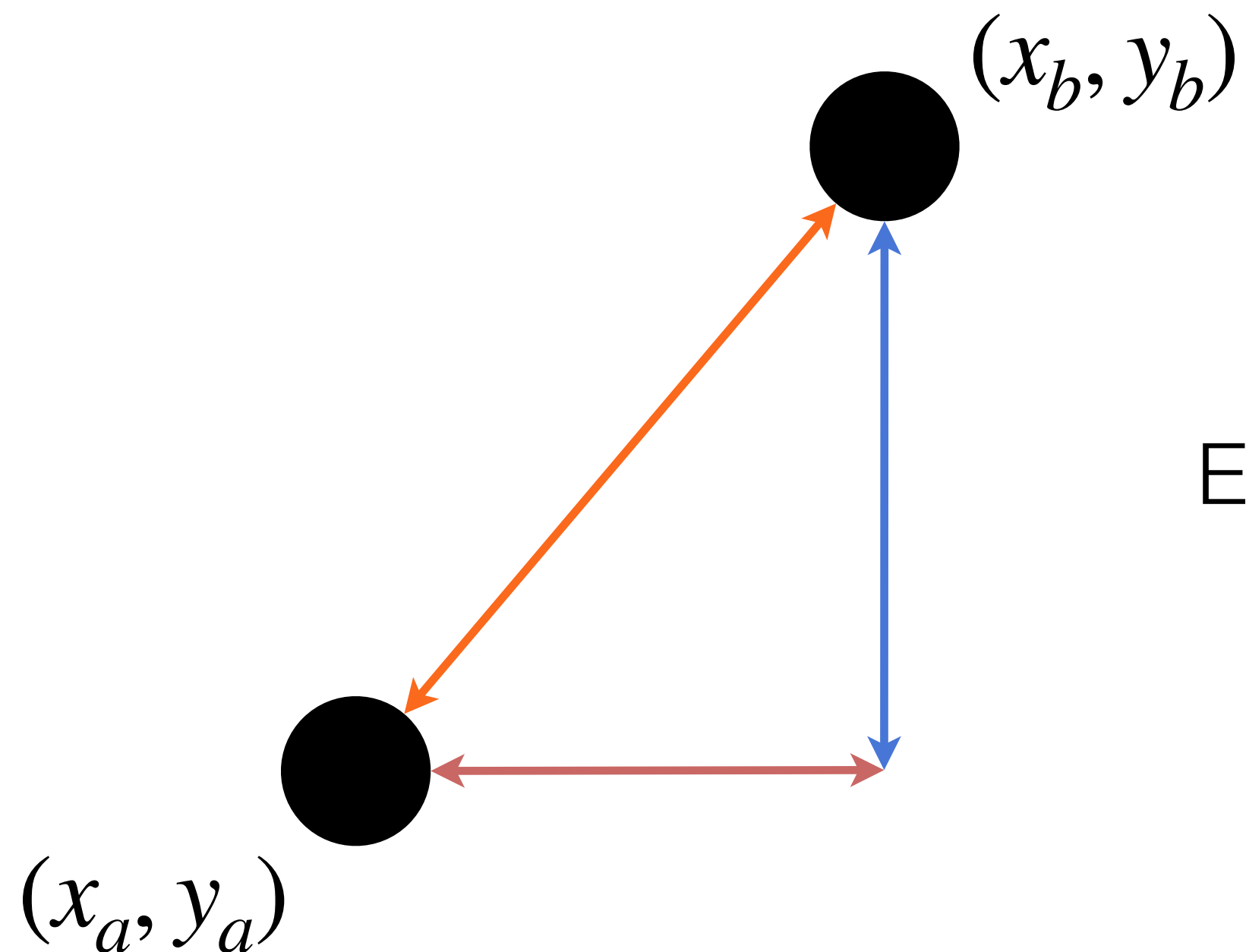
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https://commons.wikimedia.org/wiki/File:Railway_network_map_of_India_-_Schematic.svg
<https://www.airlineinformation.com/ET>
<https://www.phillymag.com/be-well-philly/2016/02/09/protected-bike-lanes-map/>
<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013292>

Spatial networks: Distance

A **spatial network** has nodes in a metric space

You can measure a distance between nodes

In practice, we will usually use the **Euclidean distance**

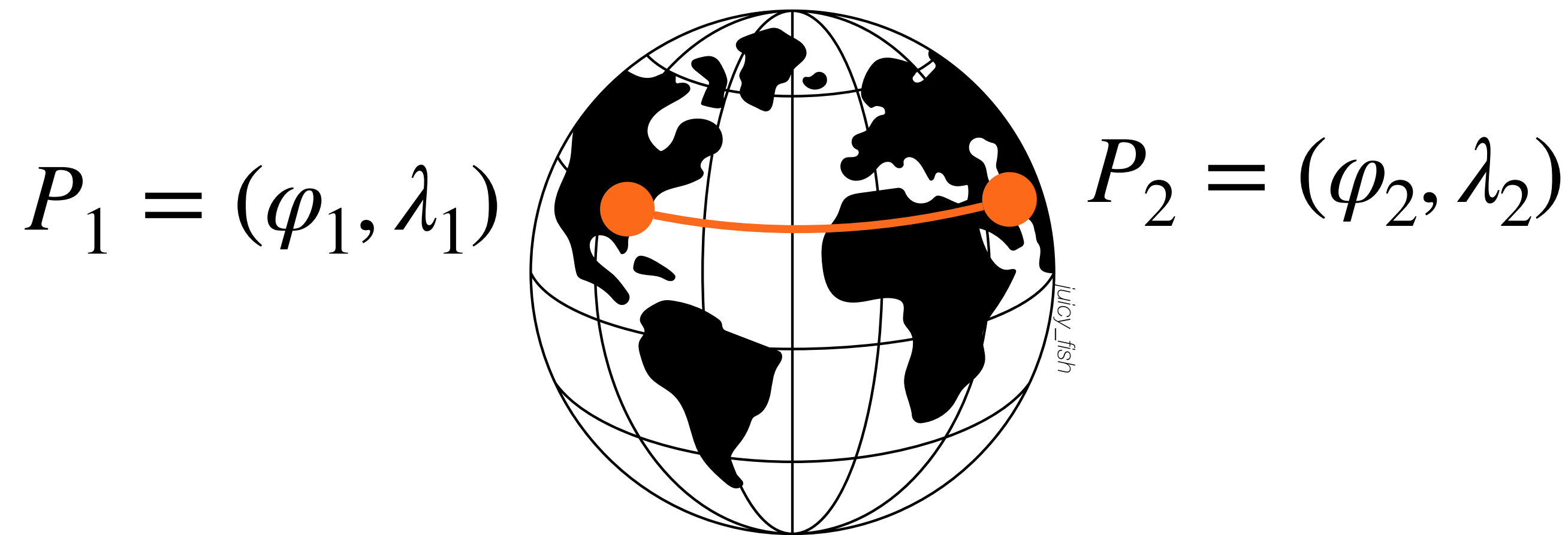


Euclidean distance (L_2 norm): $D = \sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}$

A **spatial network** has nodes in a metric space

You can measure a distance between nodes

For larger distances on a sphere use the **Haversine distance**



$$d = 2r \arcsin \left(\sqrt{\sin^2 \left(\frac{\varphi_2 - \varphi_1}{2} \right) + \cos \varphi_1 \cdot \cos \varphi_2 \cdot \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right)$$

A **spatial network** has nodes in a metric space

You can measure a distance between nodes

For example, `geopandas`, `OSMnx`, ...
already account for this; but:
ALWAYS use a
projected coordinate reference system (CRS)
for distance computations

$$d = 2r \arcsin \left(\sqrt{\sin^2 \left(\frac{\varphi_2 - \varphi_1}{2} \right) + \cos \varphi_1 \cdot \cos \varphi_2 \cdot \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right)$$

Denmark frees 32 inmates over flaws in phone geolocation evidence

Denmark imposes two-month moratorium on use of mobile phone records in trials

Jon Henley

Thu 12 Sep 2019 12.13 CEST

Share

Prefer the Guardian on Google



A man on his phone corroborate other evid

Denmark has released 32 prisoners as part of an ongoing review of 10,700 criminal cases after serious questions arose about the reliability of geolocation data obtained from mobile phone operators, local media have reported.

Nearly 40 new cases have been postponed under a two-month moratorium on the use of mobile phone records in trials, which was imposed after police found multiple glitches in the software that converts raw data from phone masts into usable evidence.

Spatial networks: Planarity

The **three utilities problem**



How can you link each house to each utility ?

The **three utilities problem**



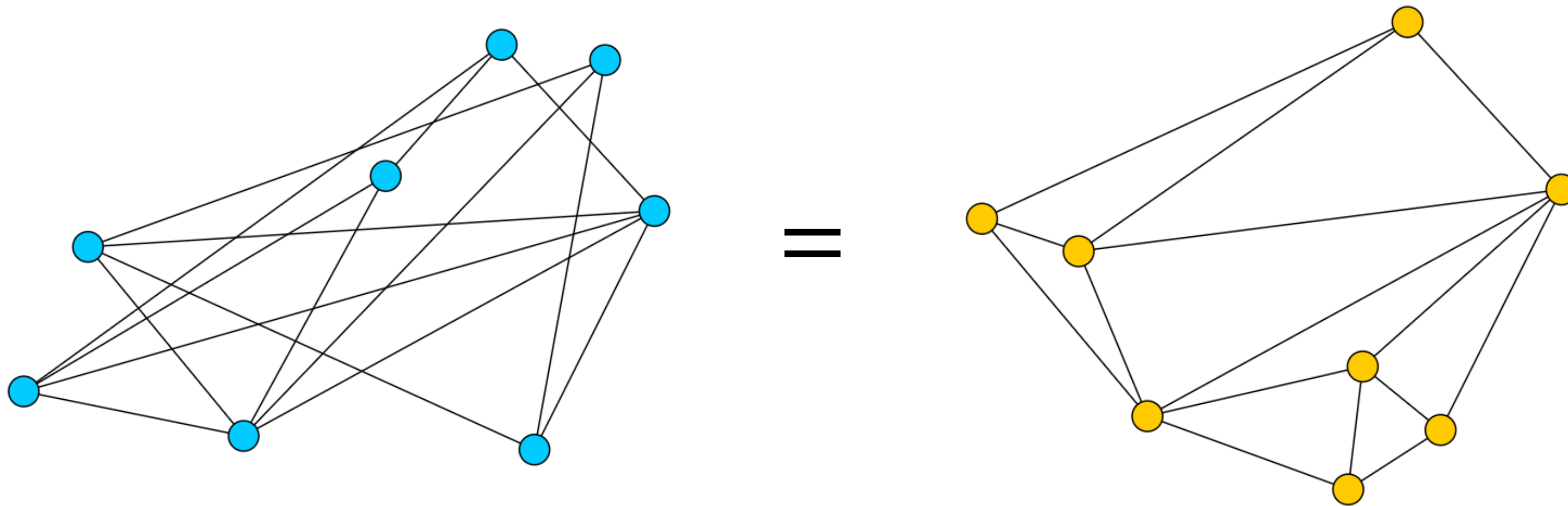
100+ years old, was solved in 1930 by Kuratowski:

You can't

(Kuratowski, 1930)

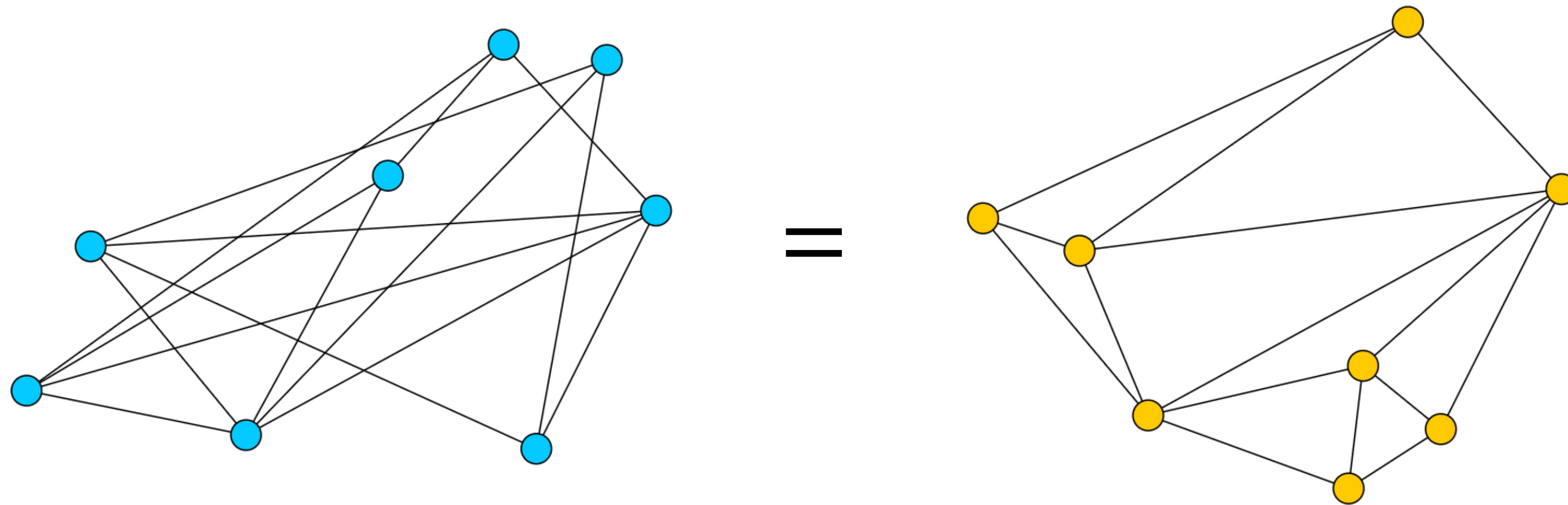
A graph is **planar** if it can be **embedded in the plane**

Planar: You can move the nodes around
until no links cross



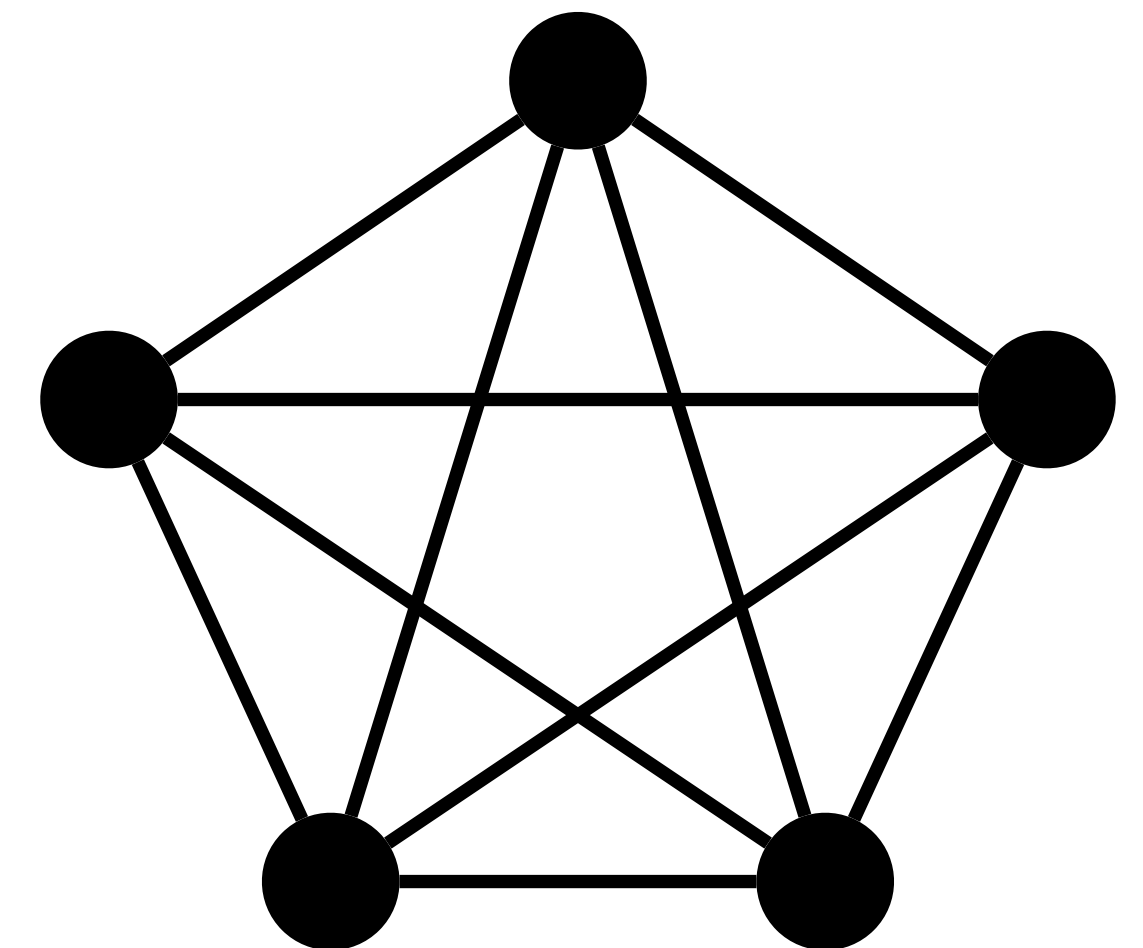
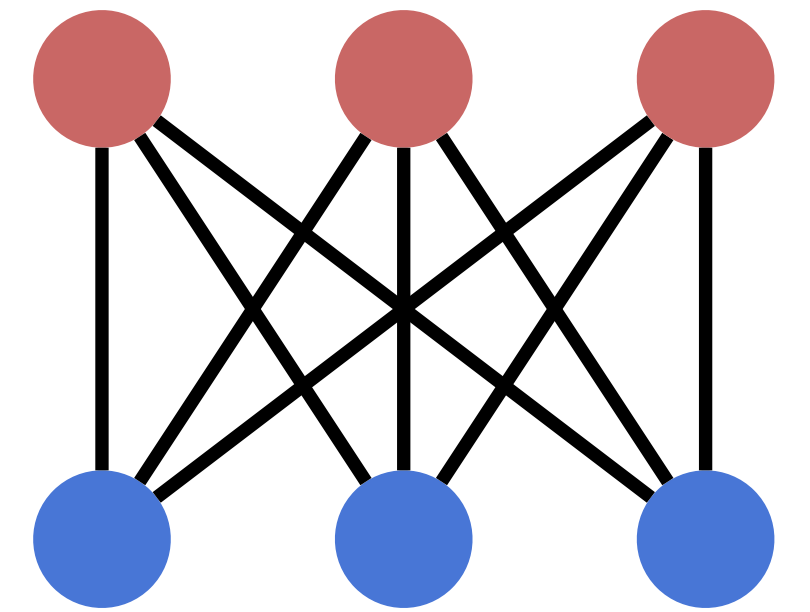
A graph is **planar** if it can be **embedded in the plane**

Planar: You can move the nodes around until no links cross



Non-planar: You can't

Utility graph



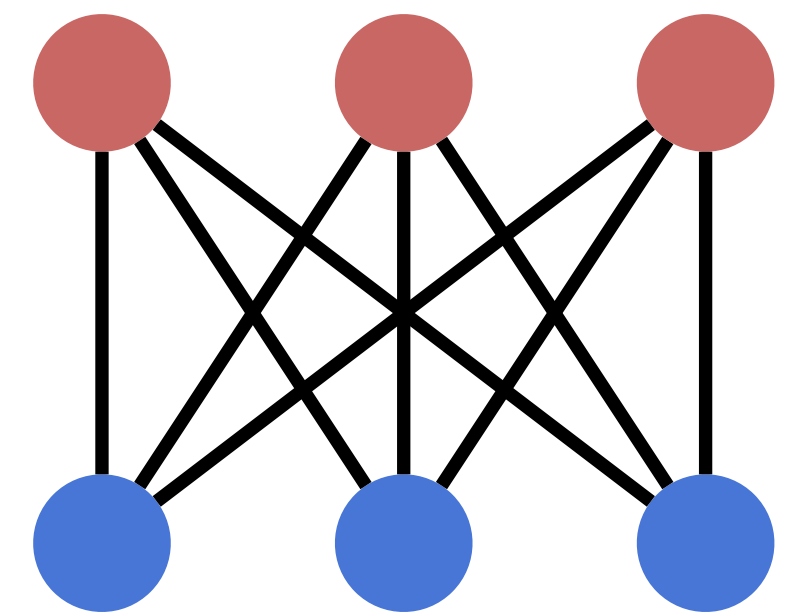
Kuratowski's theorem gives an easy way to check planarity

Kuratowski's theorem

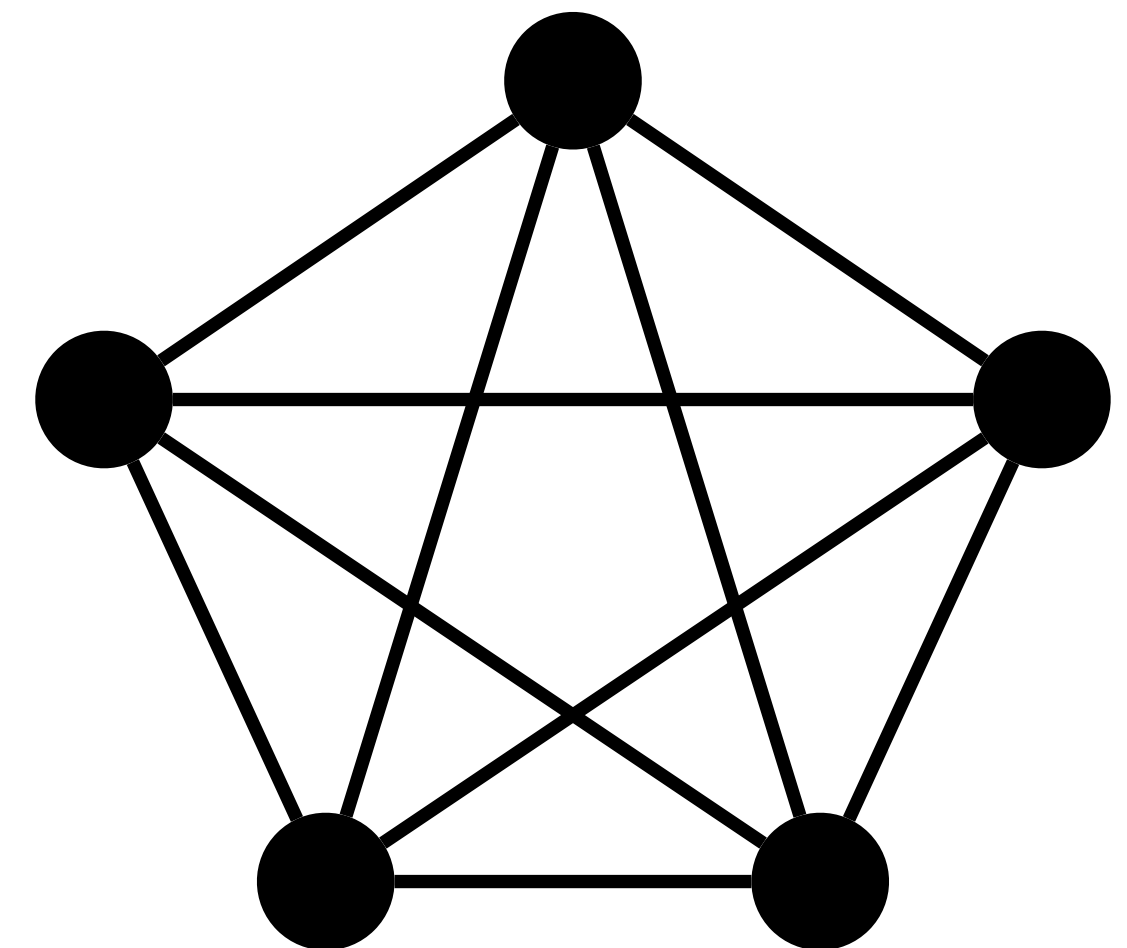
A graph is planar if and only if it does not contain (a subgraph that is a subdivision of) K_5 or $K_{3,3}$

(Kuratowski, 1930)

$K_{3,3}$



K_5



Kuratowski's theorem gives an easy way to check planarity

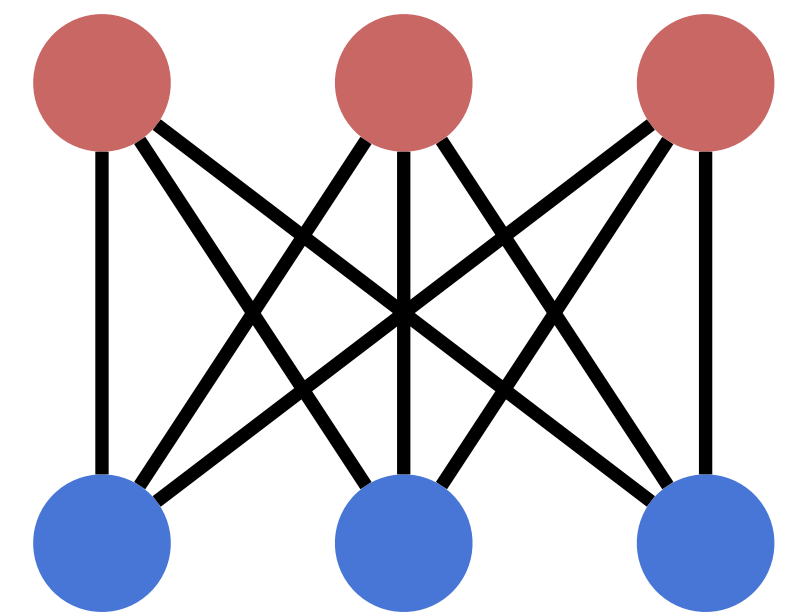
Kuratowski's theorem

A graph is planar if and only if it does not contain (a subgraph that is a subdivision of) K_5 or $K_{3,3}$

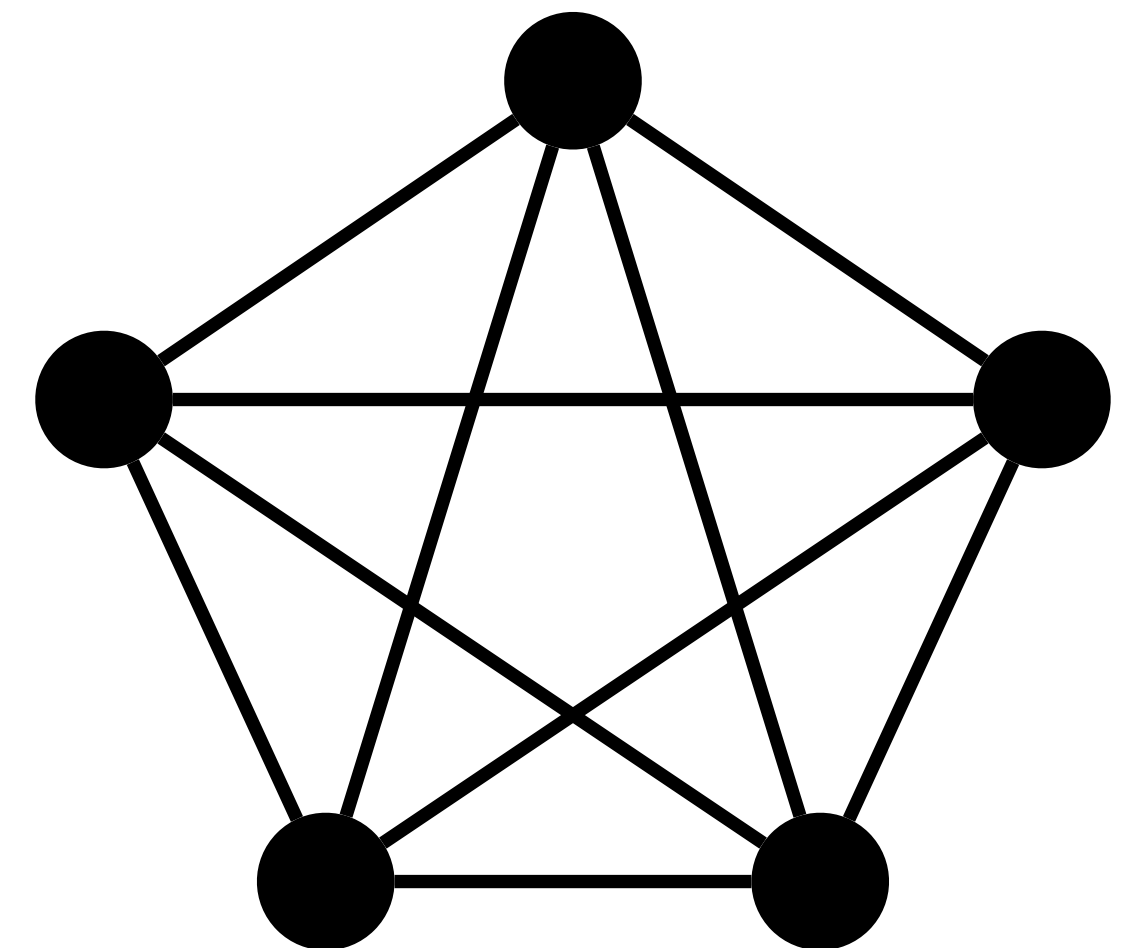
(Kuratowski, 1930)

Allows testing planarity in linear time.

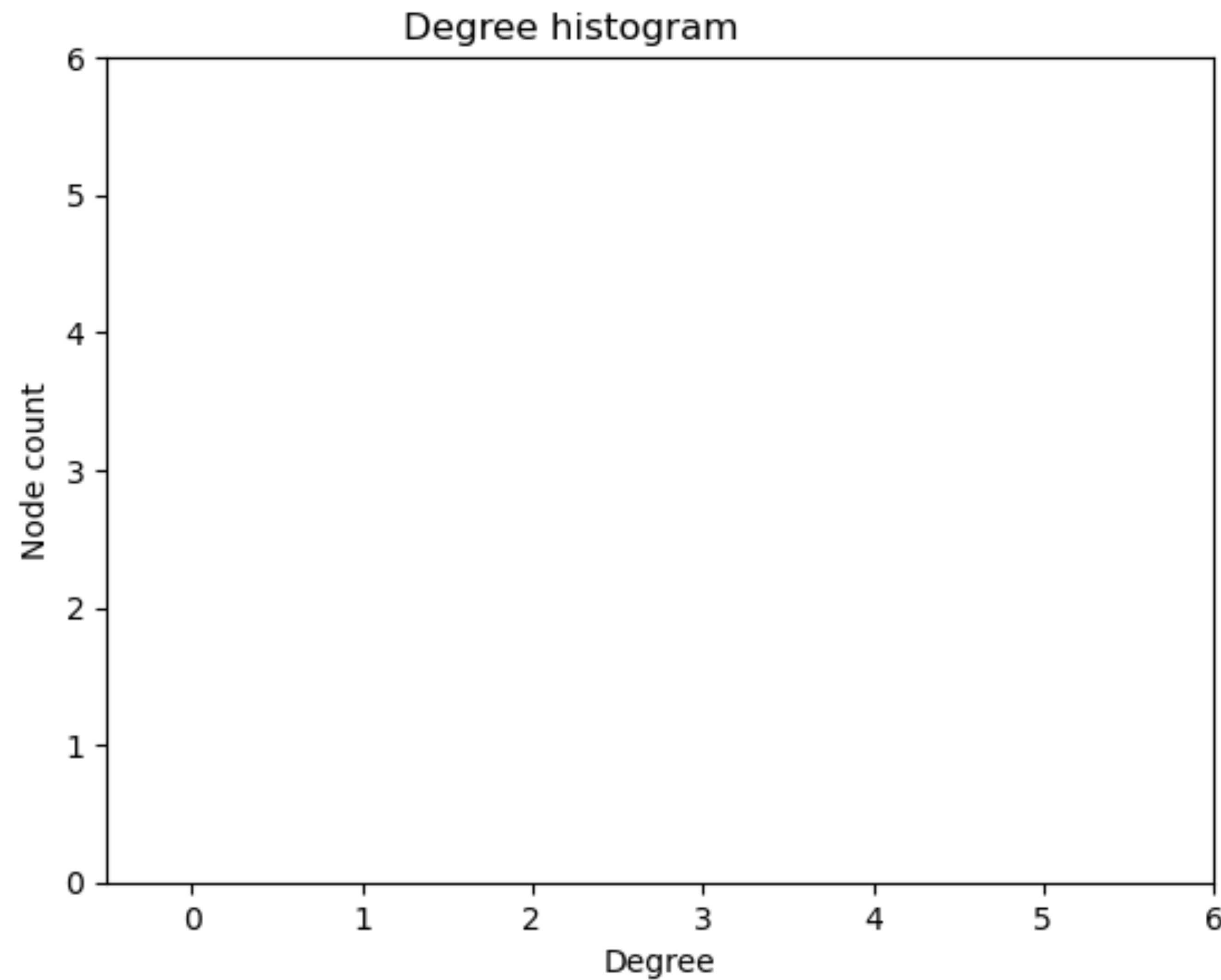
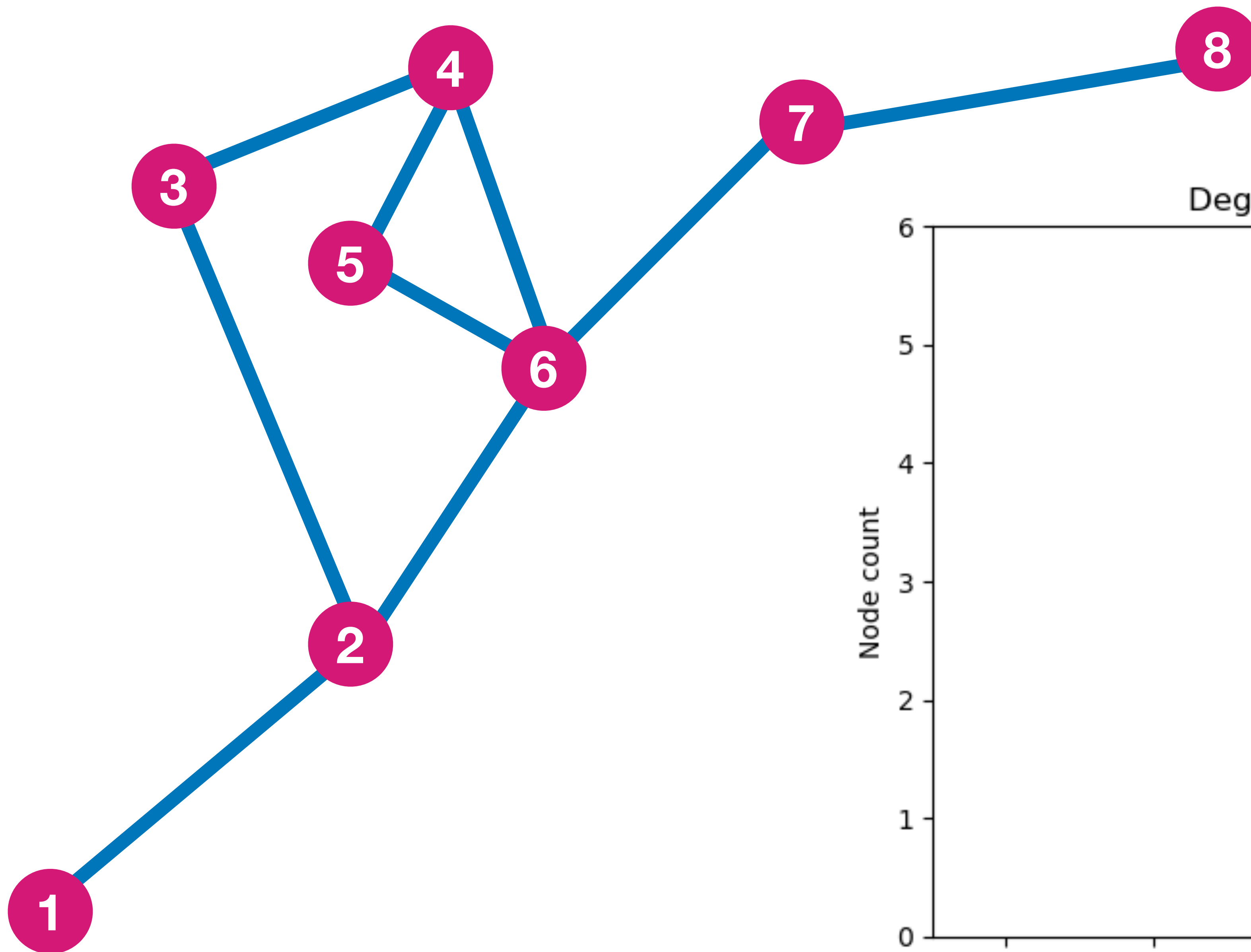
$K_{3,3}$



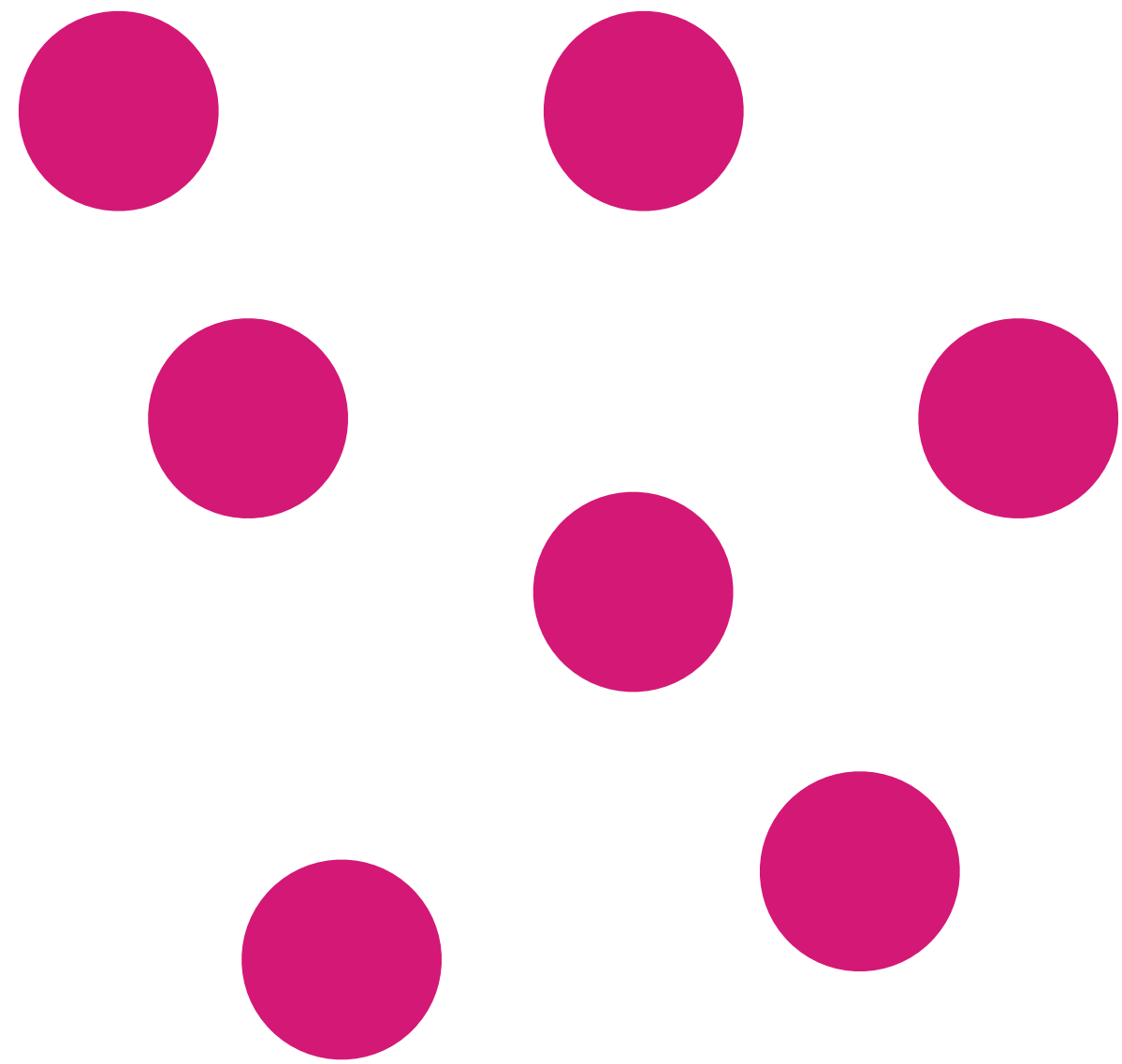
K_5



Spatial networks: Degree distributions



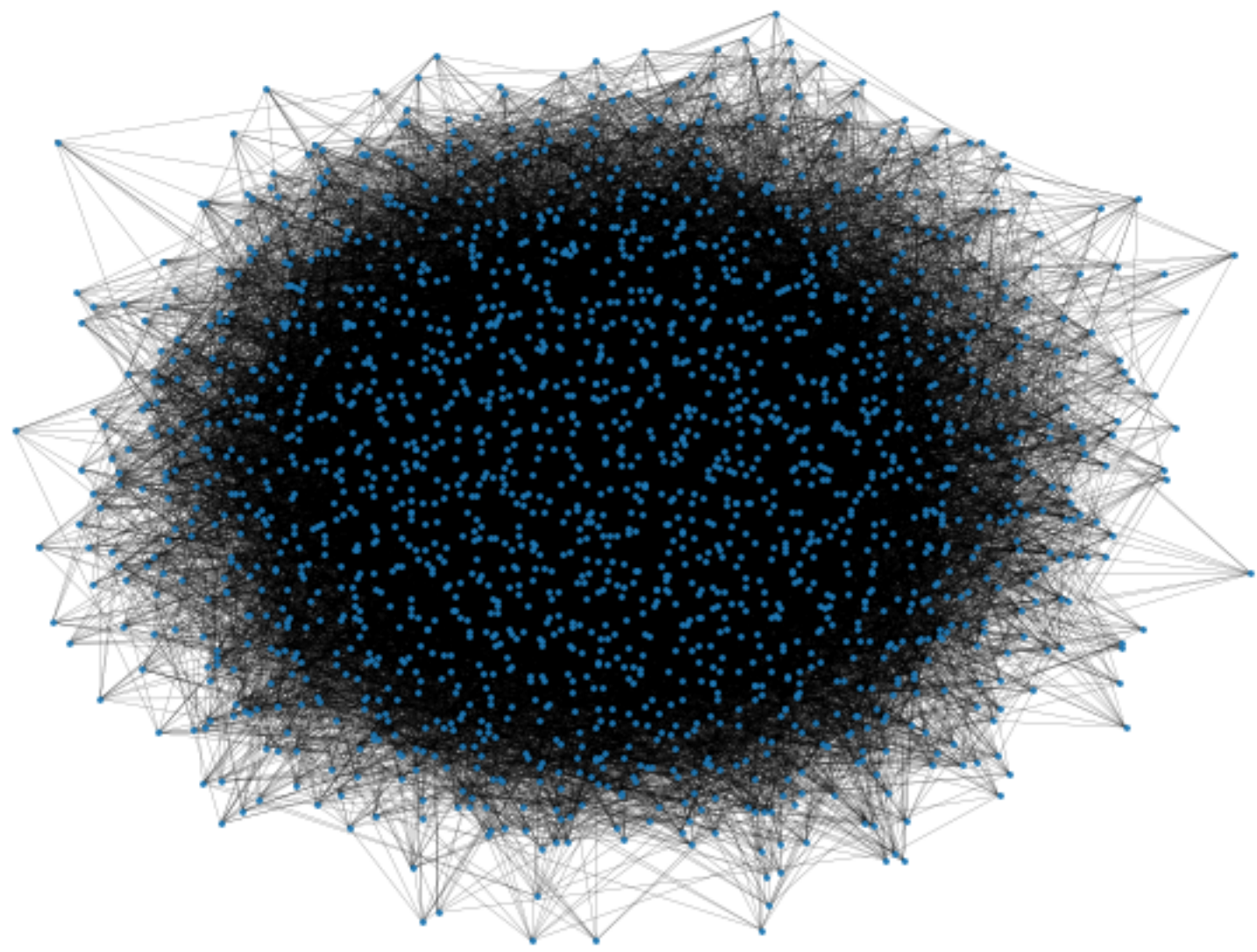
$G(N,p)$: Erdős-Rényi (random) network model



+

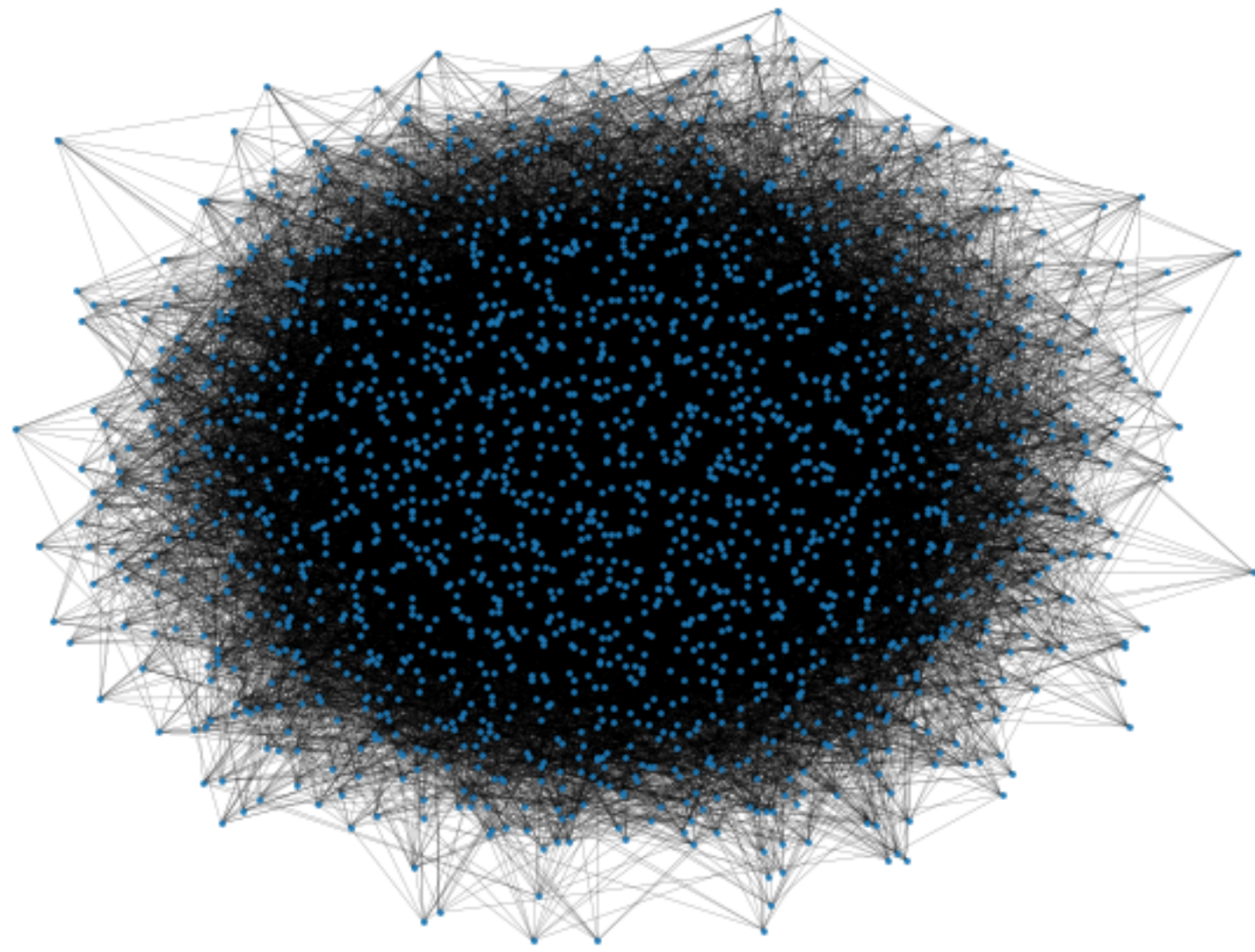


maybe! (“coinflip” for each potential node pair)



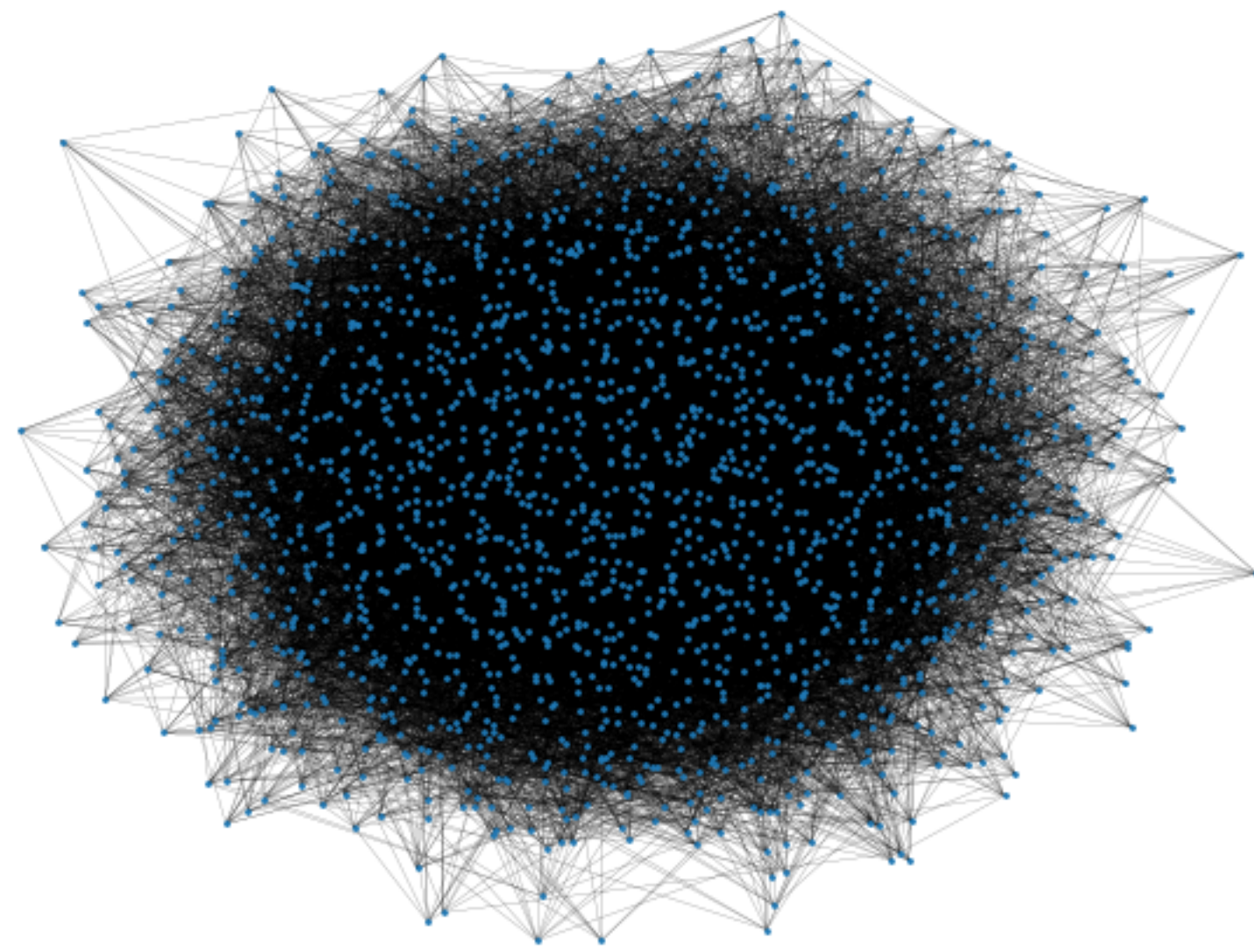
N=2000

What degree distribution do we expect in $G(N,p)$?

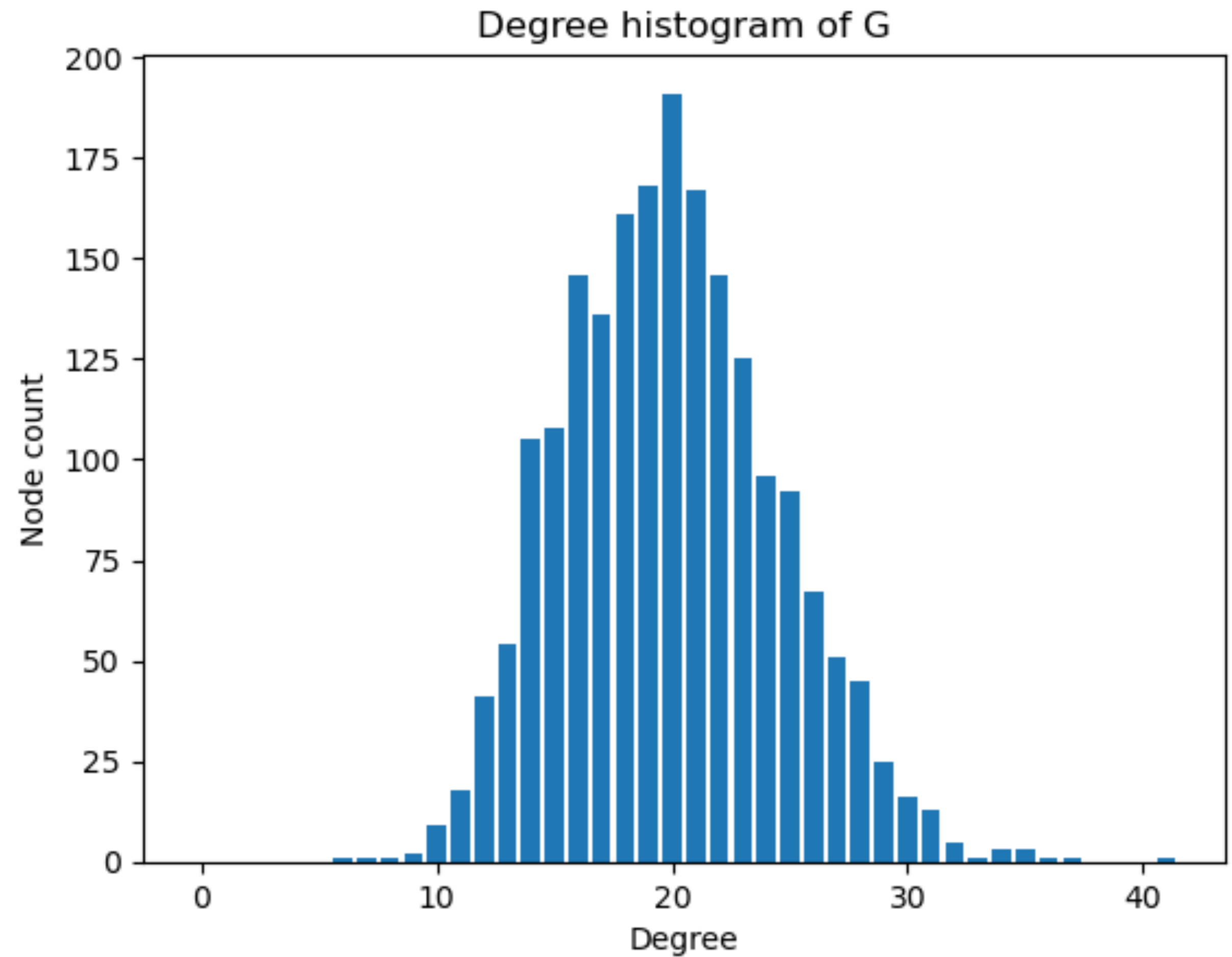


$N=2000$

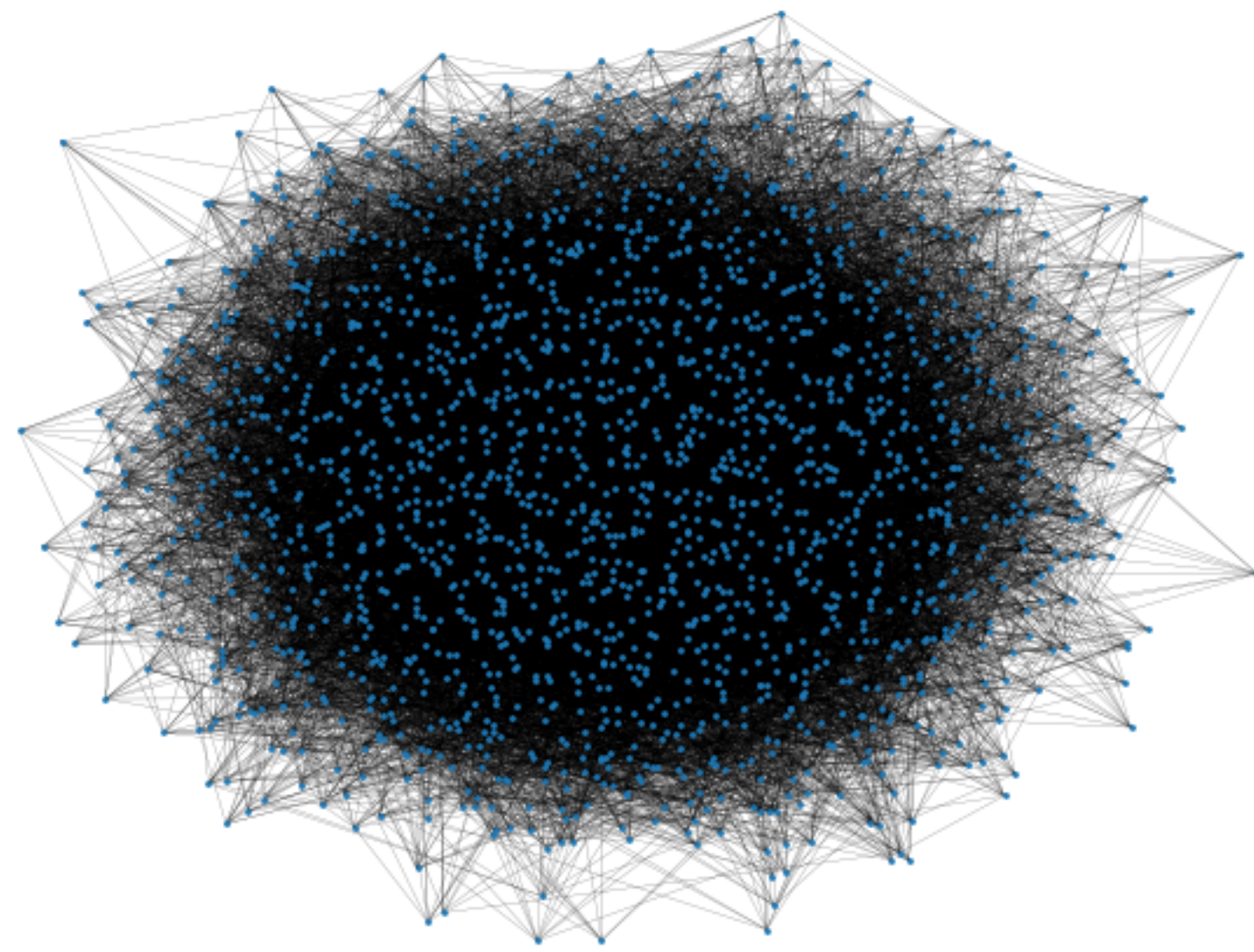
What degree distribution do we expect in $G(N,p)$?



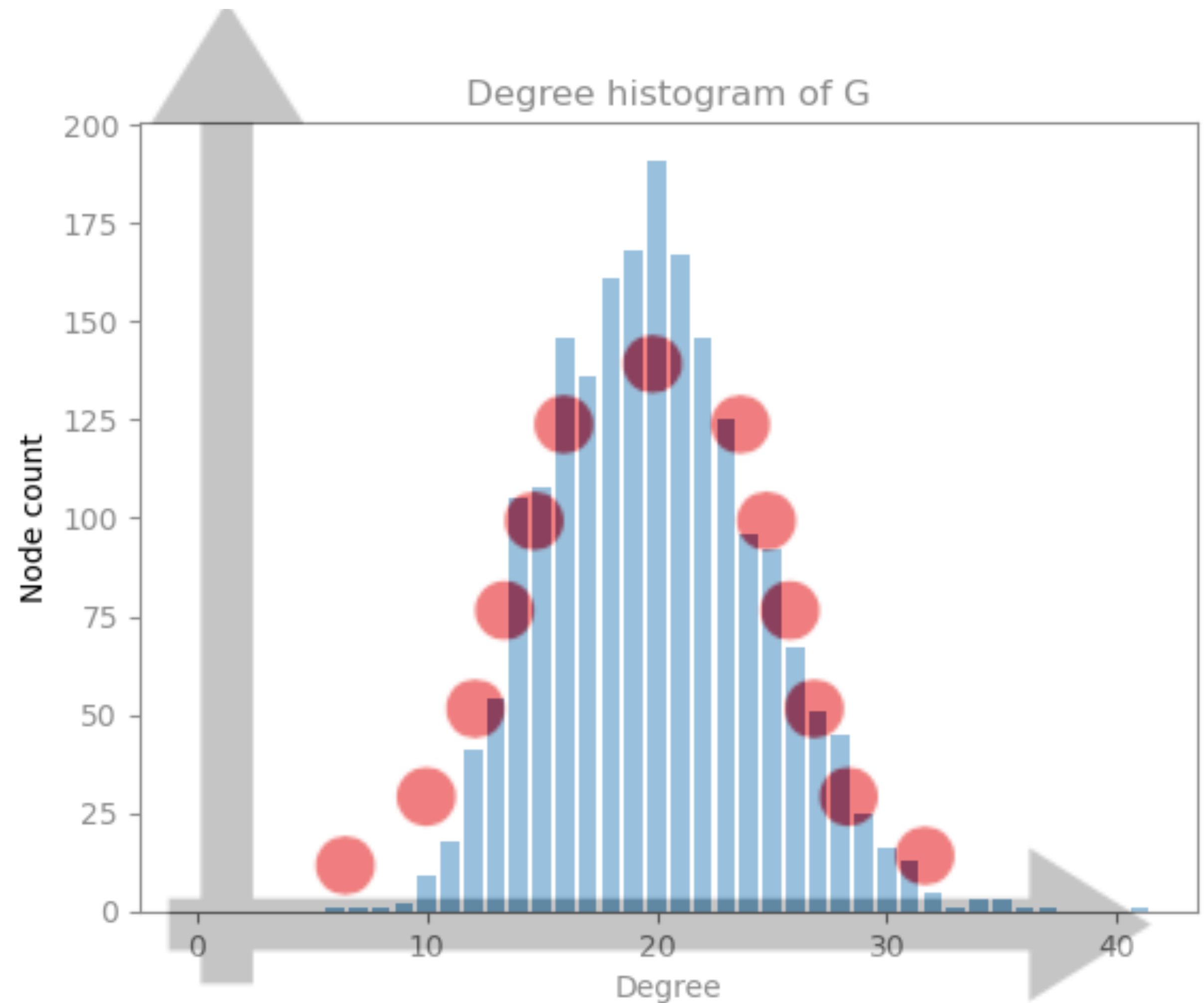
$N=2000$



What degree distribution do we expect in $G(N,p)$?



$N=2000$

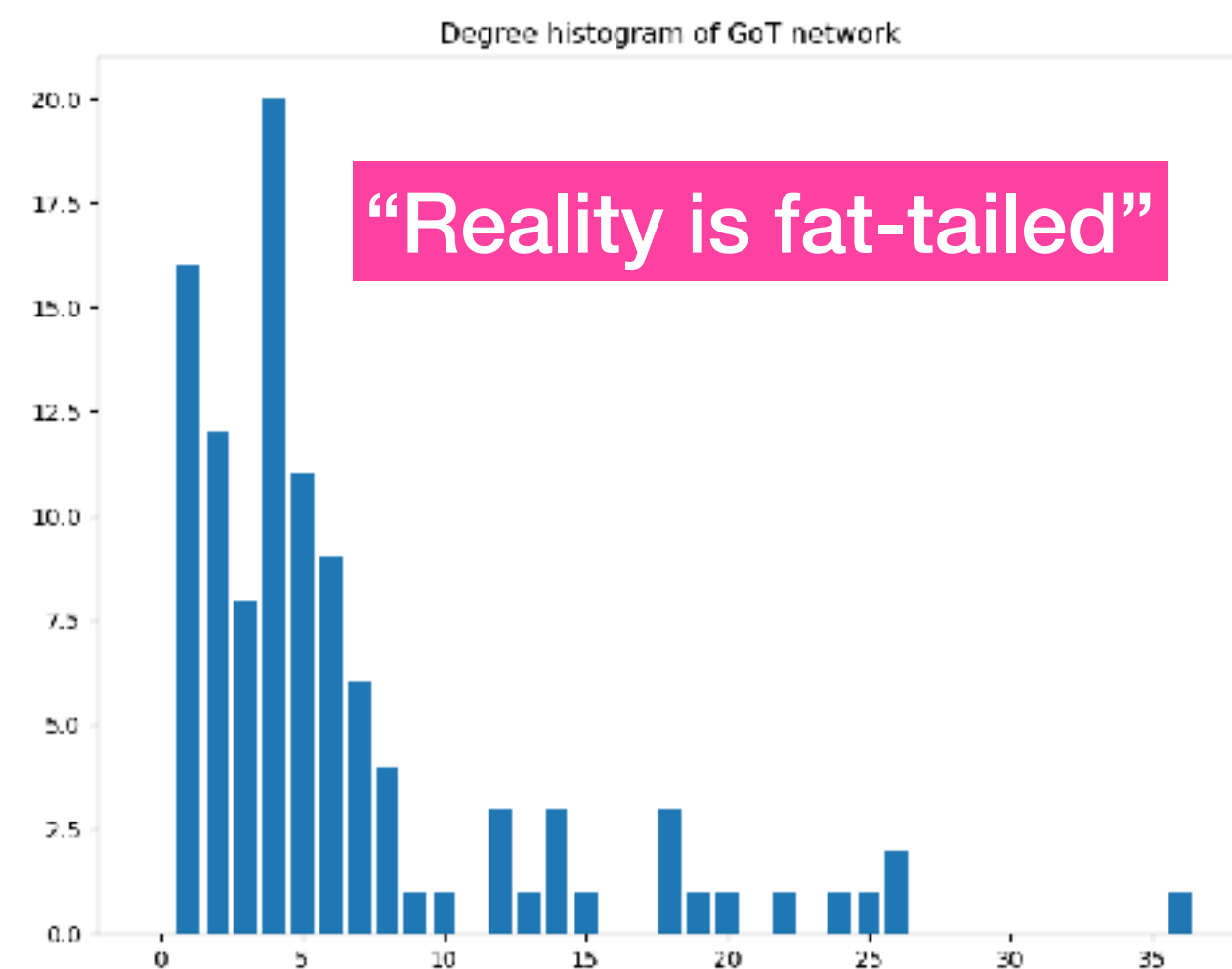
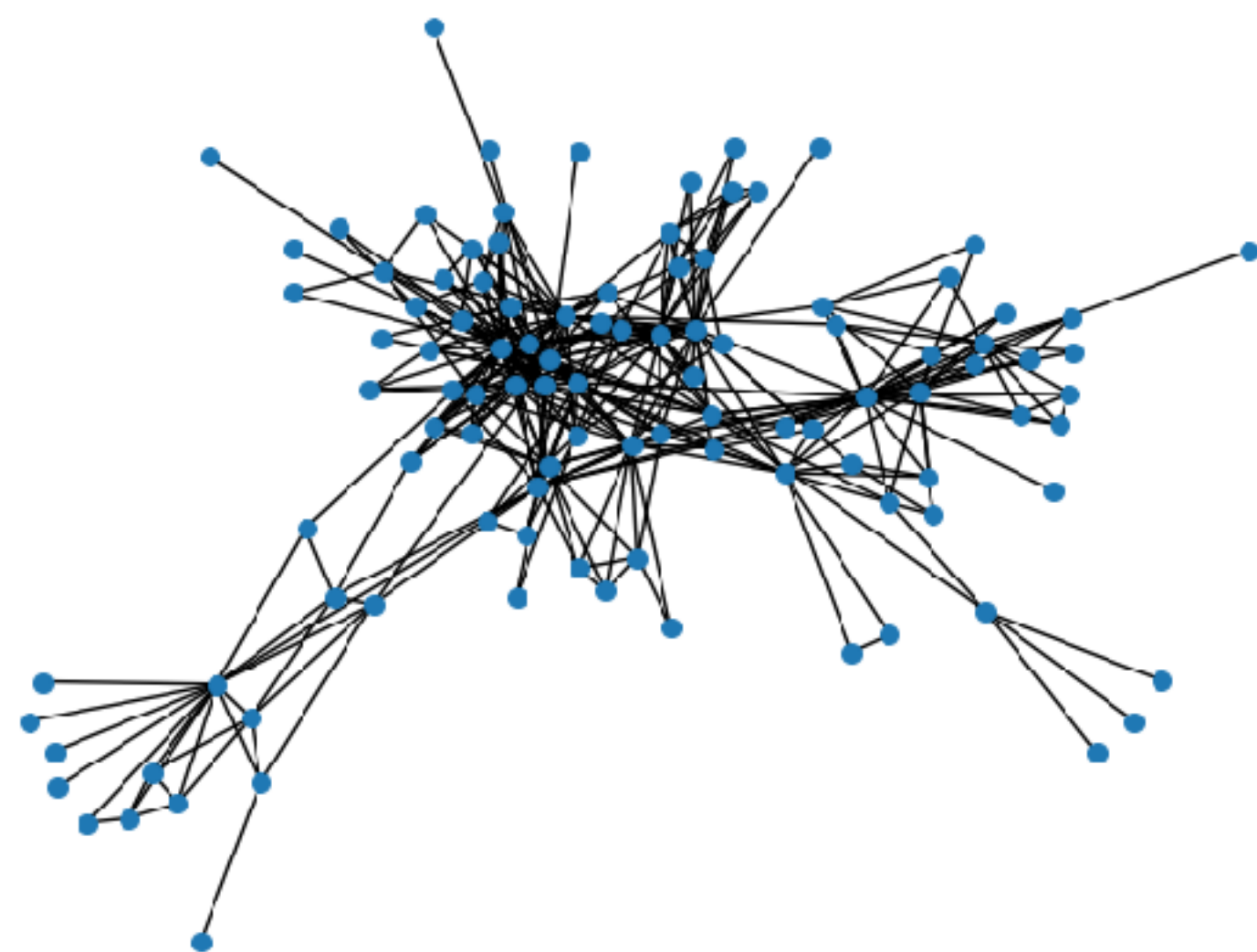


Binomial

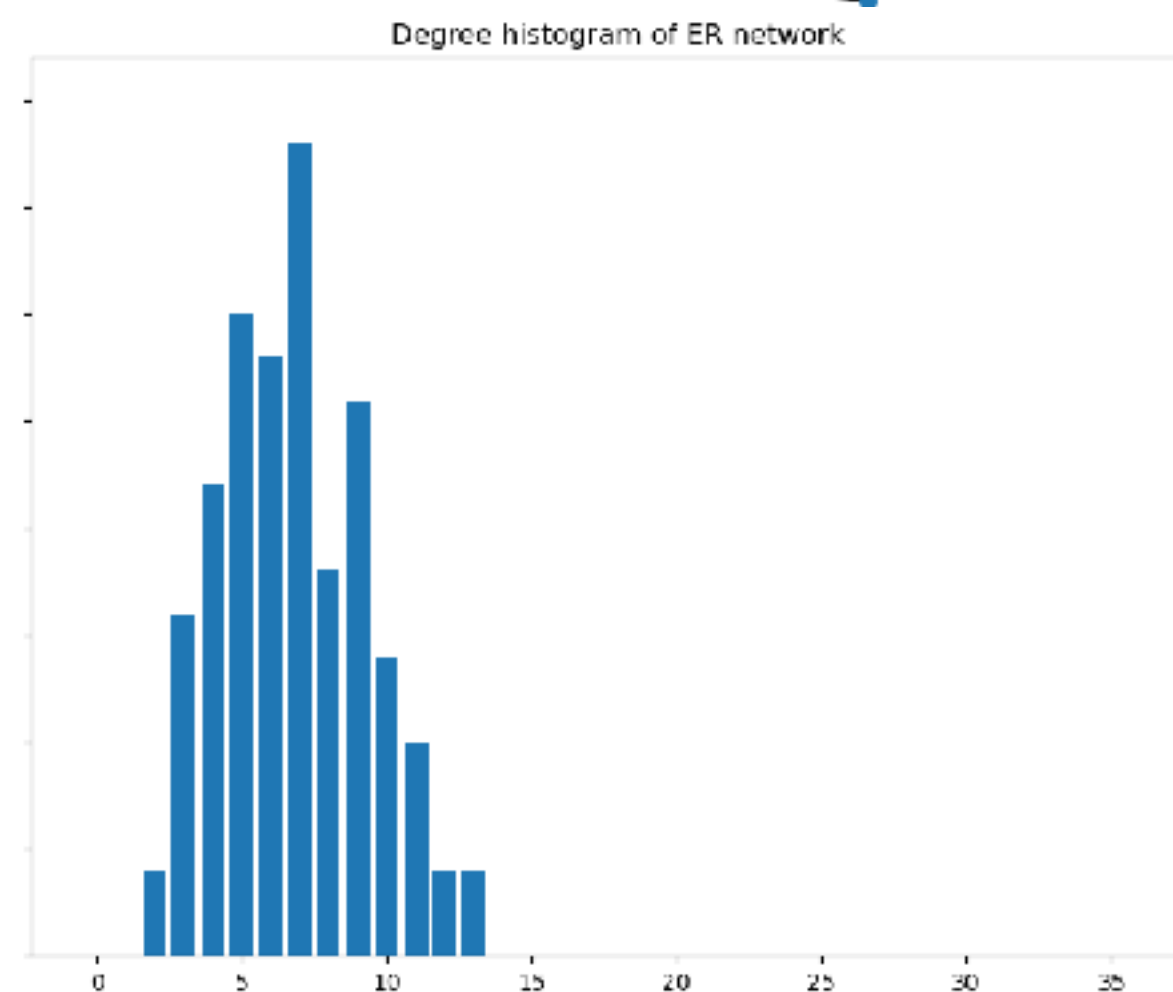
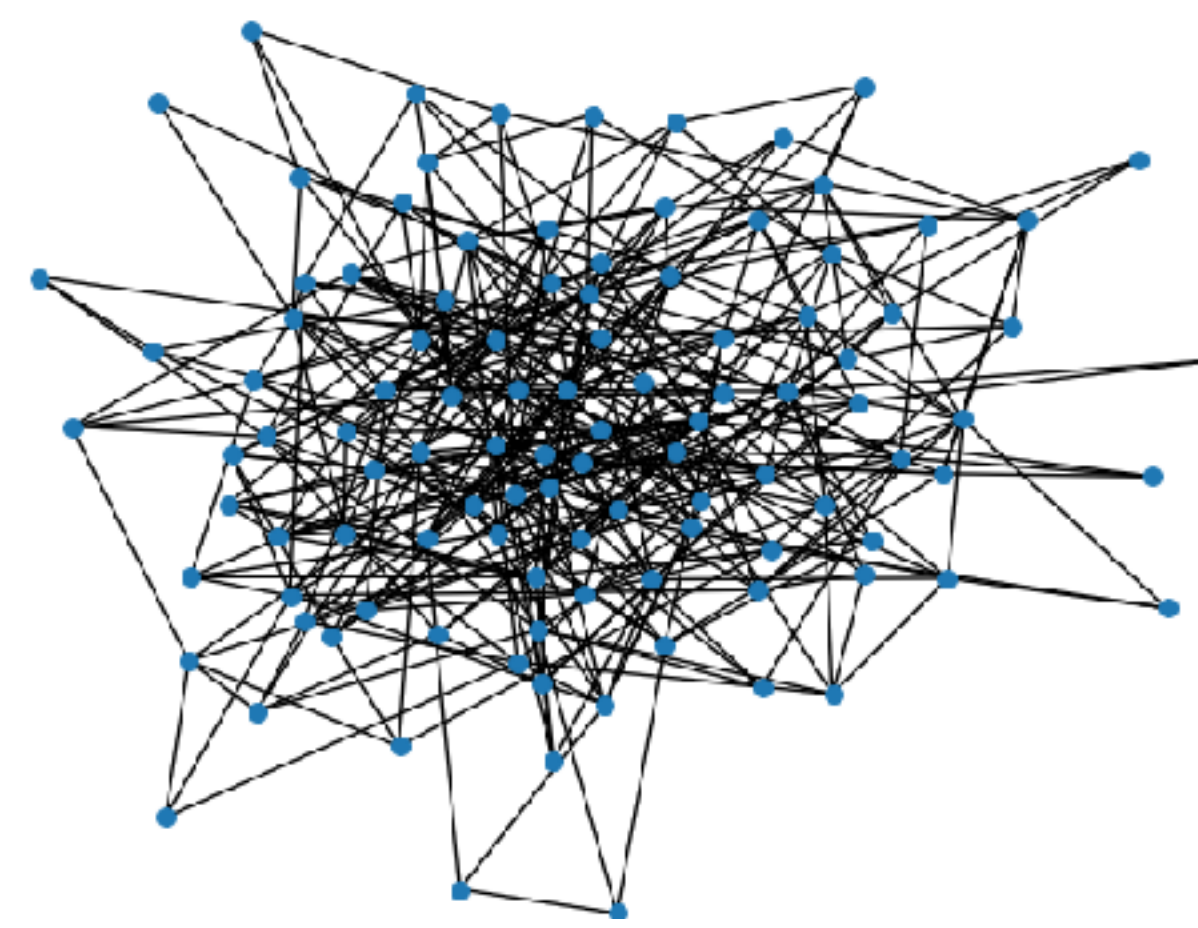
How can we compare networks?

Look at their degree distributions!

Game of Thrones Network

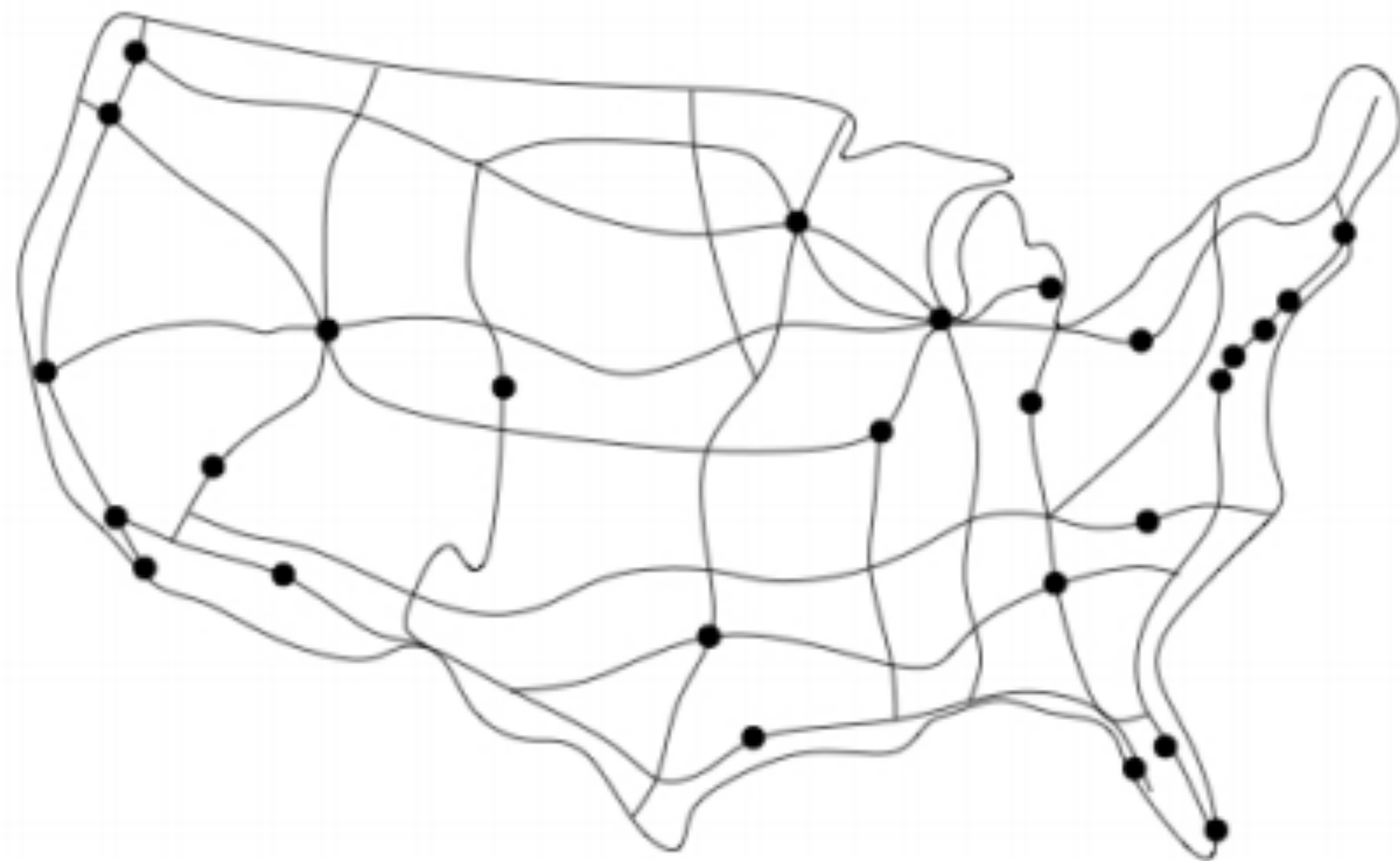
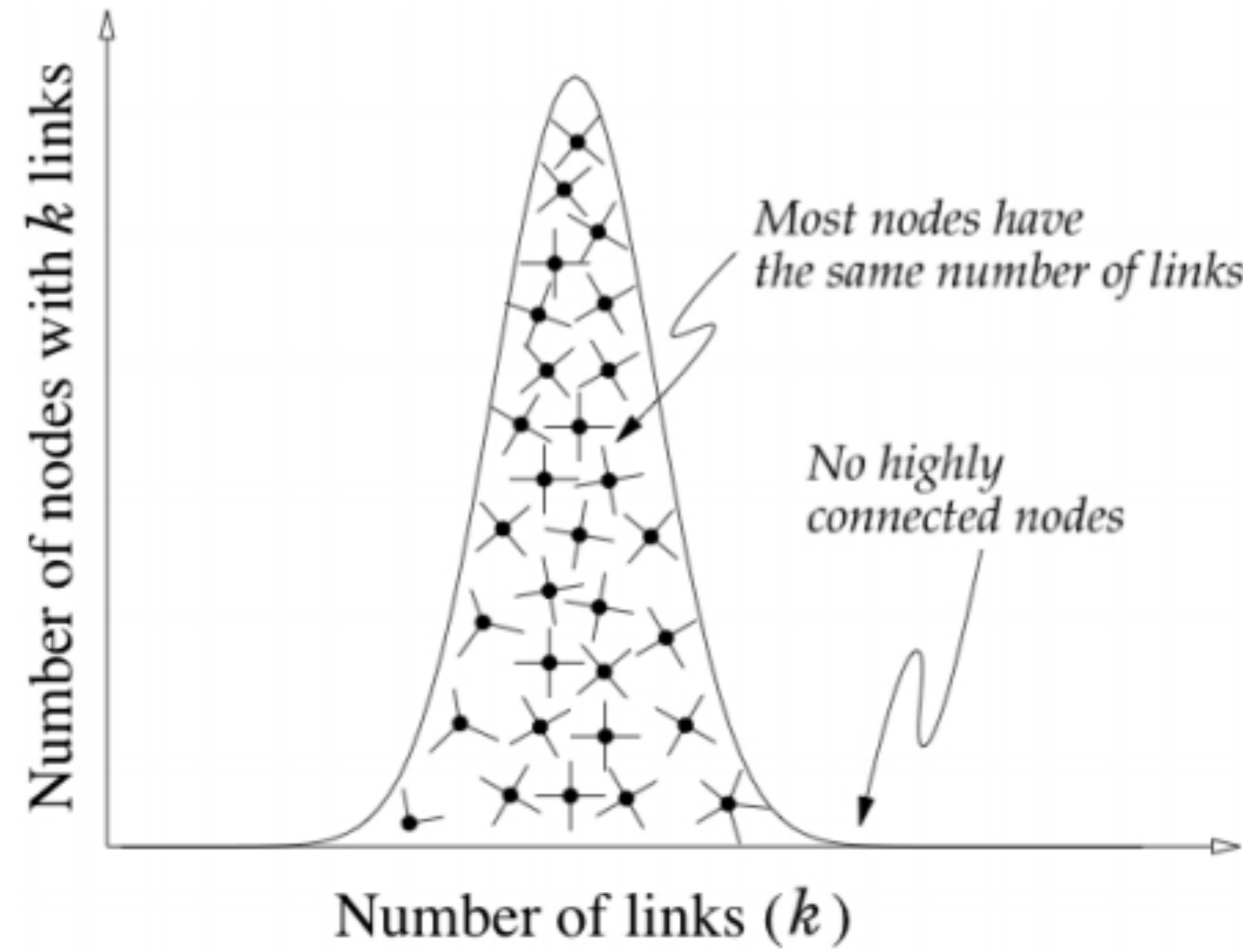


Erdős-Rényi network

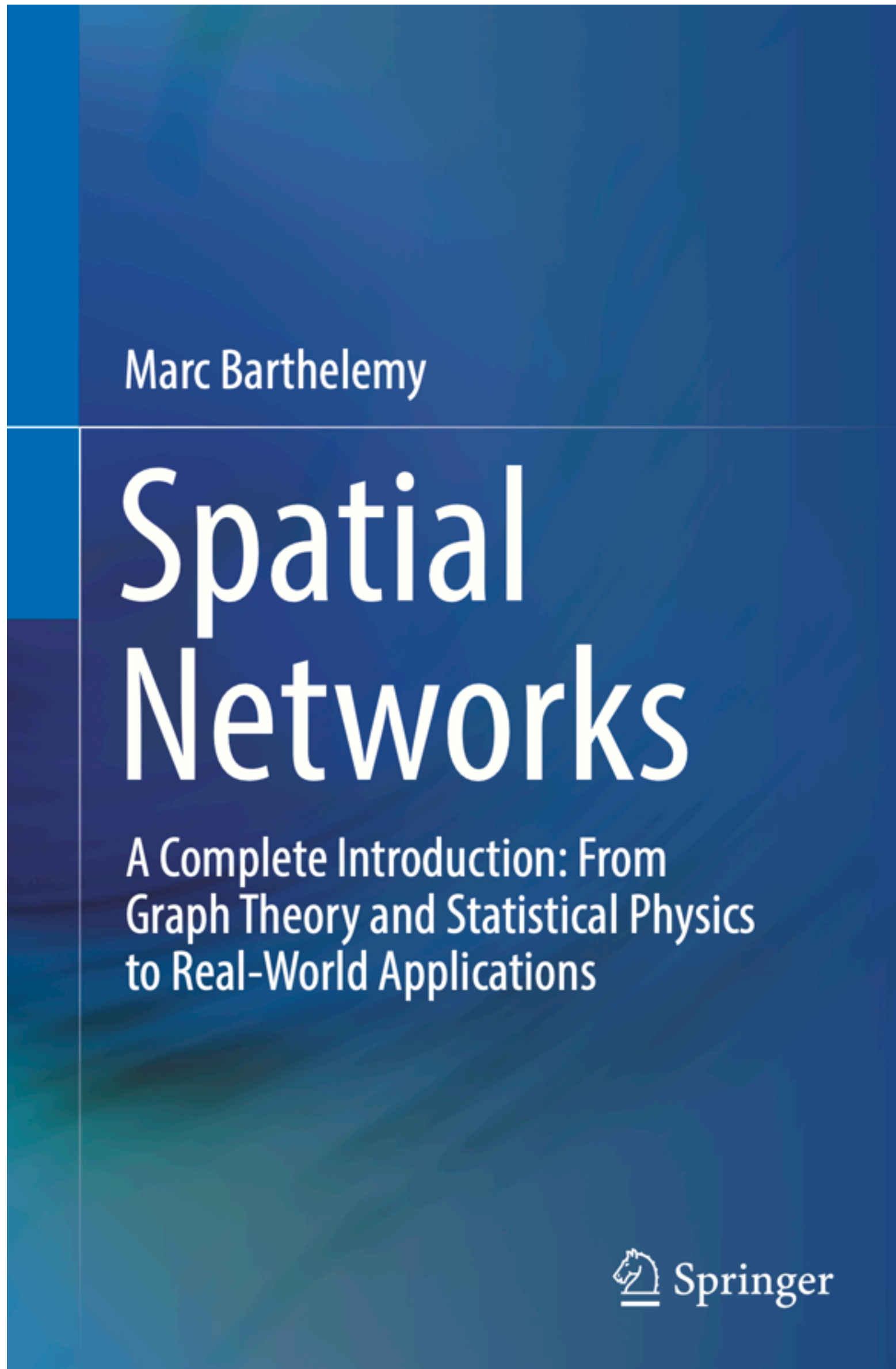


**However! Spatiality is a serious
constraint:**

Thin-tailed network



Streets



Chapter 4

Simple Measures



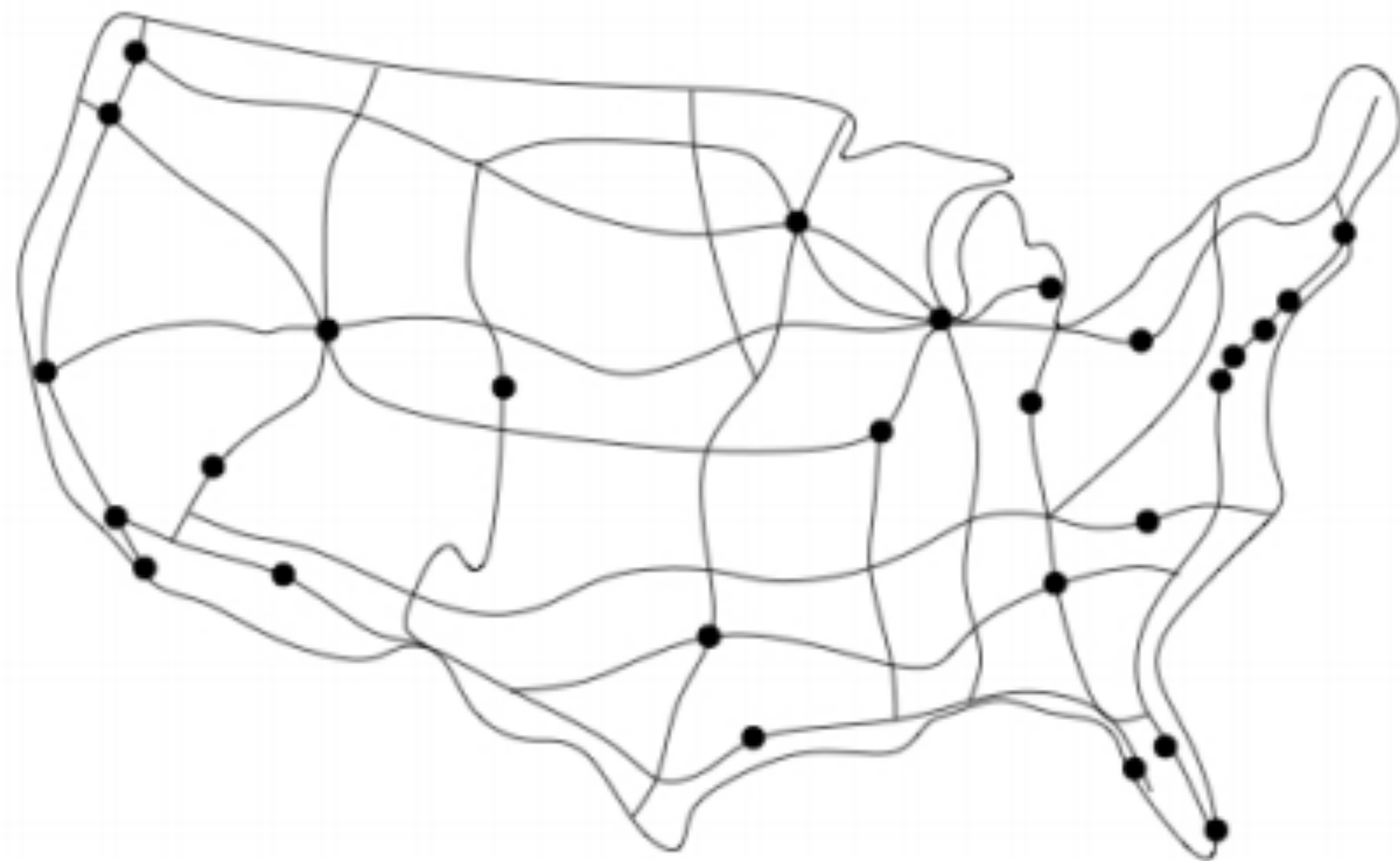
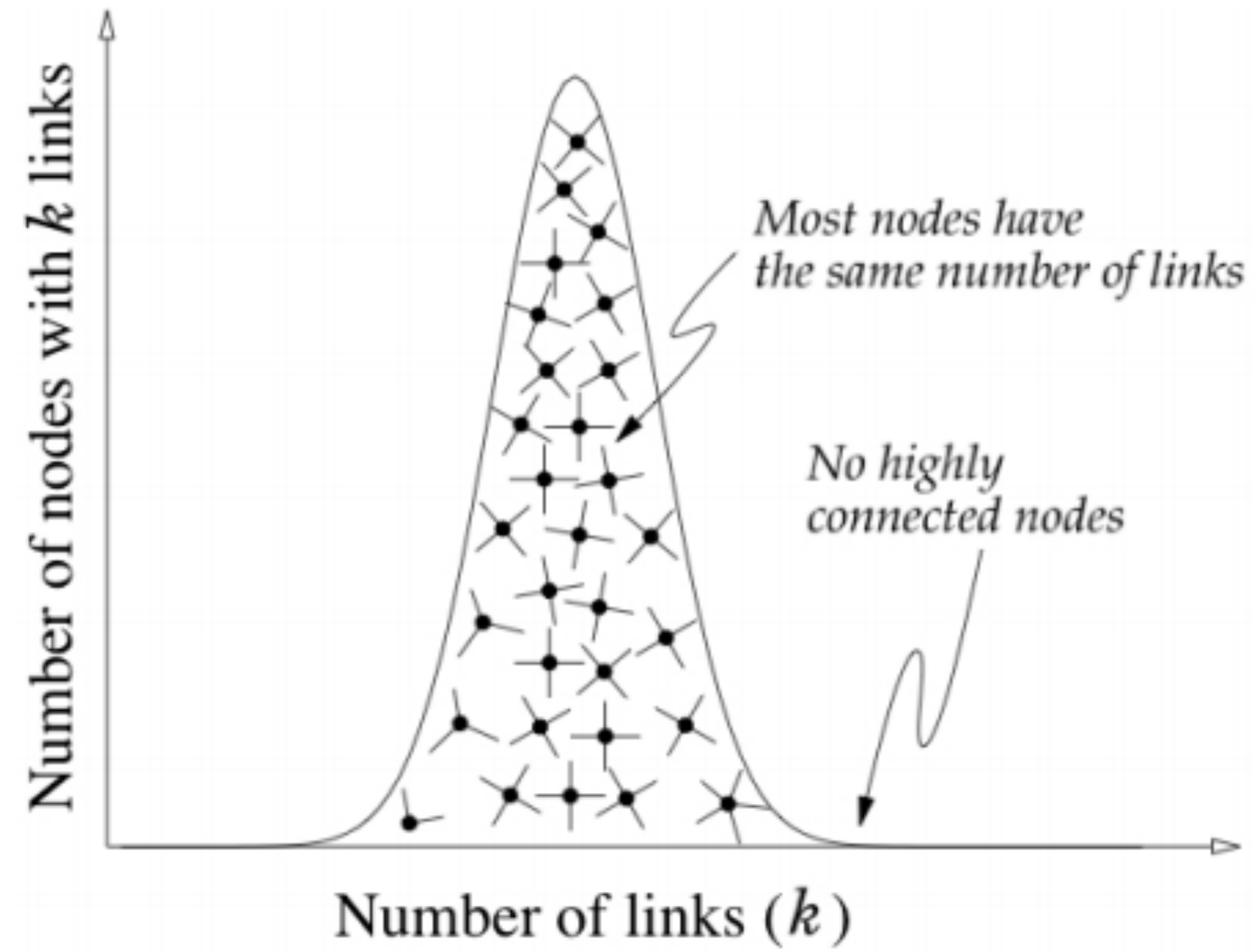
Many studies on complex networks focus on how to characterize them and what are the most relevant measures for understanding their structure. In particular, the degree distribution and the existence of the second moment for an infinite network was shown to be critical when studying dynamical processes on networks. These behaviors are strongly connected to degree fluctuations and to the existence of hubs. In the case of spatial networks, the physical constraints are usually large and prevent the appearance of such hubs. These constraints also impact other quantities that are non-trivial for complex networks but that become irrelevant for spatial networks. We review here these measures that are essentially useless for spatial networks and we then discuss useful, simple measures that were mostly introduced in the context of quantitative geography.

4.1 Irrelevant Measures for Spatial Networks

Quantities that depend very much on the spatial constraints turn out to be irrelevant for spatial networks in general. The prime example is the degree distribution which in many complex networks was found to be a broad law and in some cases well fitted by a power law of the form

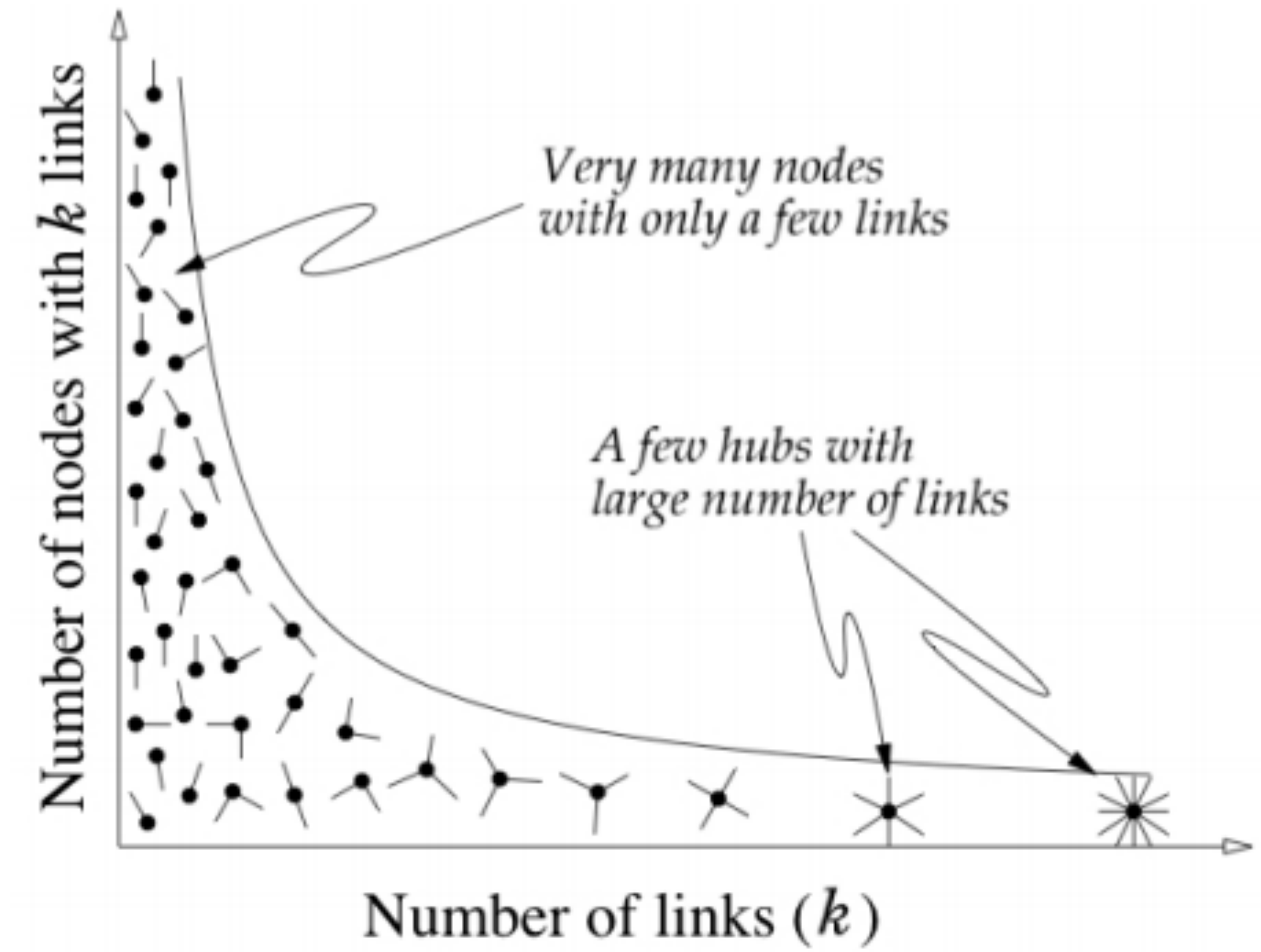
$$P(k) \sim k^{-\gamma} \quad (4.1)$$

Thin-tailed network



Streets

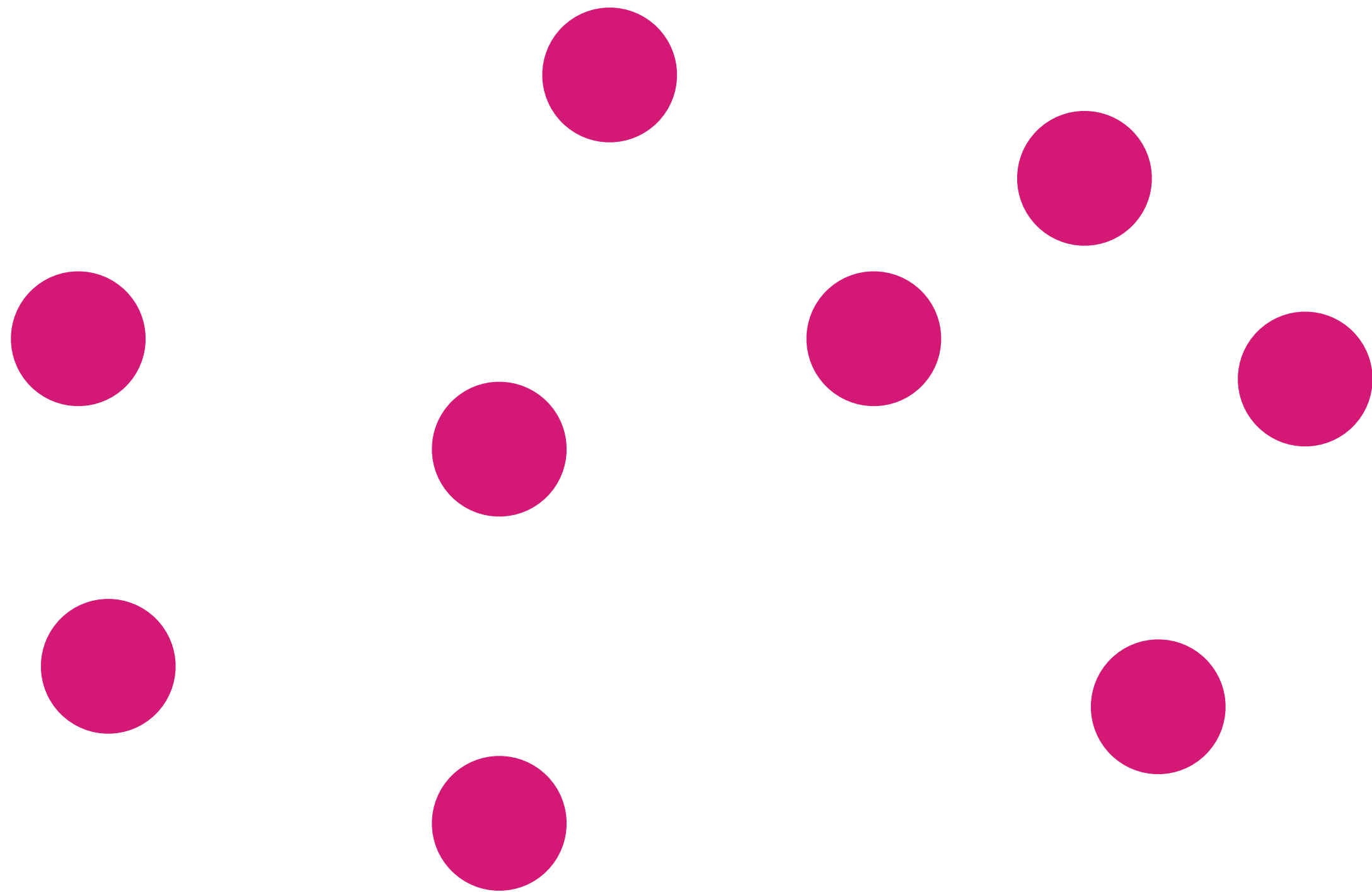
Heavy-tailed network



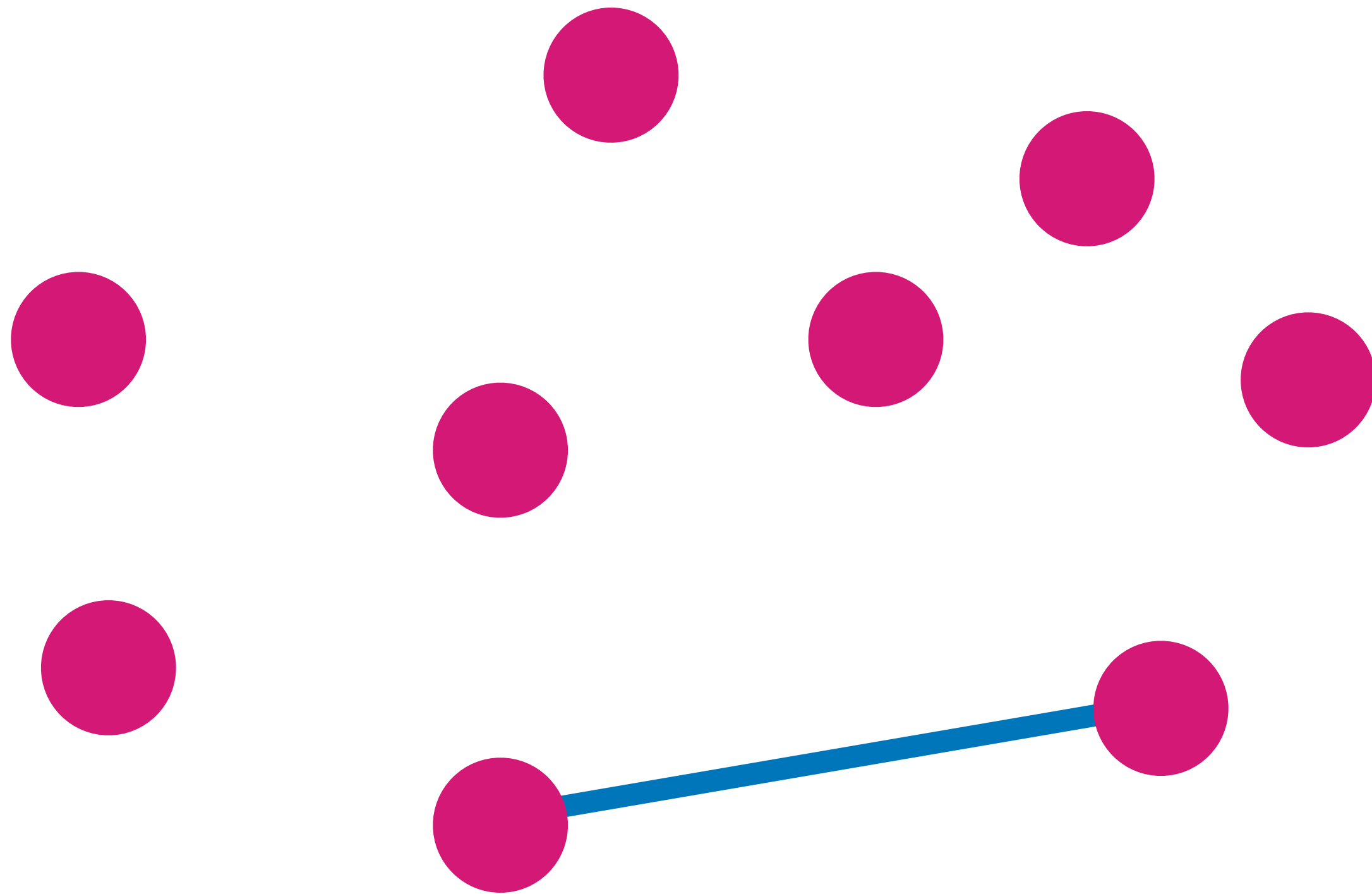
Airlines connections

Triangulations & “optimal” graphs

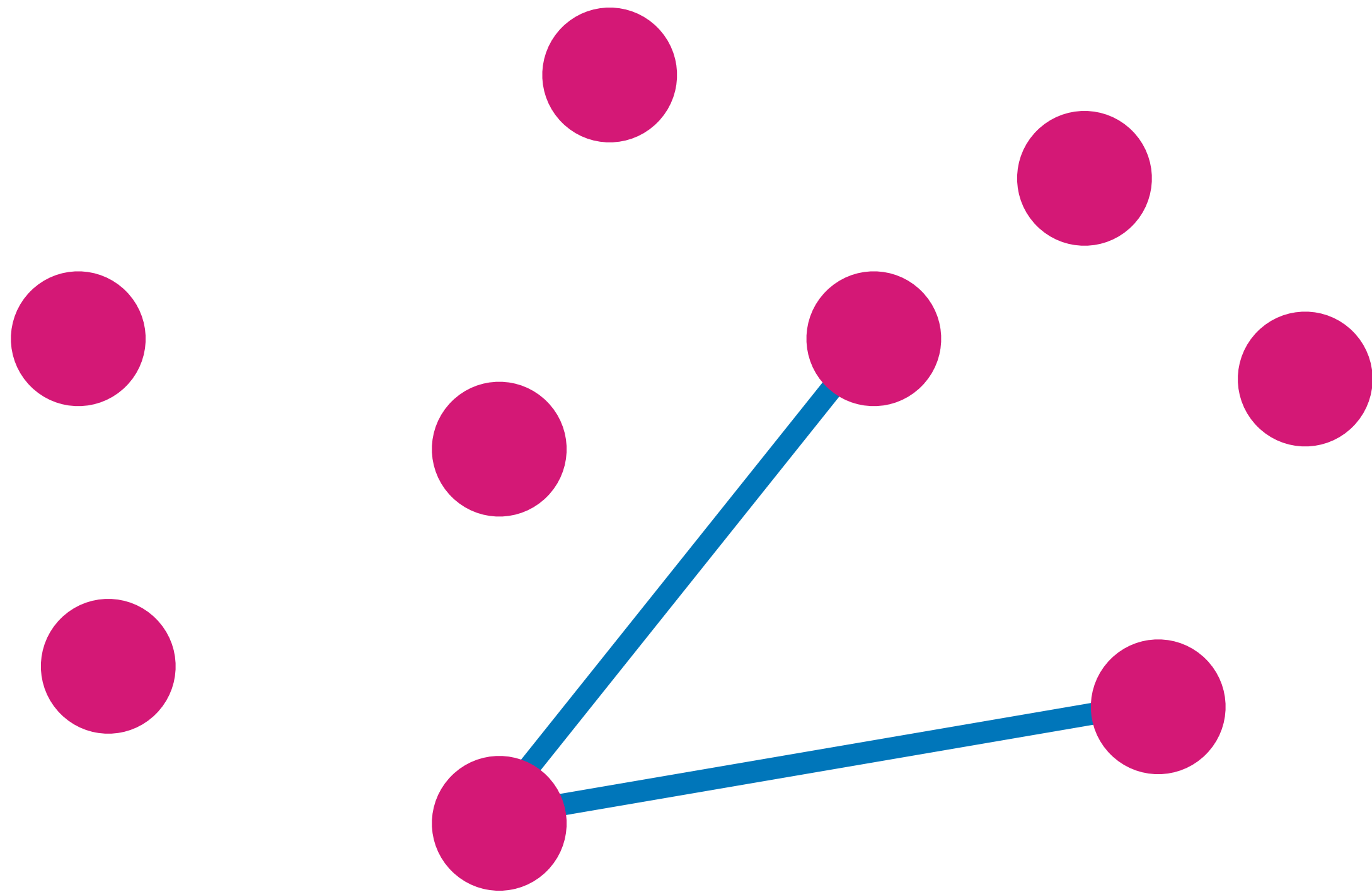
Draw as many non-crossing edges as you can!



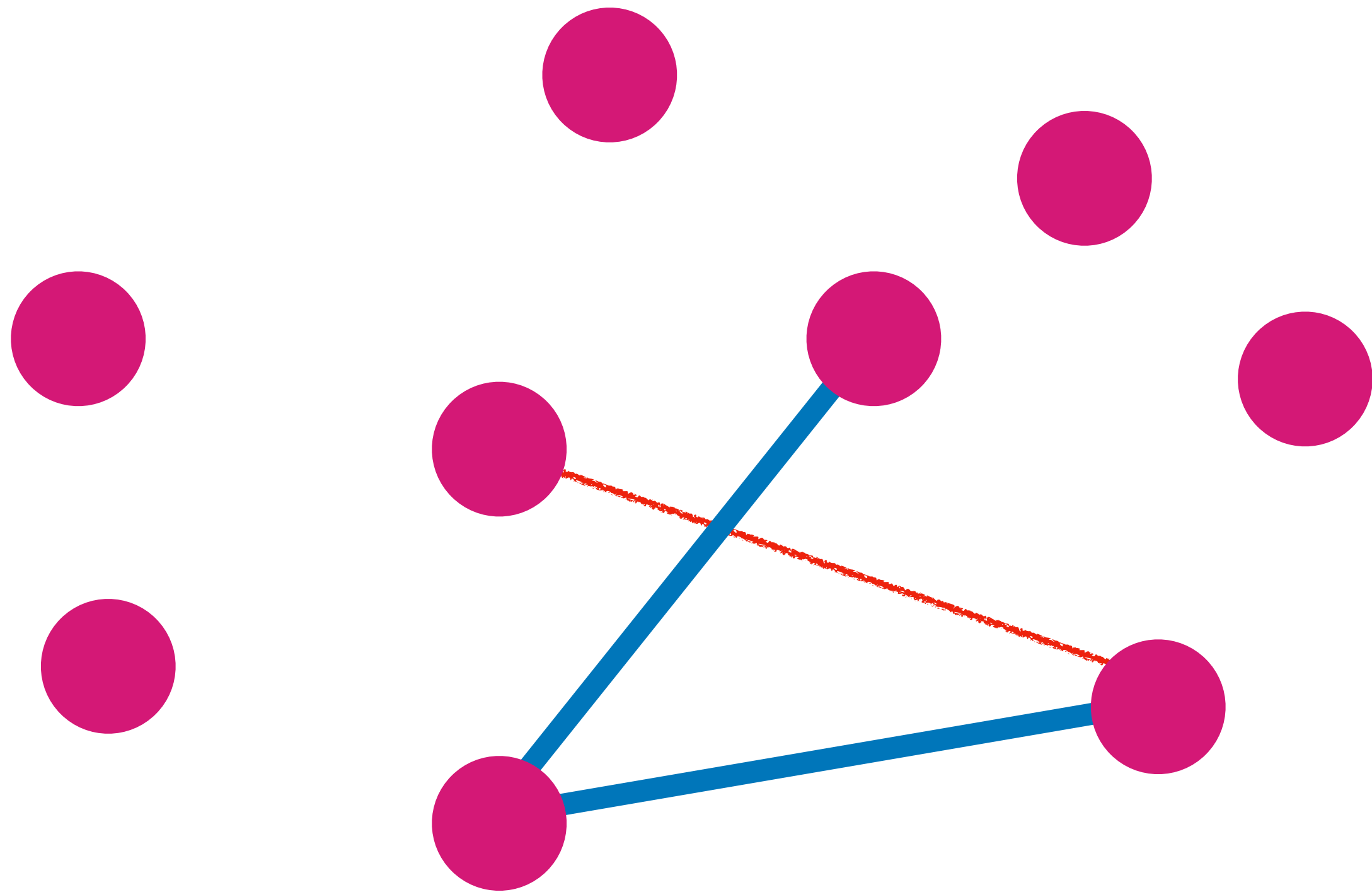
Draw as many non-crossing edges as you can!



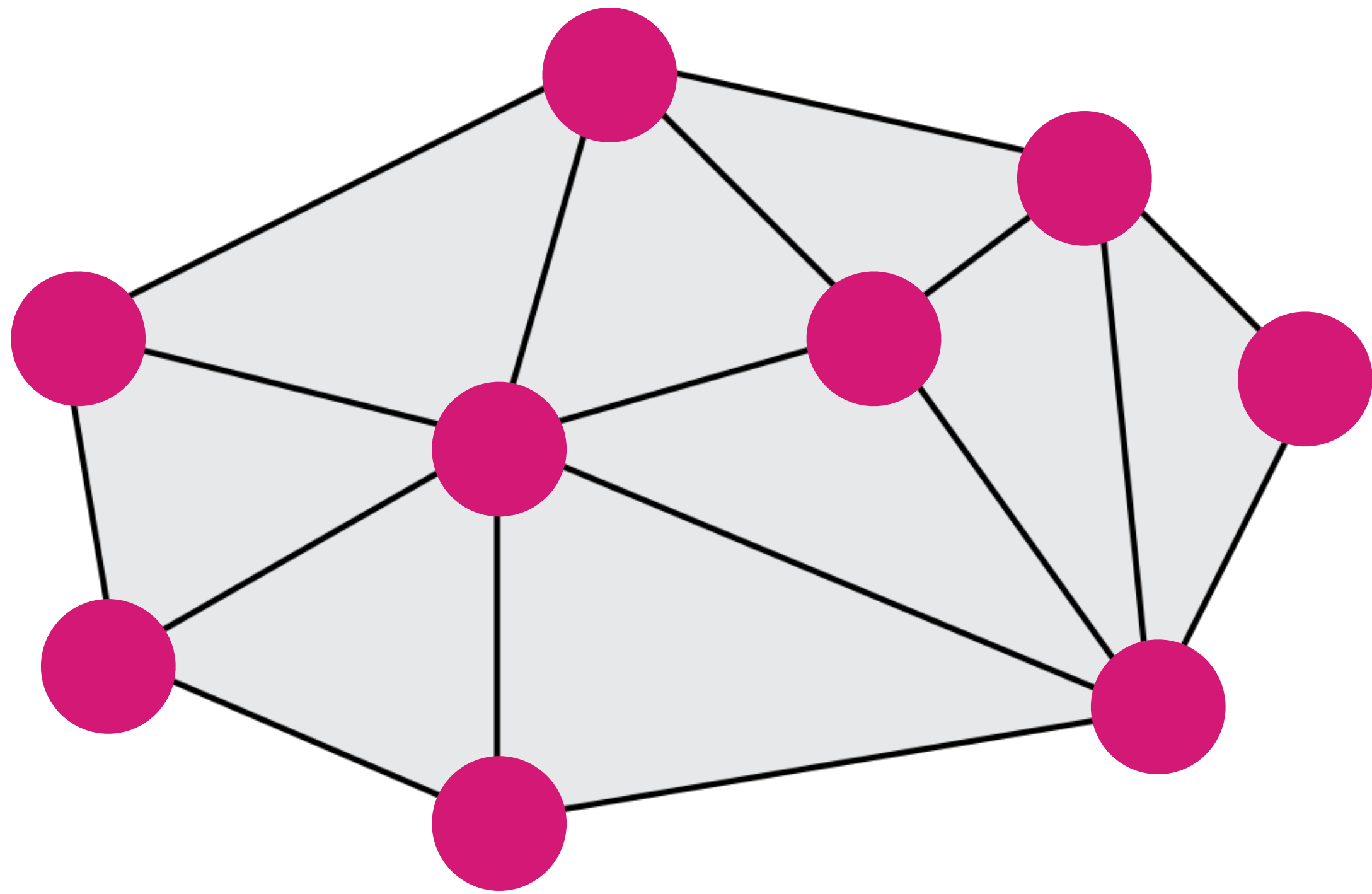
Draw as many non-crossing edges as you can!



Draw as many non-crossing edges as you can!

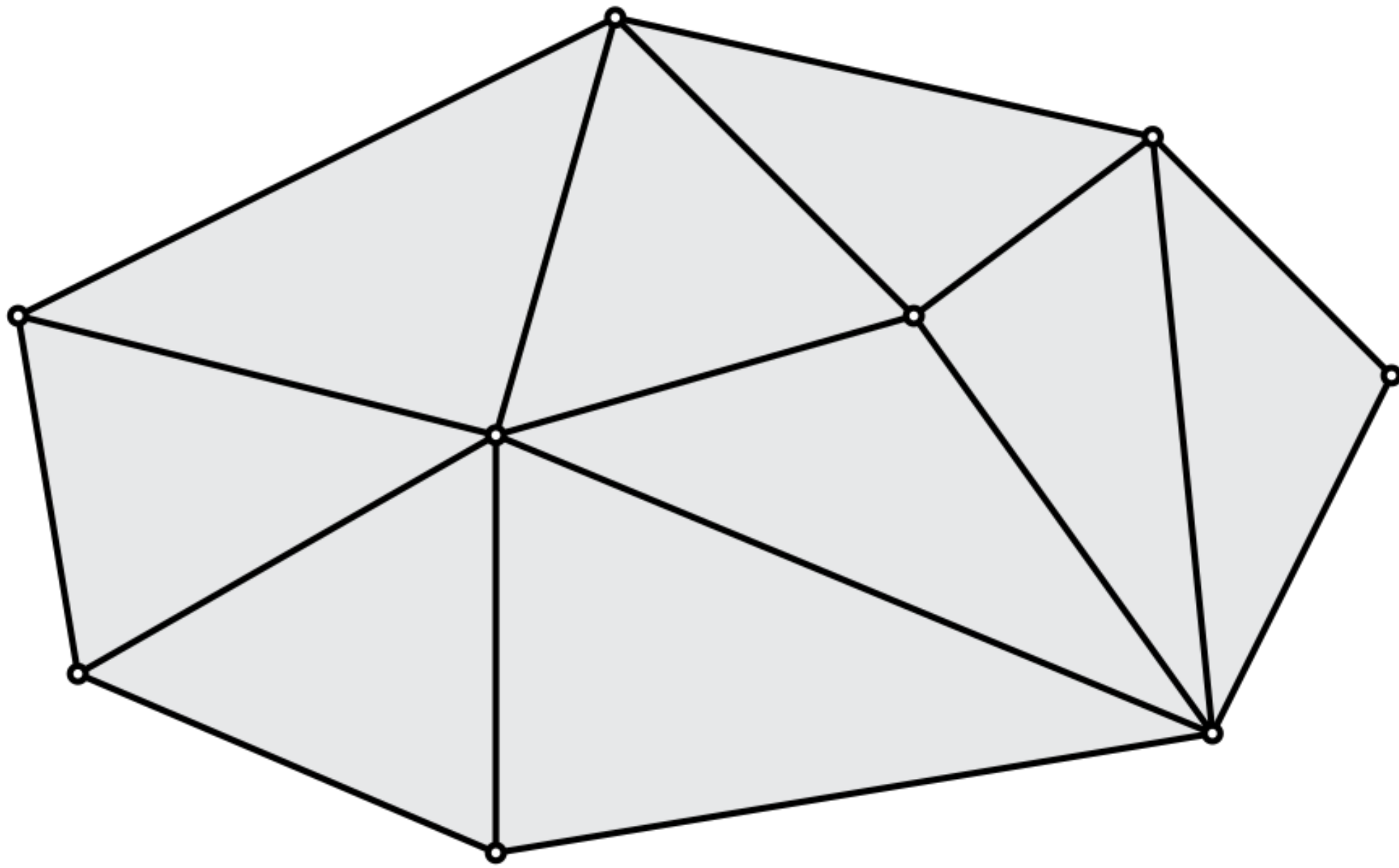


A point **triangulation** is a maximal set of non-crossing edges



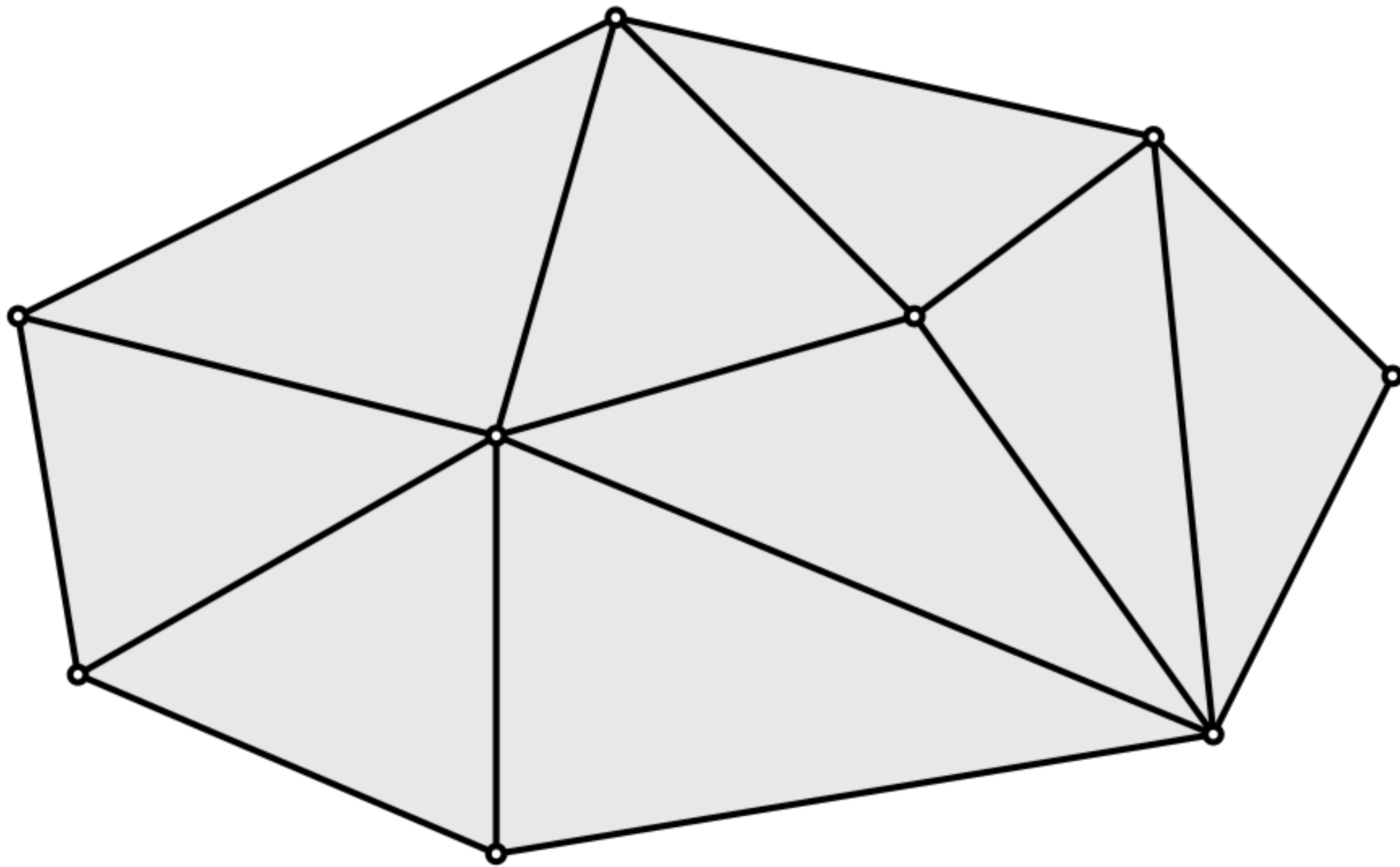
It is planar

A point **triangulation** is a maximal set of non-crossing edges

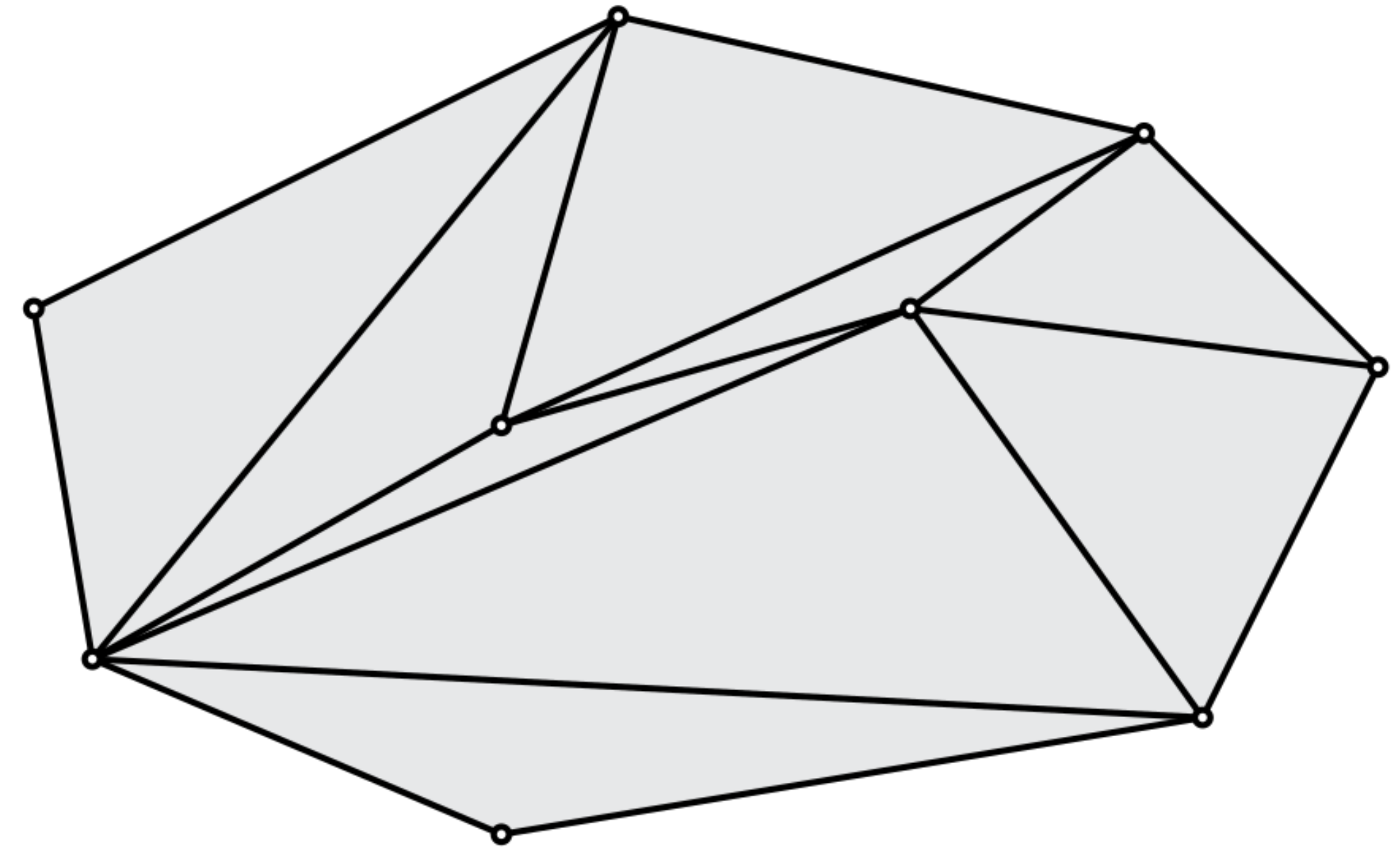


It is planar

A point **triangulation** is a maximal set of non-crossing edges



It is planar

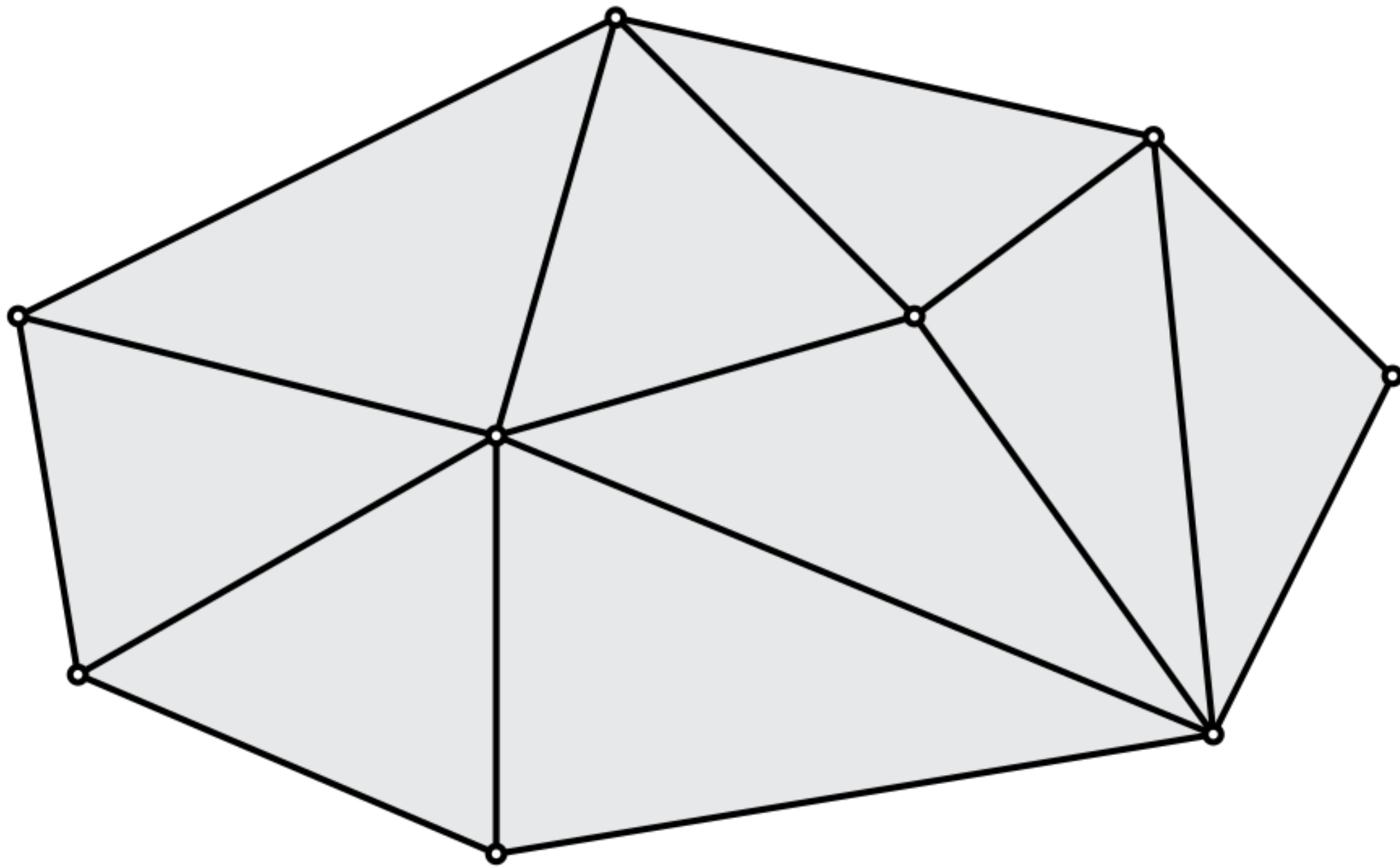


It is not unique

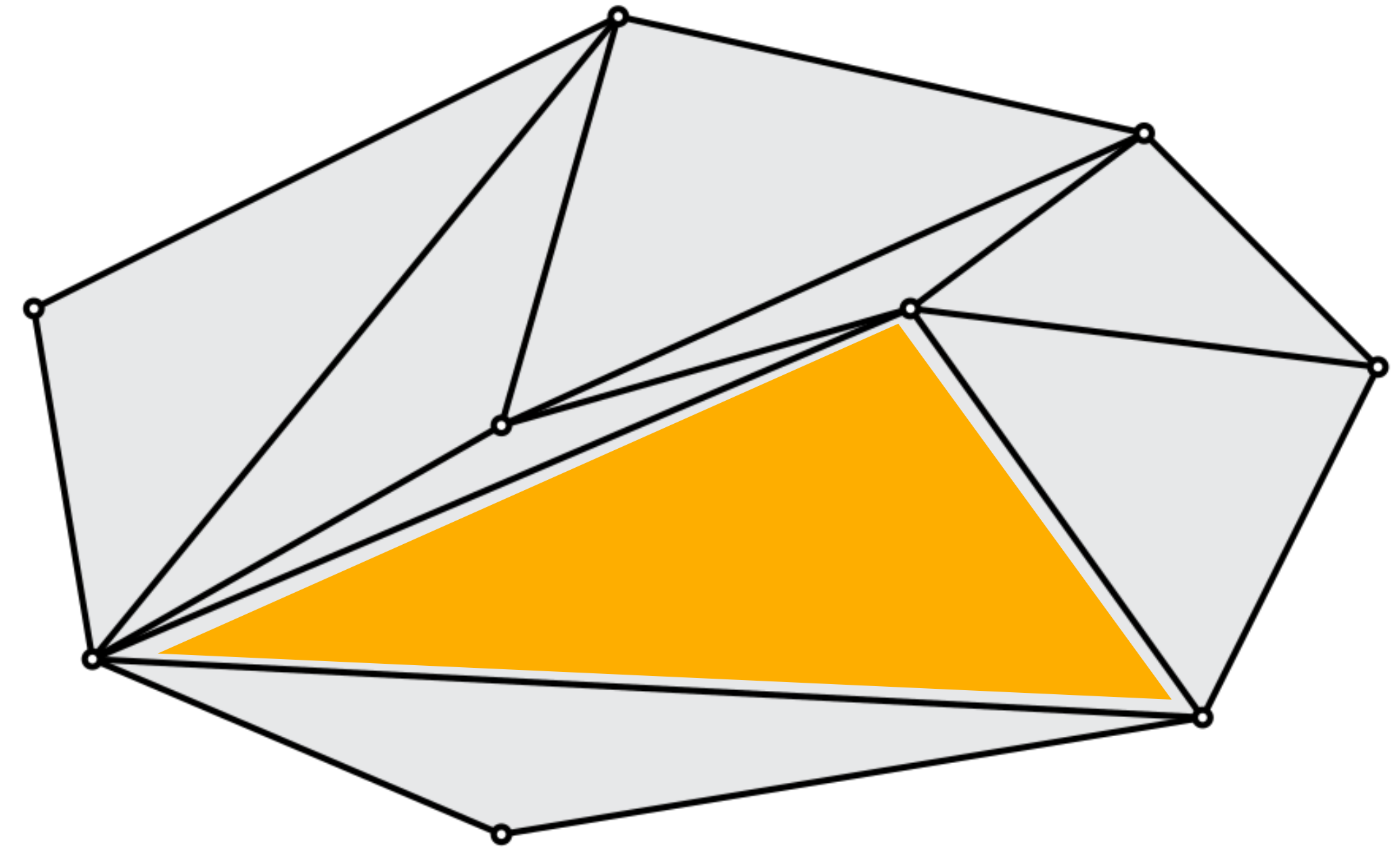
A point **triangulation** is a maximal set of non-crossing edges

graph faces

non-overlapping cells covering
the embedding plane

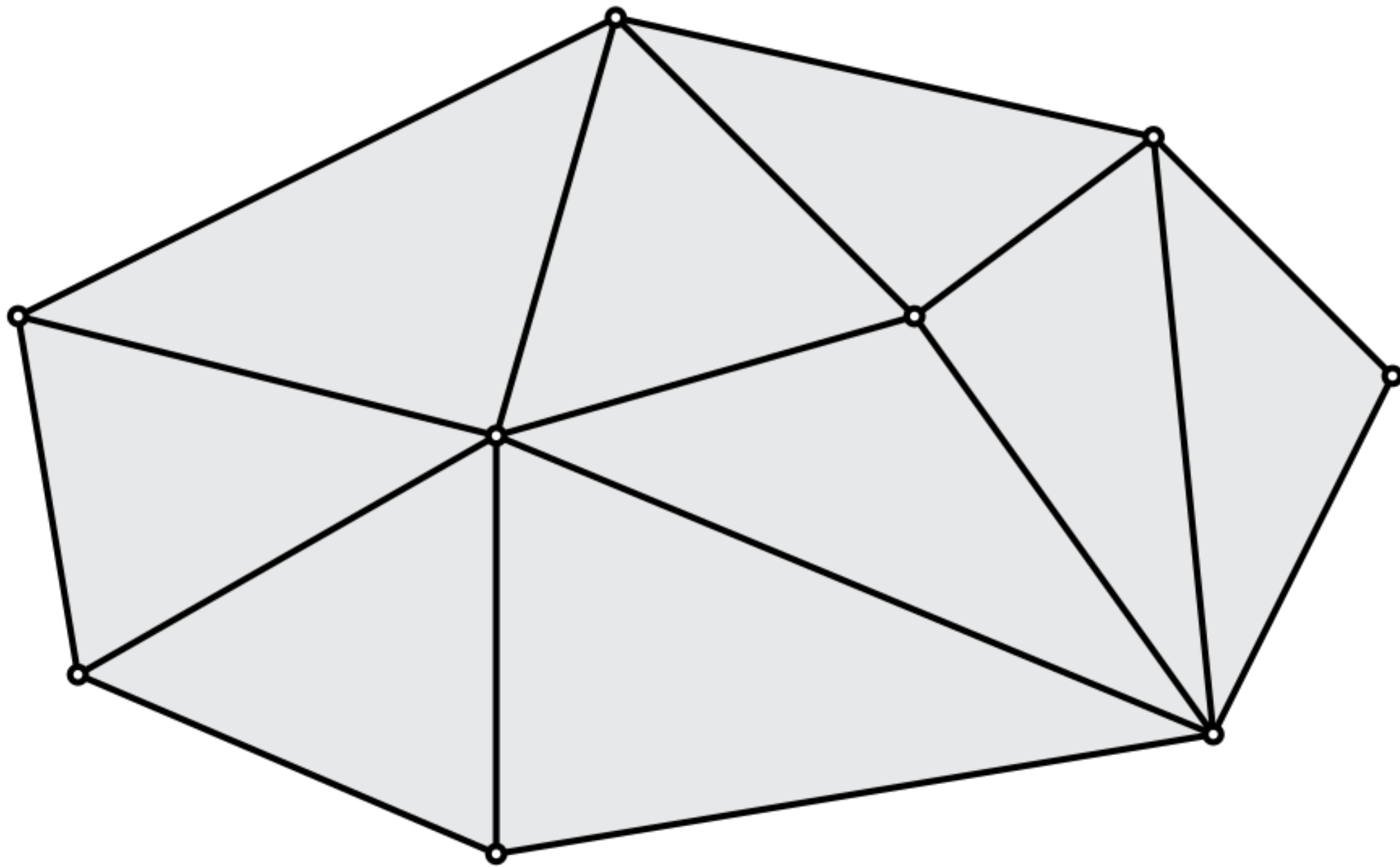


It is planar

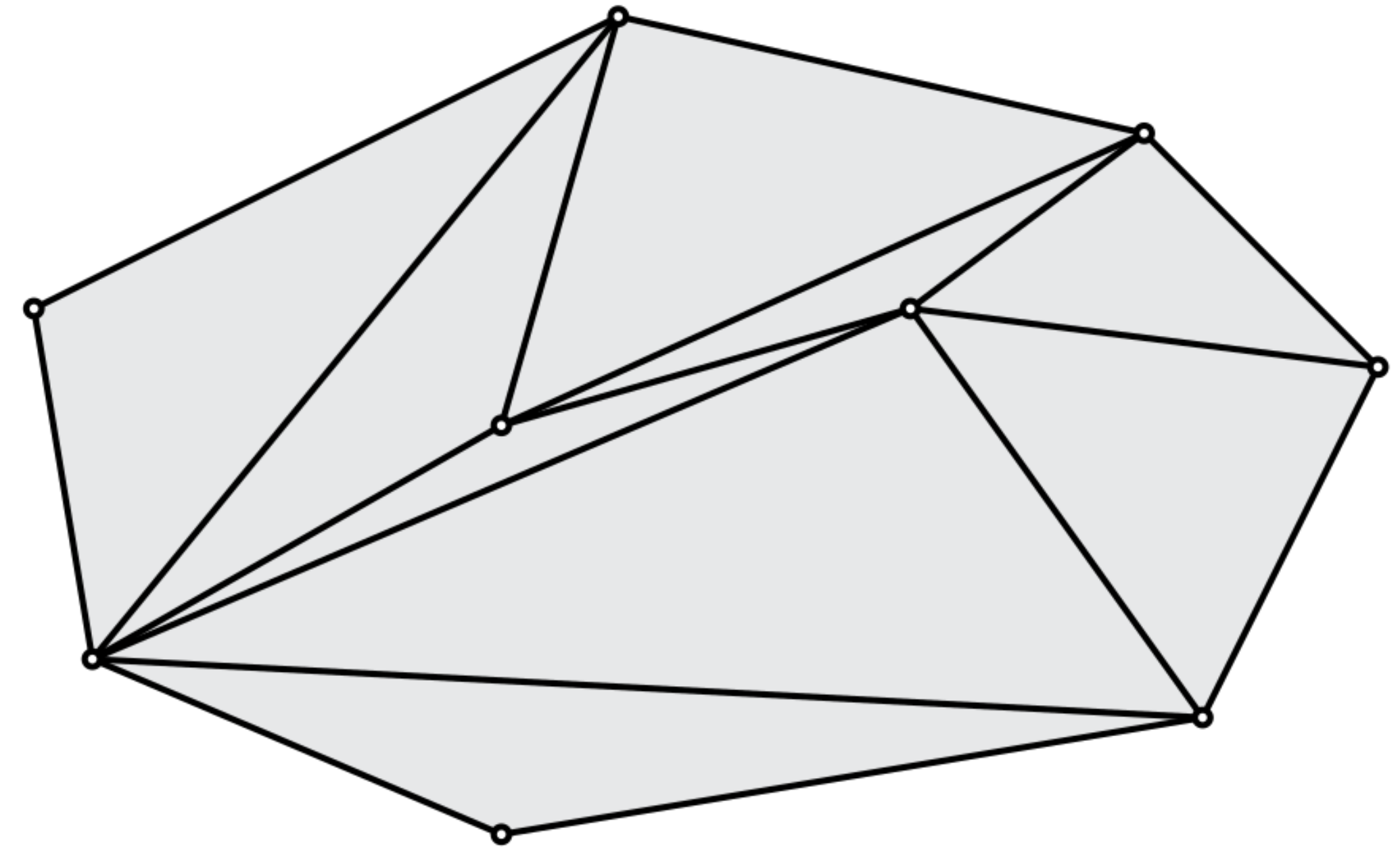


It is not unique

A point **triangulation** is a maximal set of non-crossing edges



It is planar

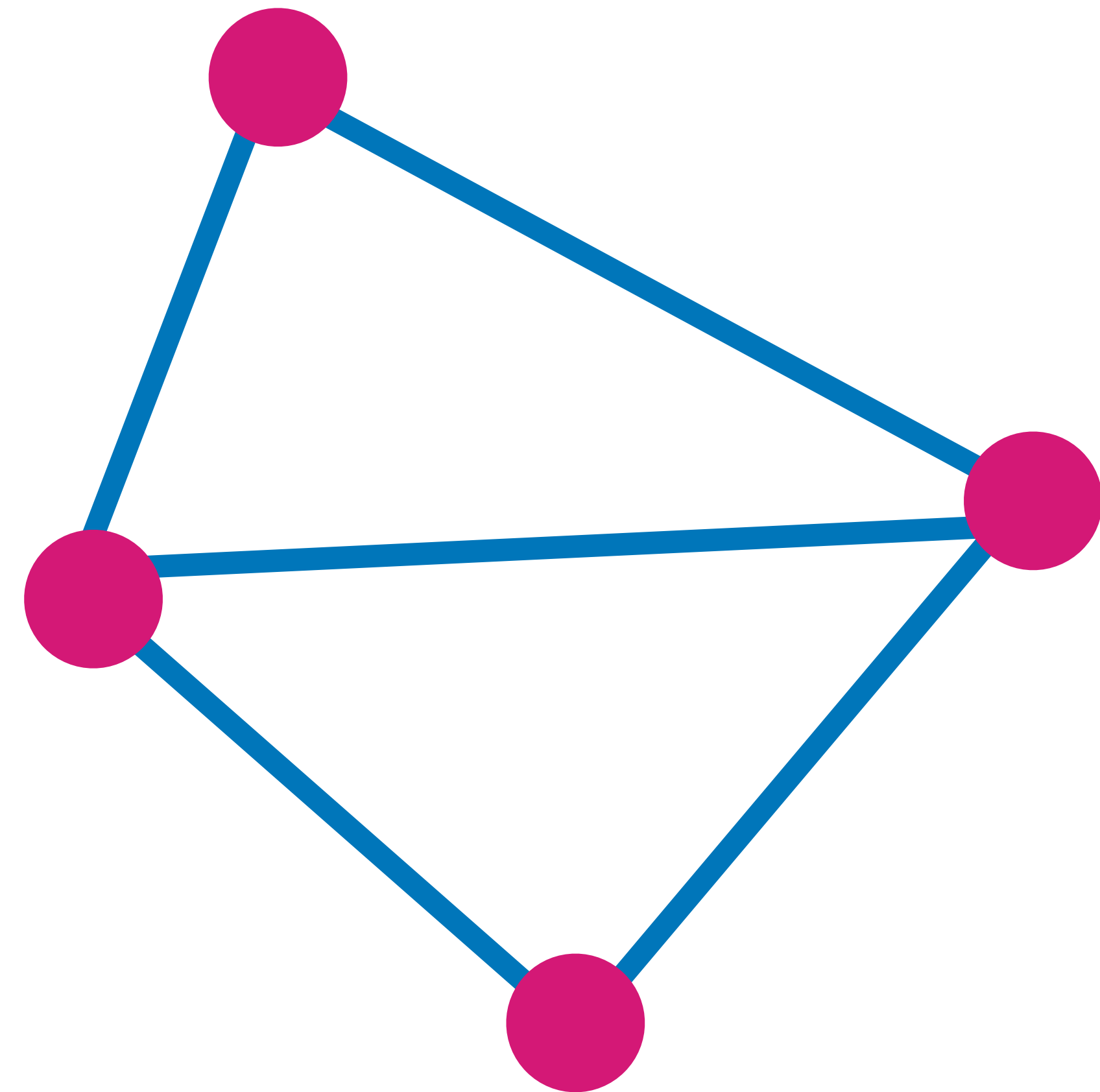
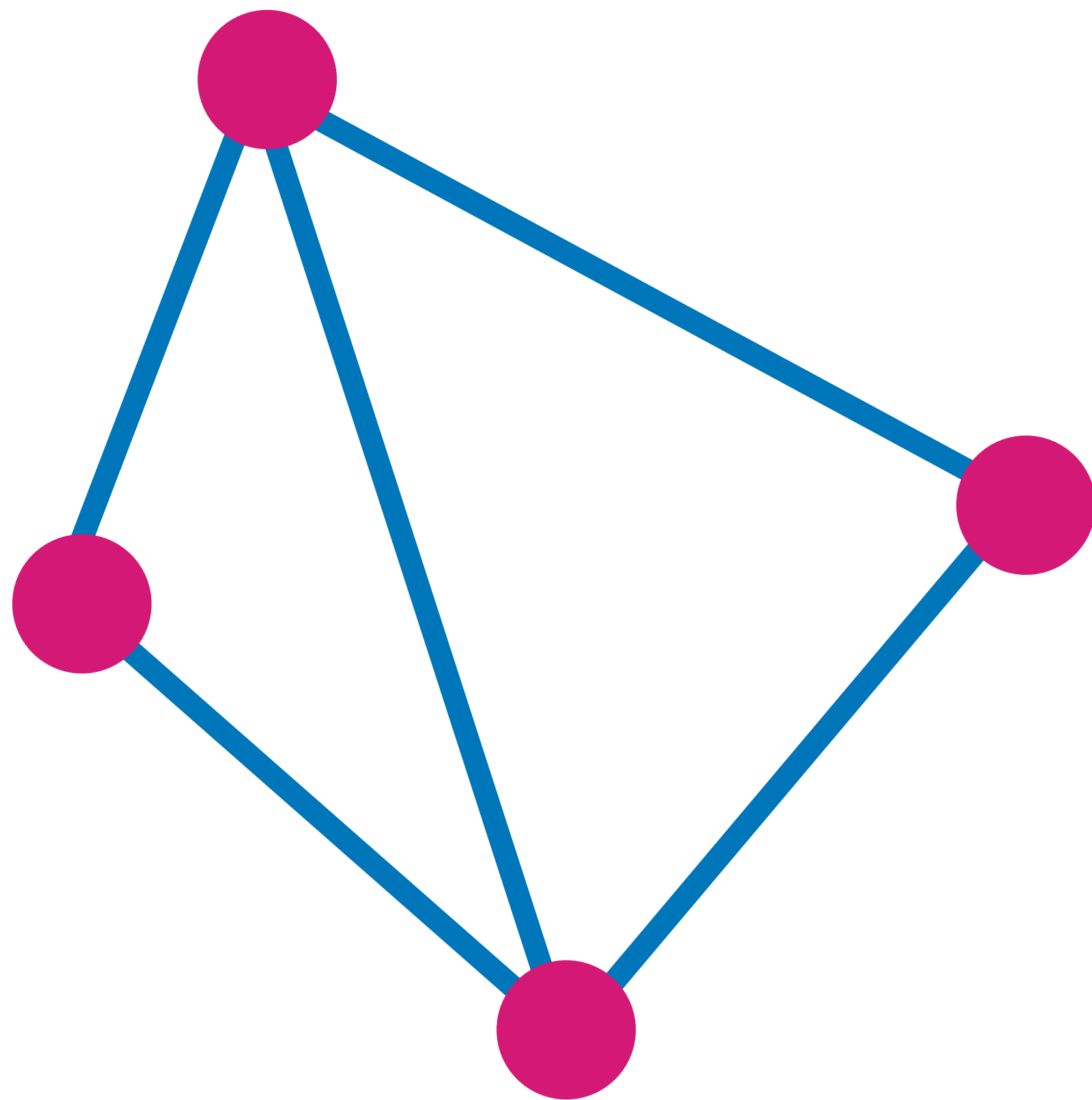


It is not unique

How to avoid “sliver” faces?

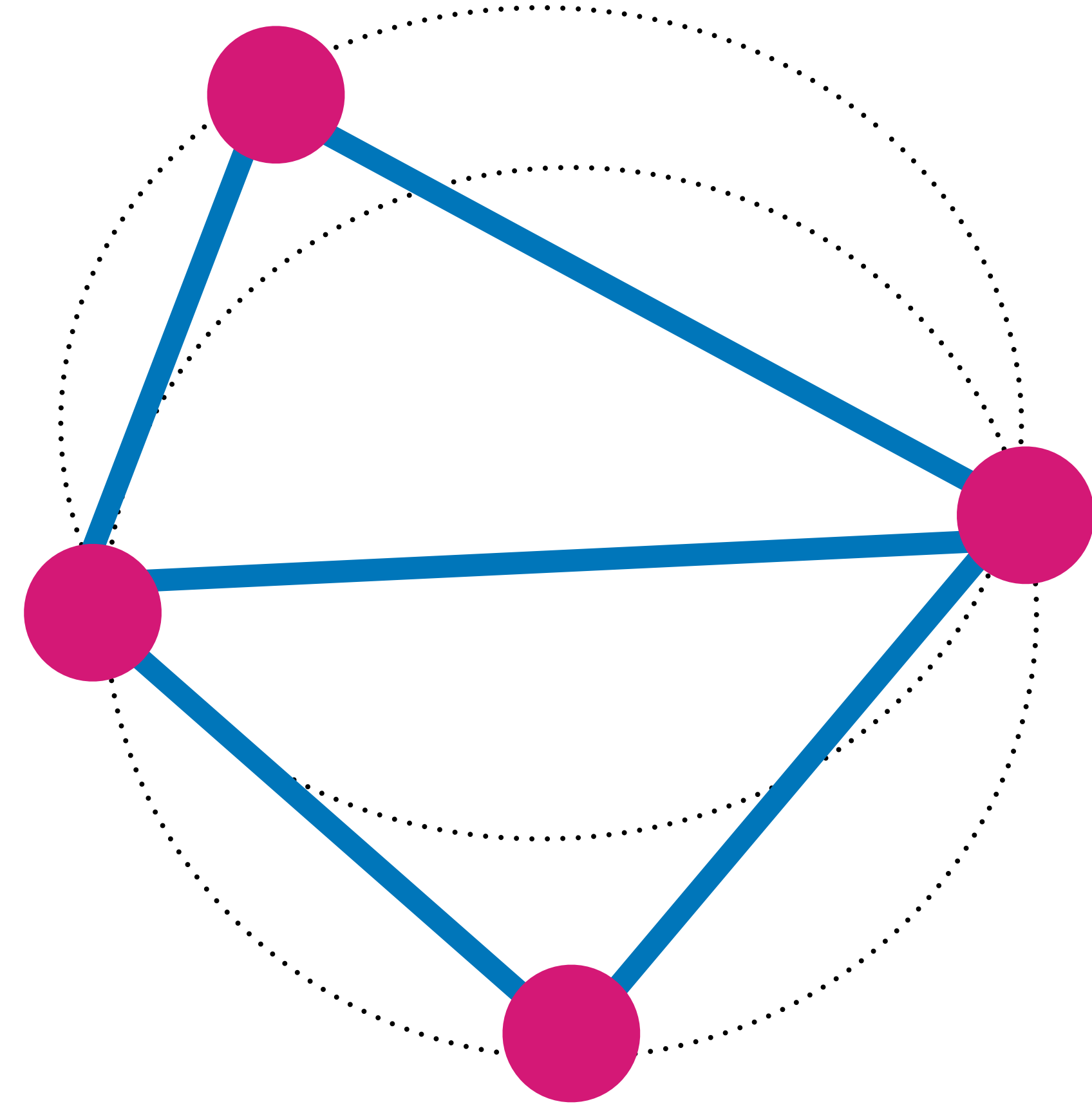
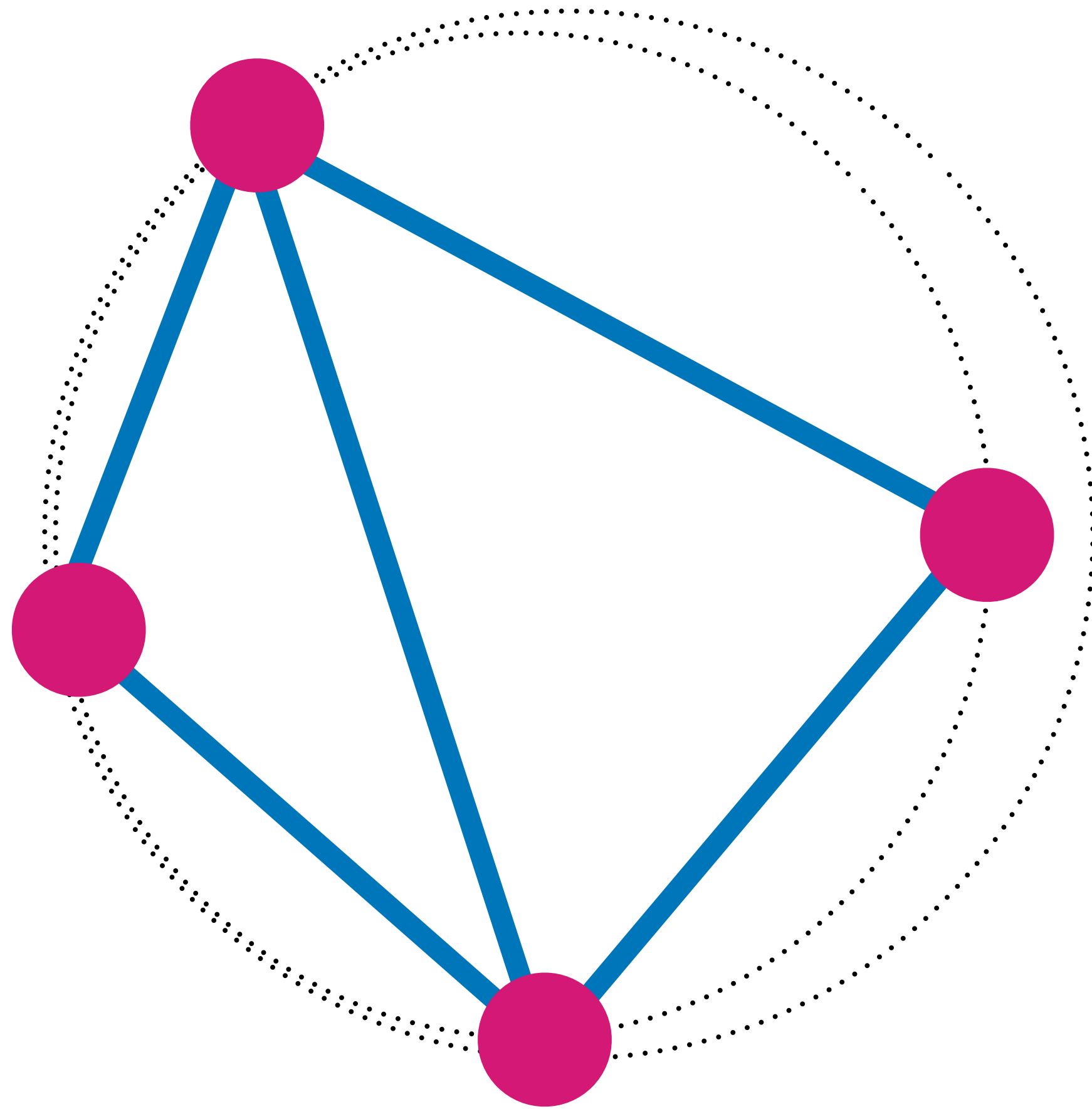
A **Delaunay triangulation** maximizes the minimum of all angles

No point can be inside the circumcircle of any triangle



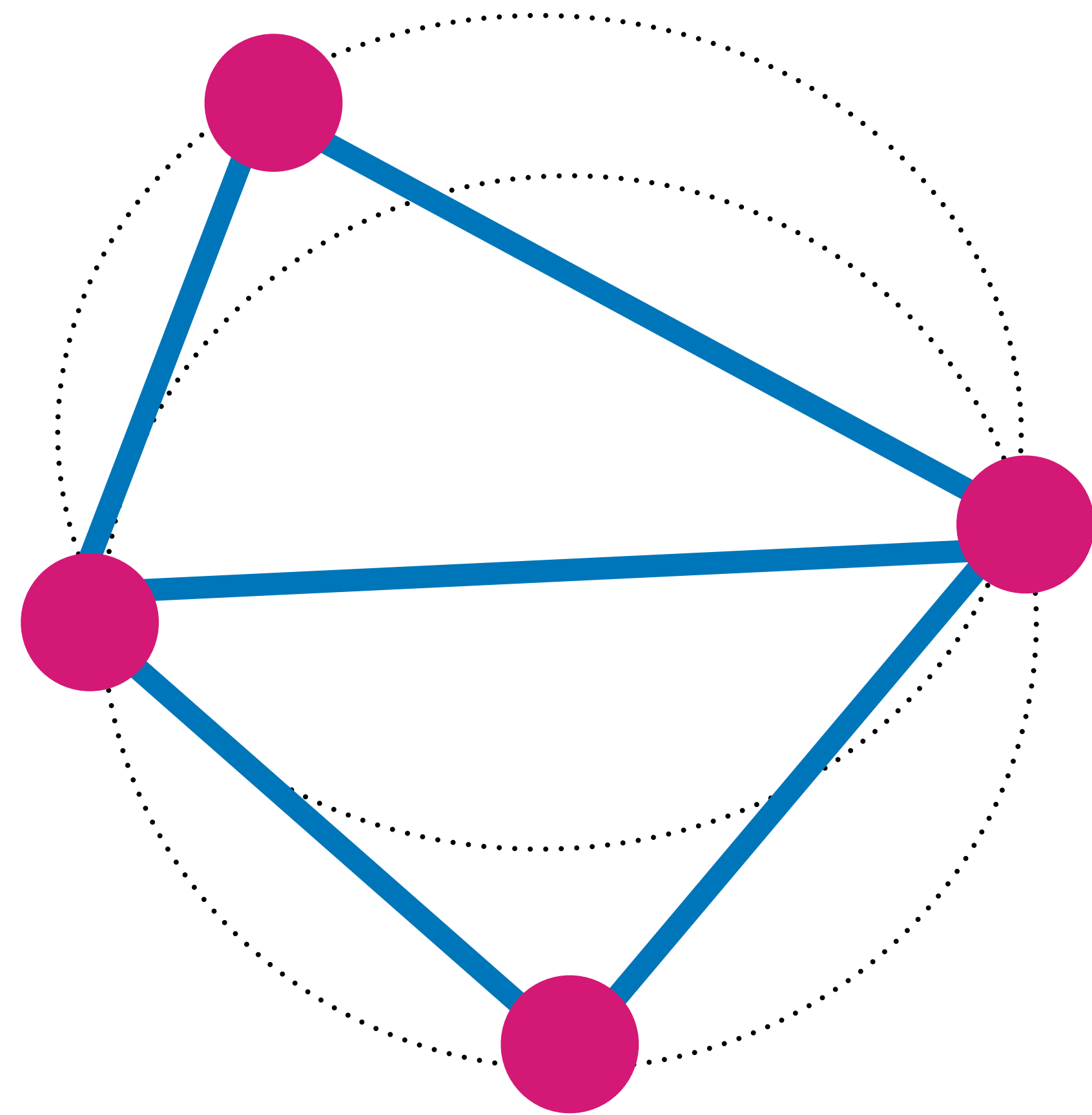
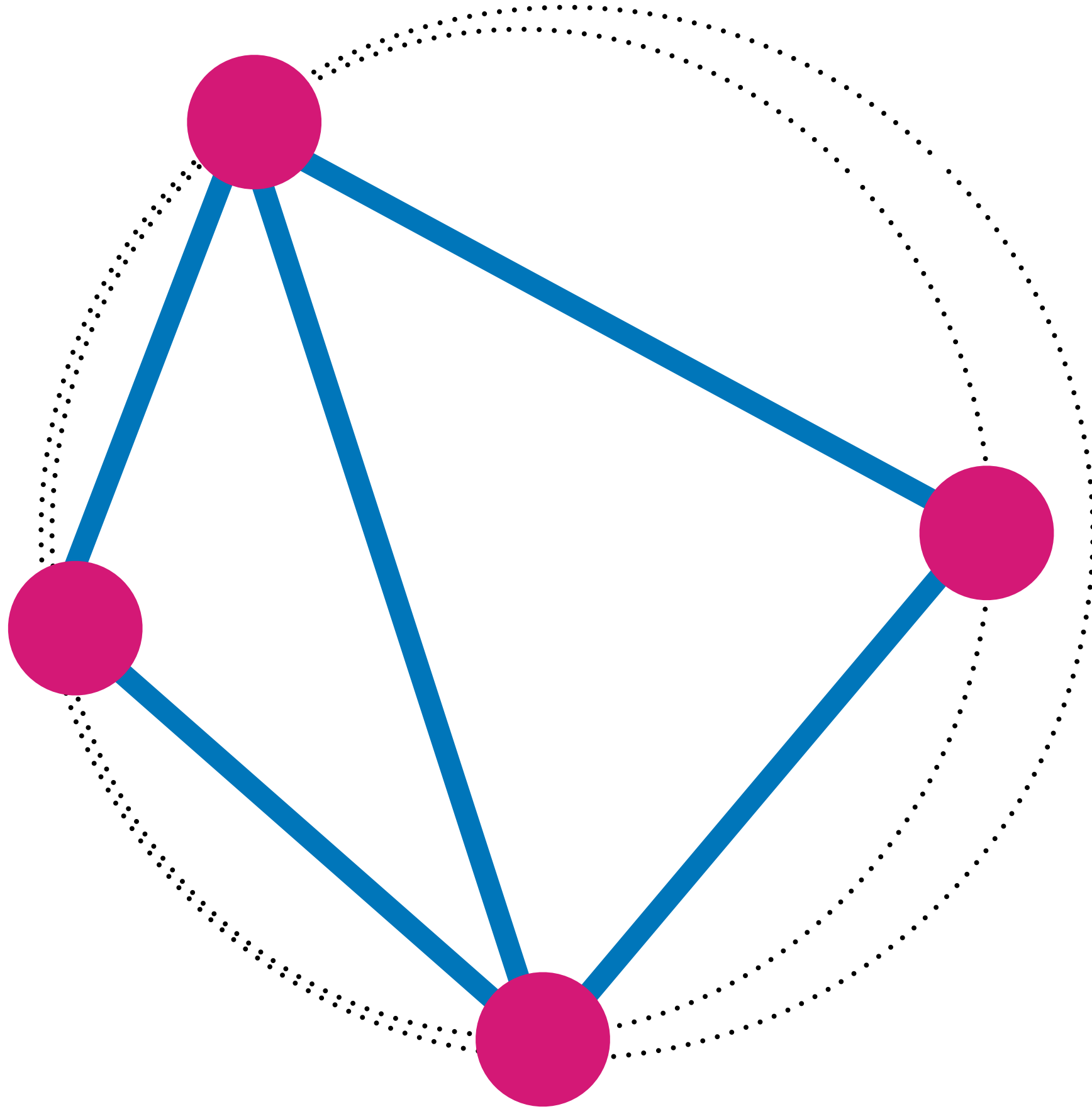
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A **Delaunay triangulation** maximizes the minimum of all angles

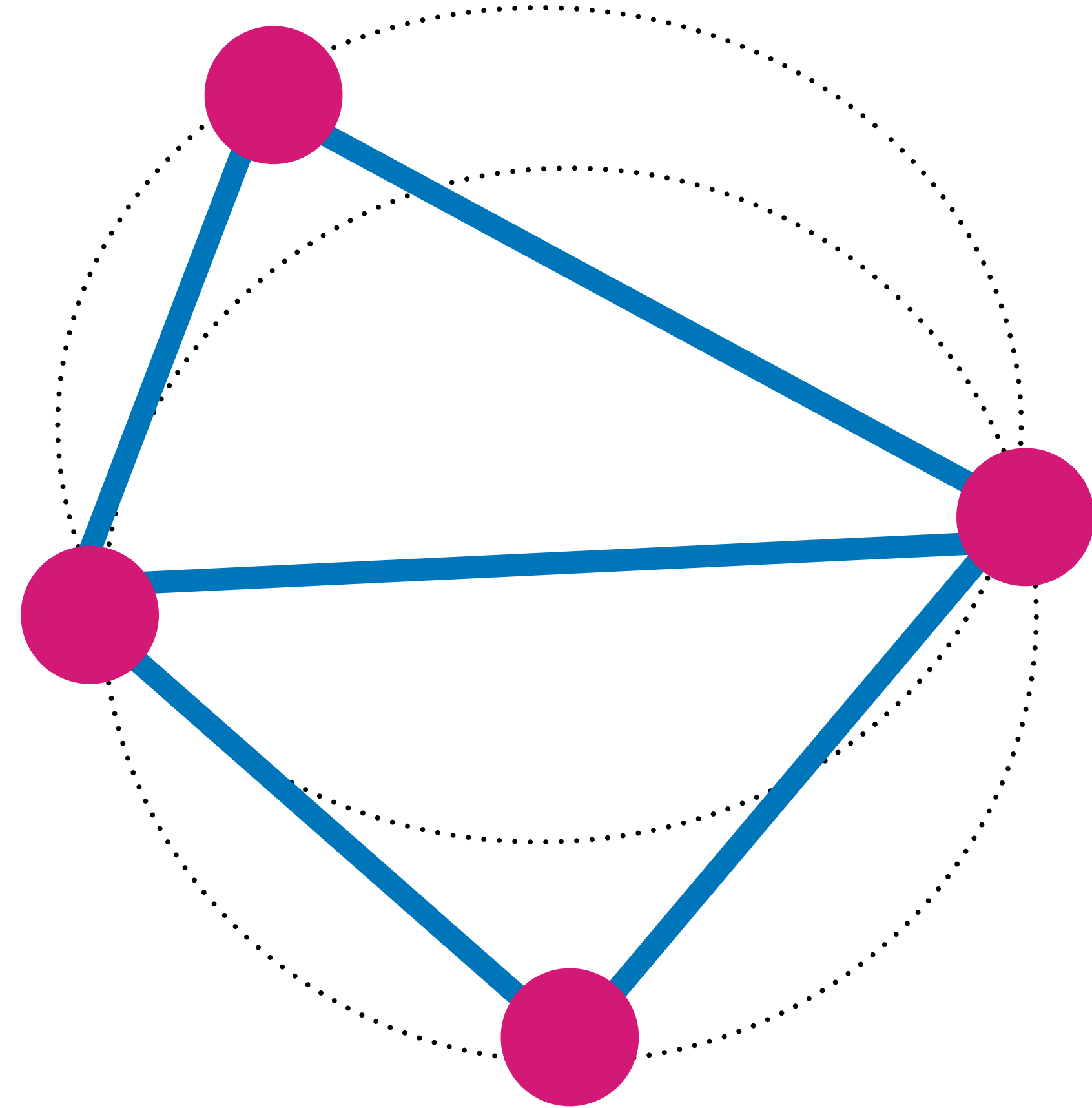
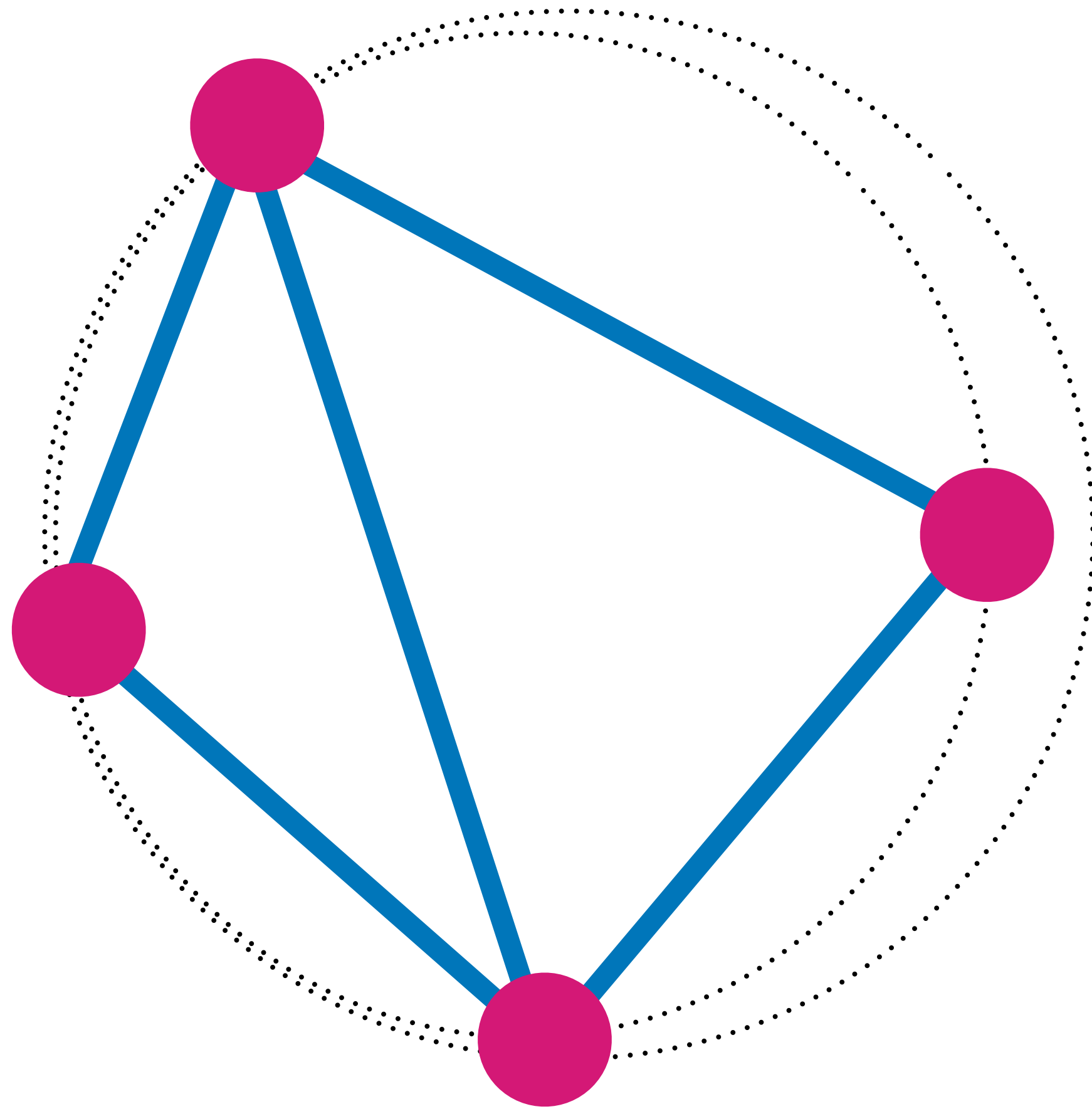
No point can be inside the circumcircle of any triangle



Which one is a Delaunay triangulation?

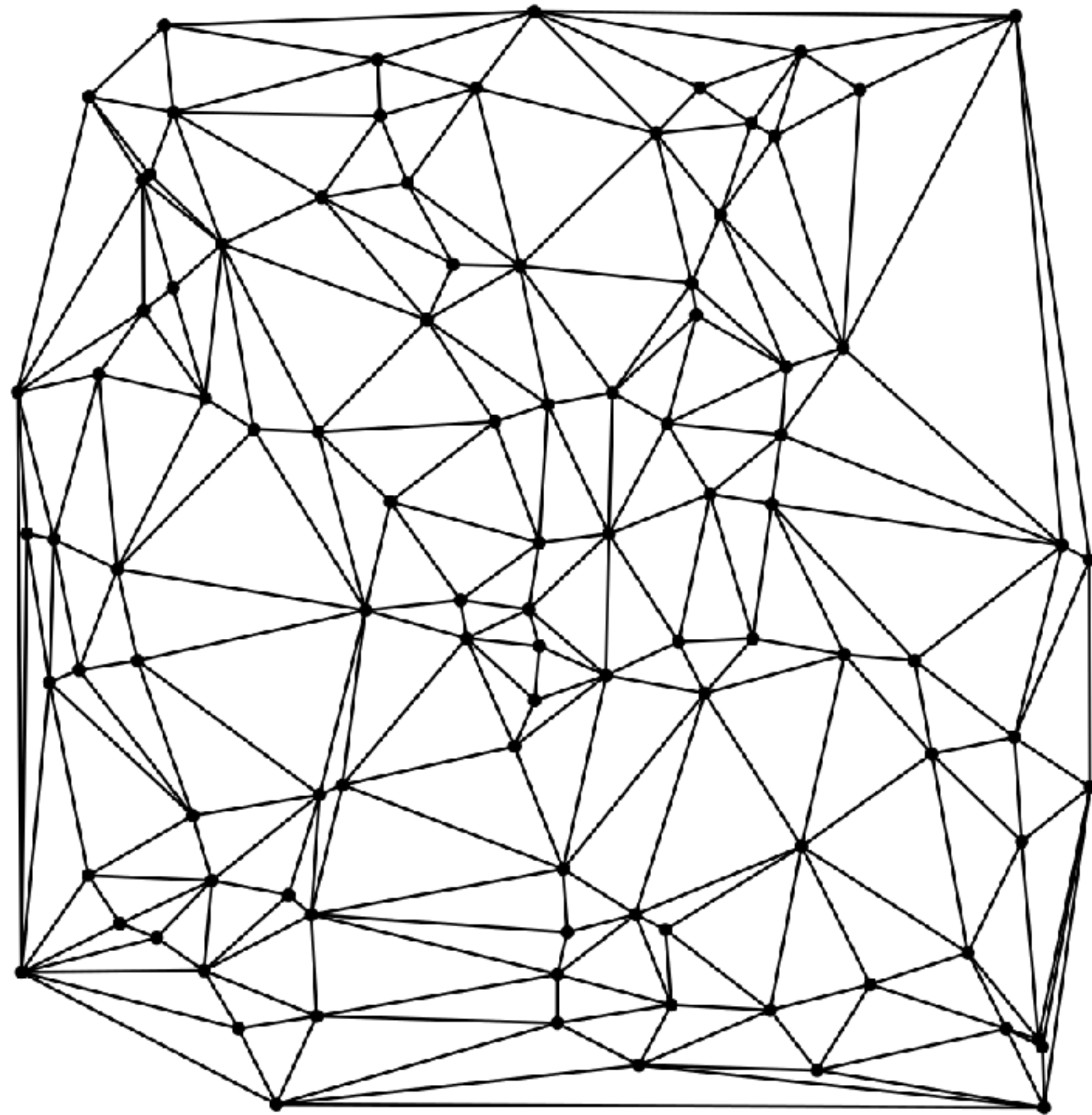
A **Delaunay triangulation** maximizes the minimum of all angles

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A **Delaunay triangulation** maximizes the minimum of all angles

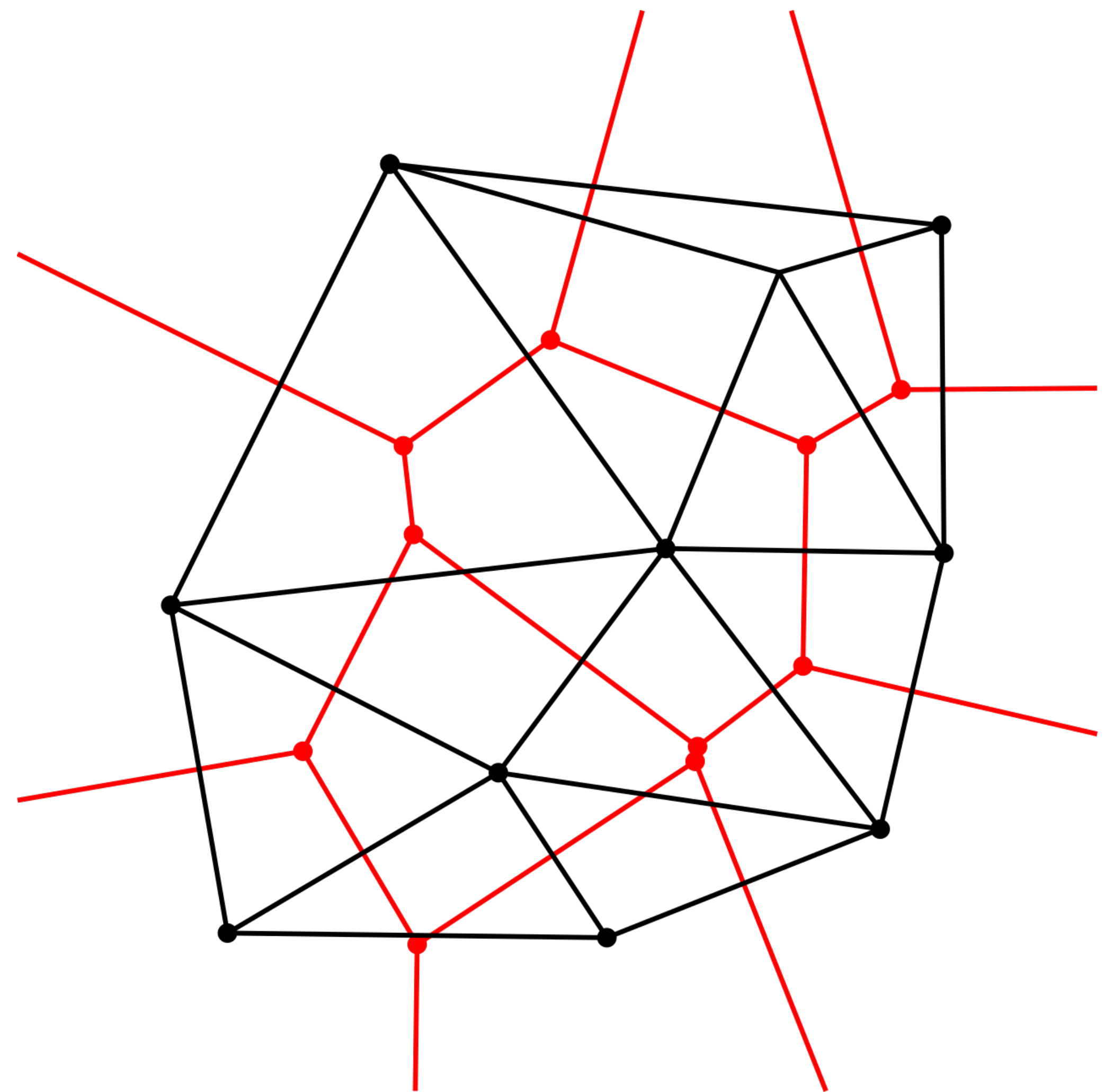
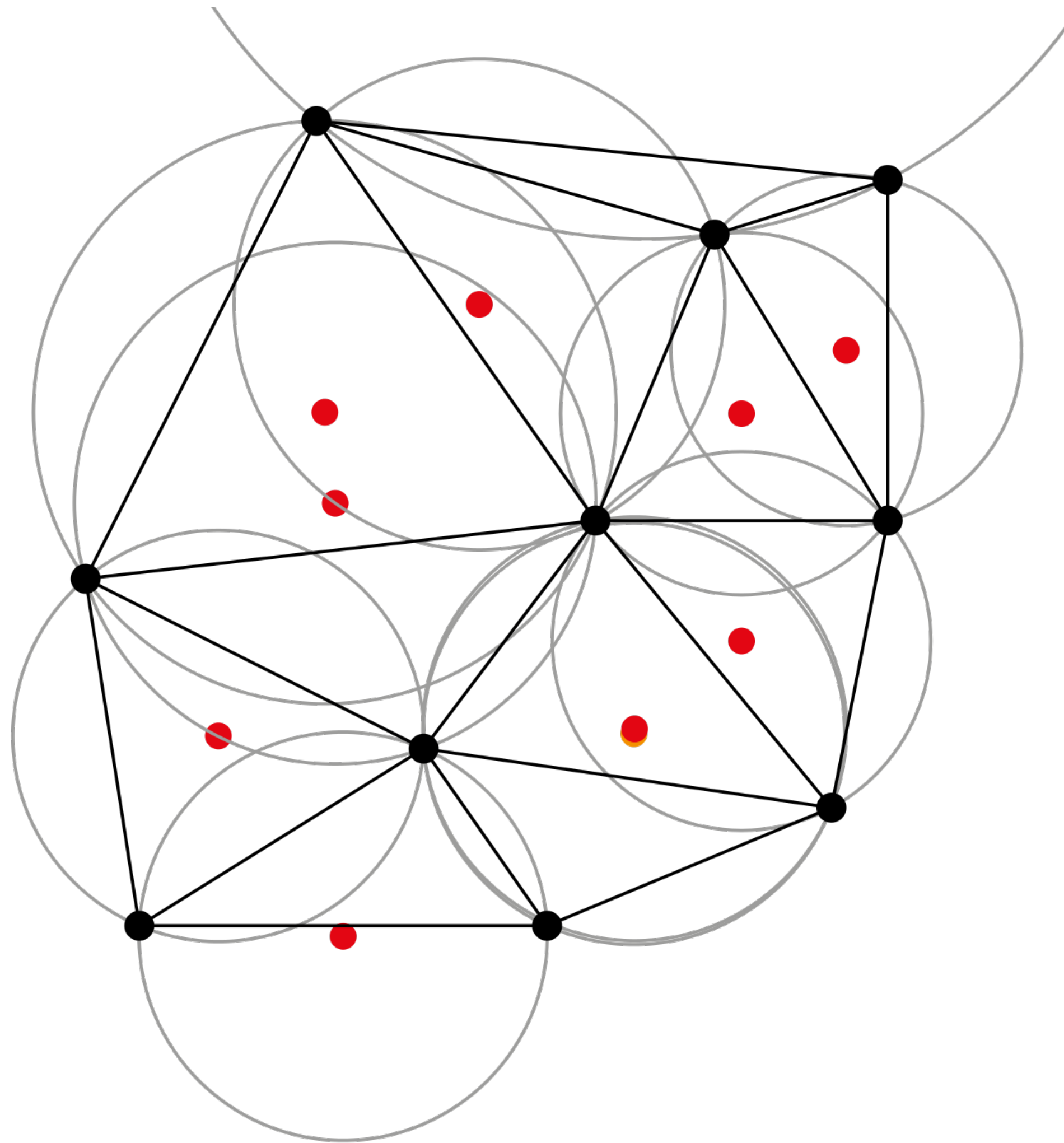
No point can be inside the circumcircle of any triangle



A Delaunay triangulation of a random set of 100 points in a plane.

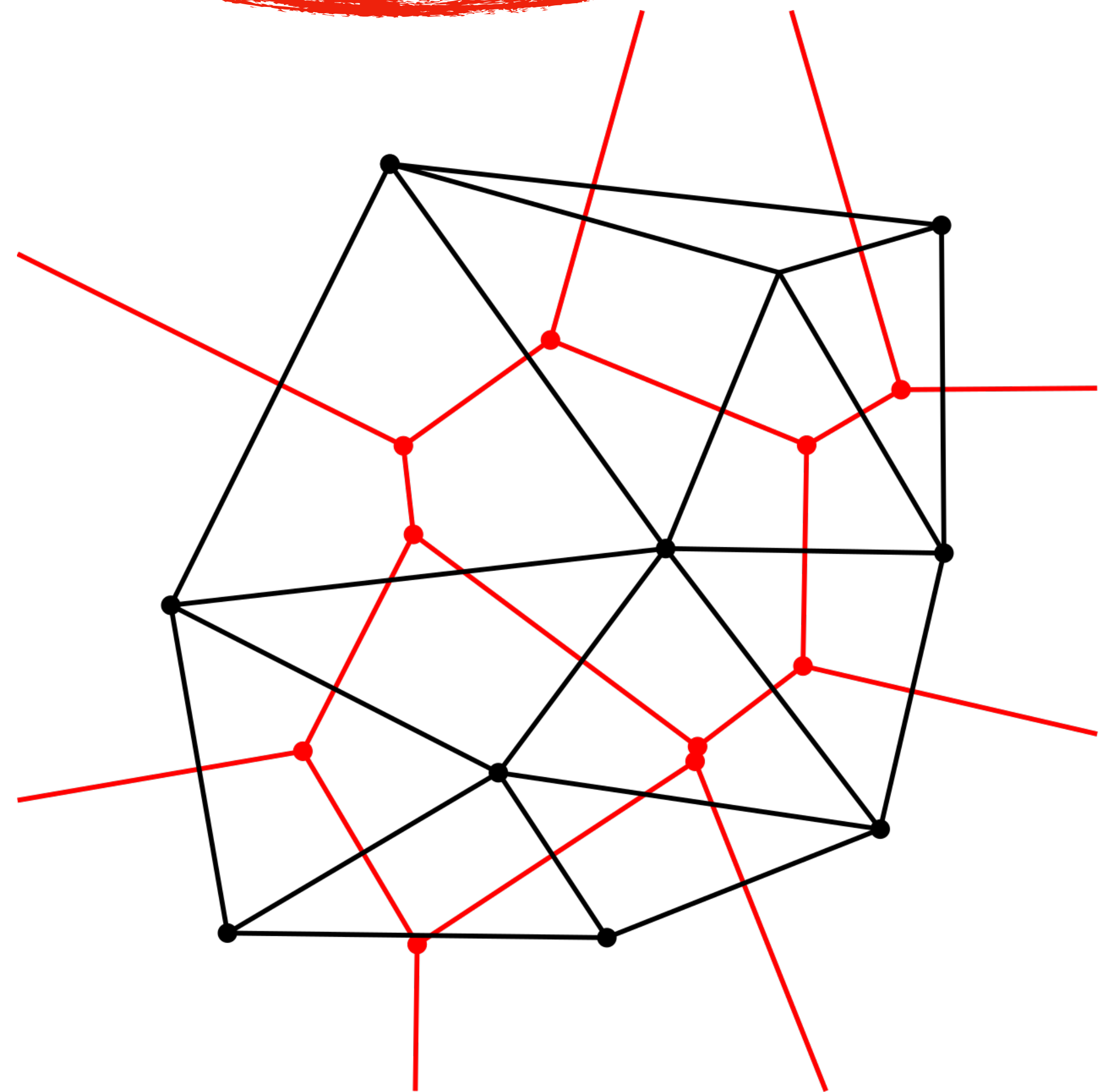
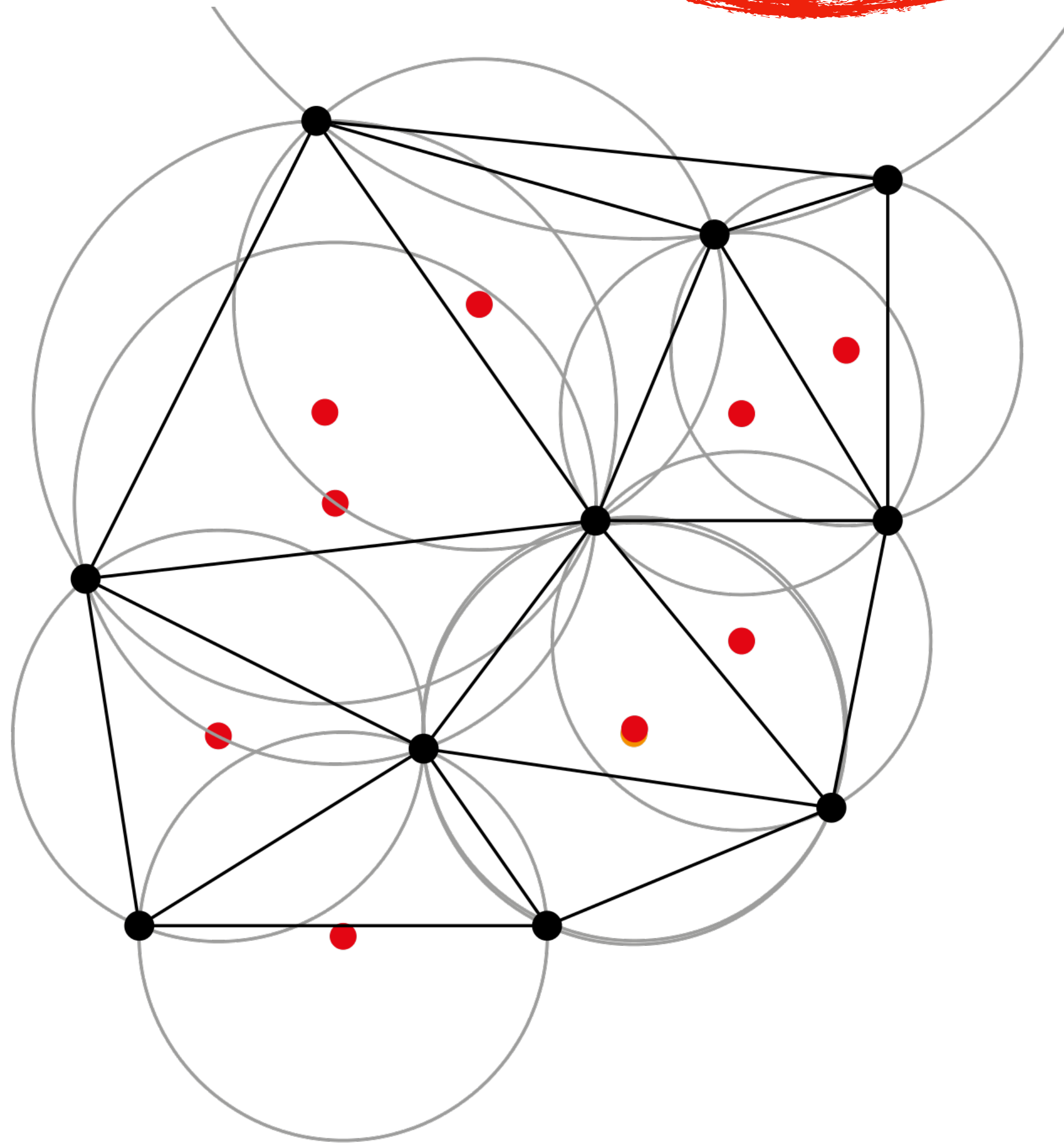
A **Delaunay triangulation** maximizes the minimum of all angles

It is the dual graph of the **Voronoi diagram**

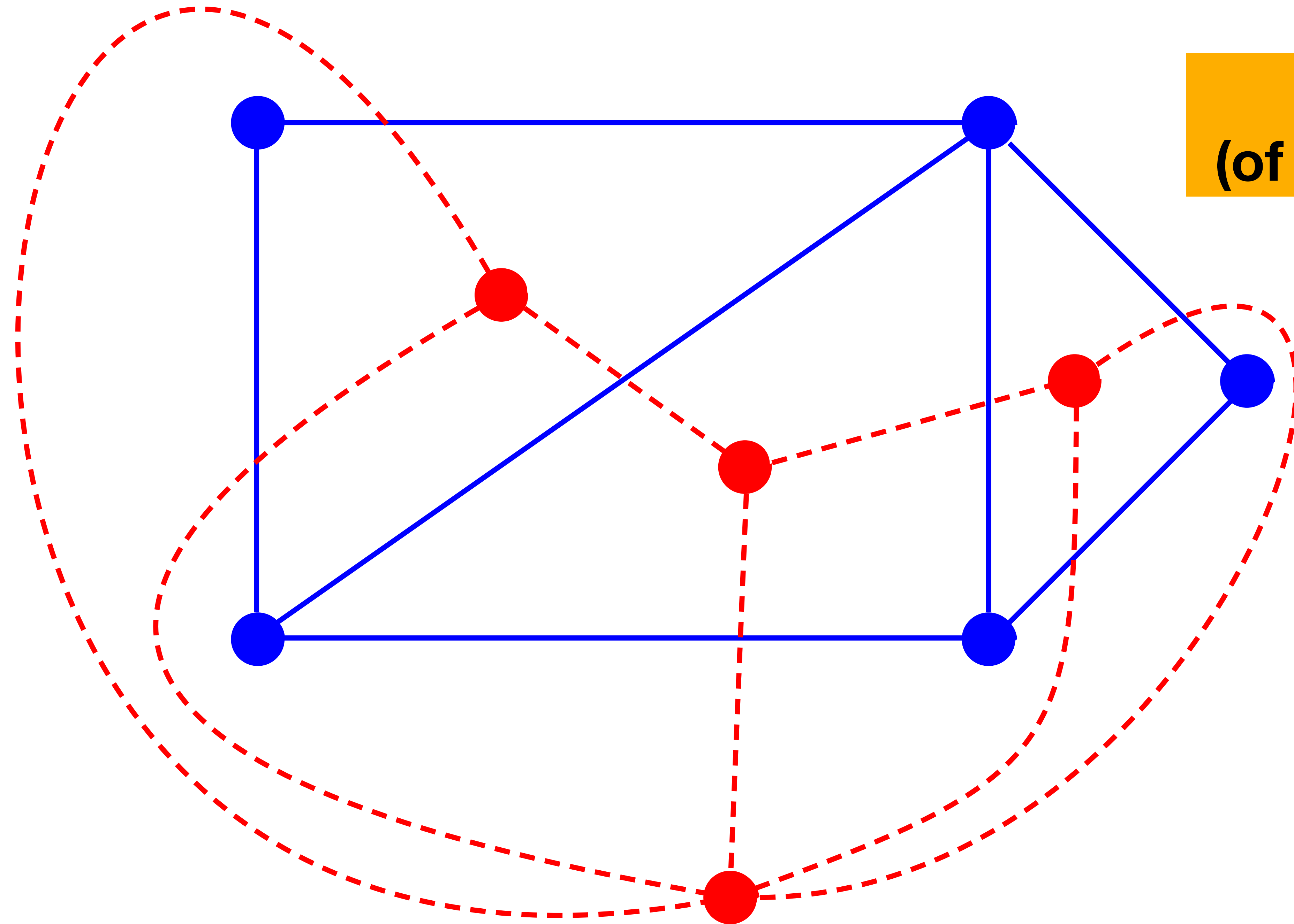


A **Delaunay triangulation** maximizes the minimum of all angles

It is the **dual graph** of the **Voronoi diagram**



Dual graph (of a planar graph)

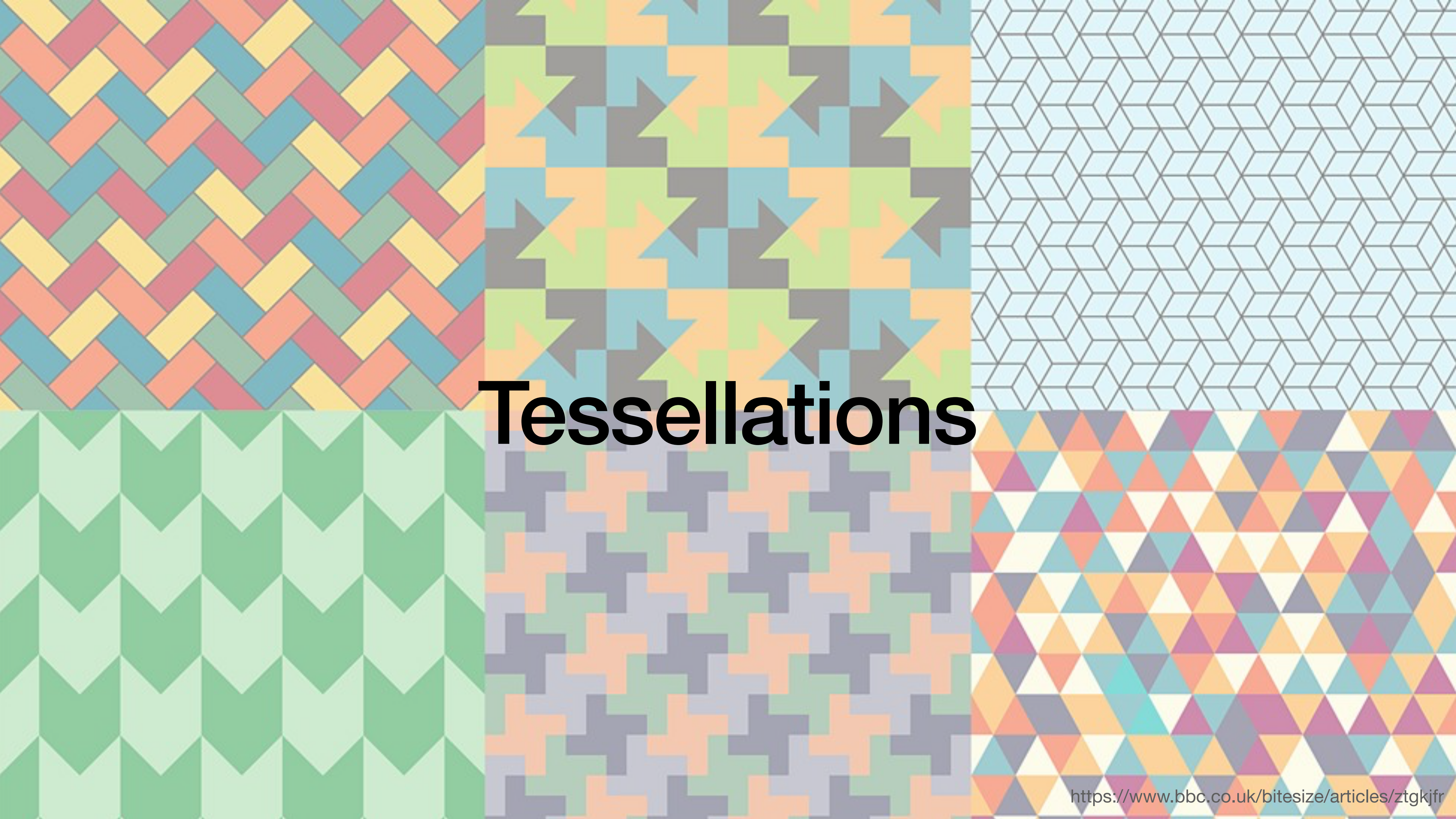


...has one node
for each face;

and one edge for
each adjacent
face pair

Voronoi tessellation (of the plane)

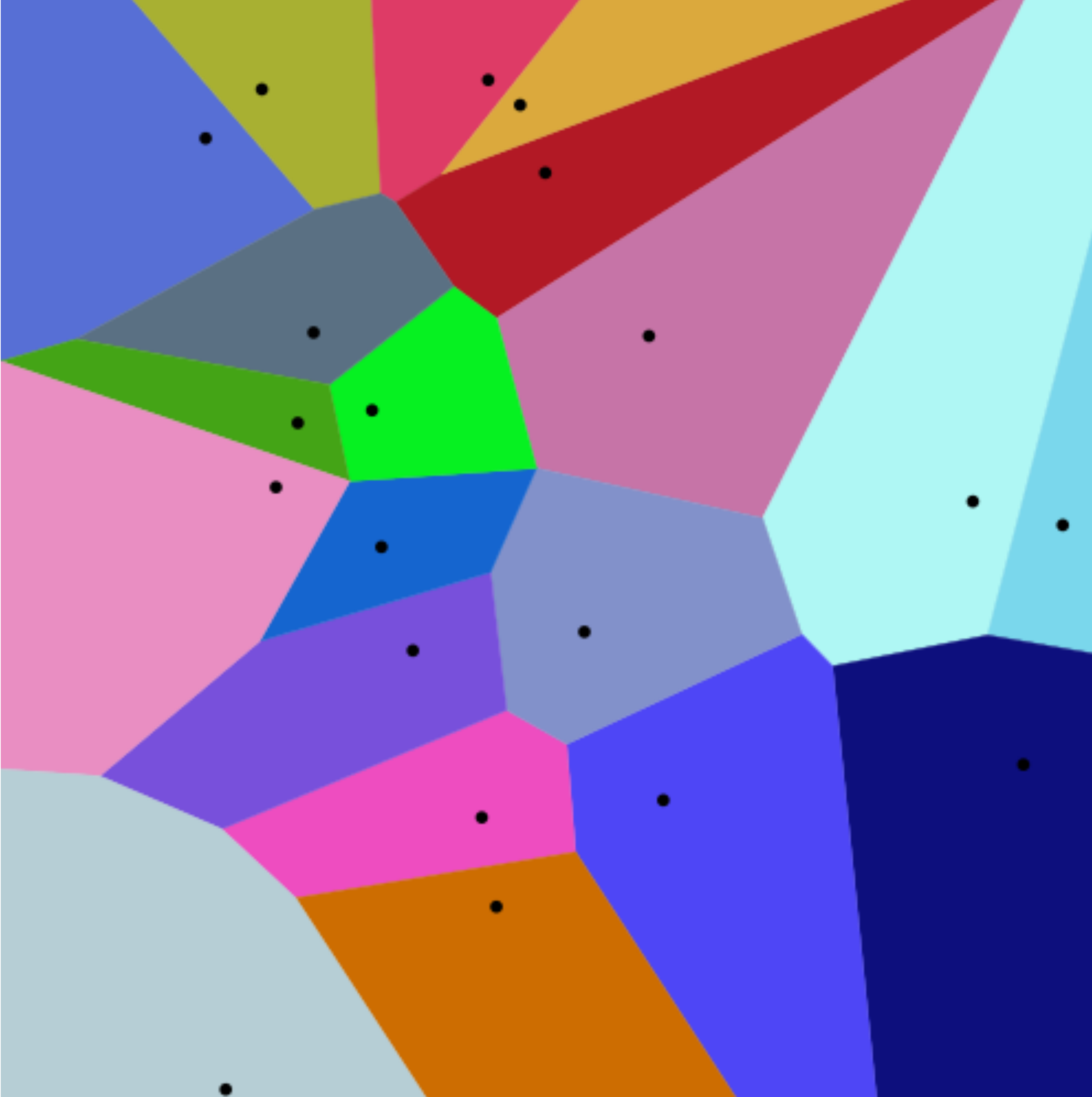




Tessellations

Voronoi tessellation (of the plane)



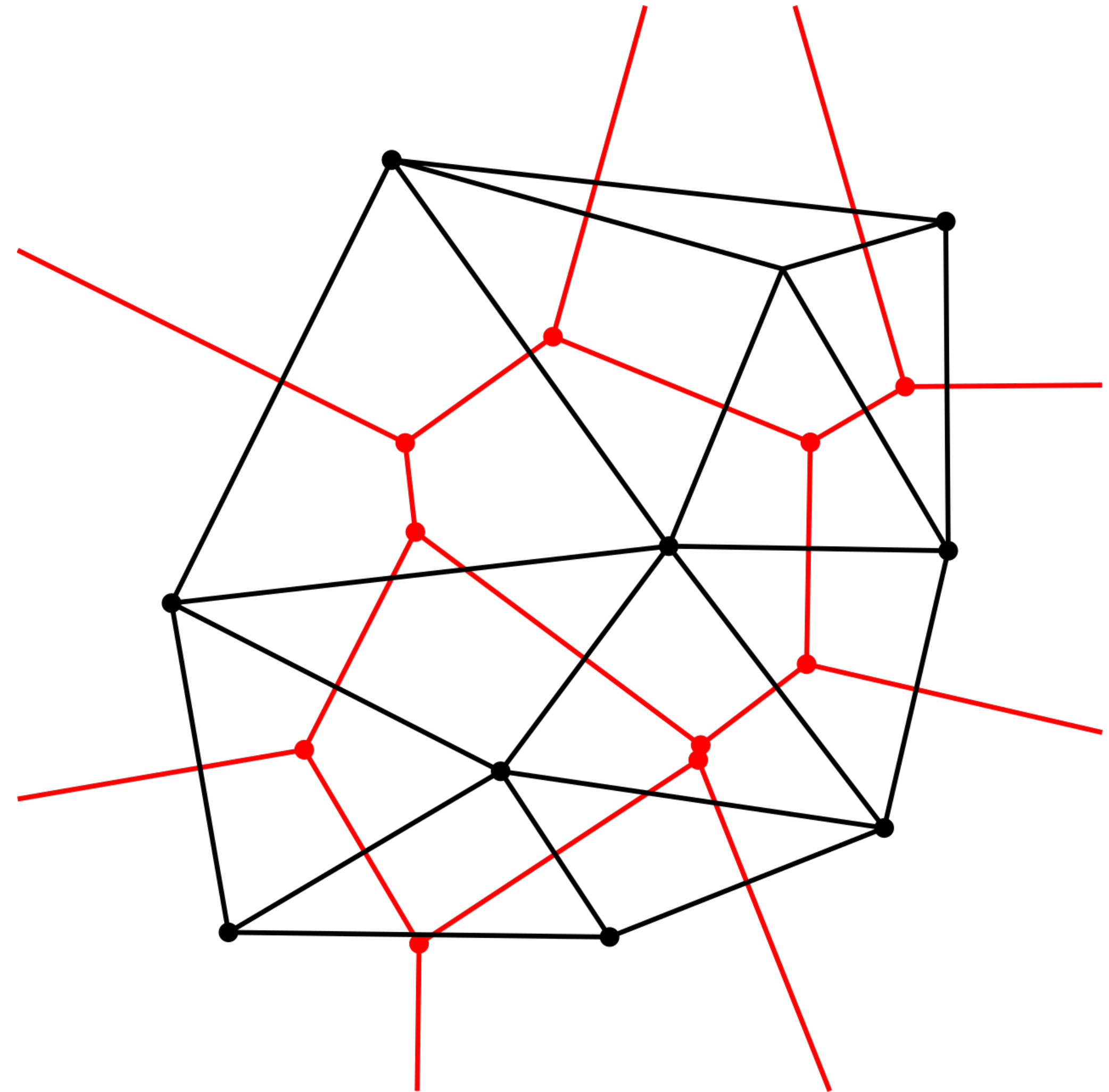
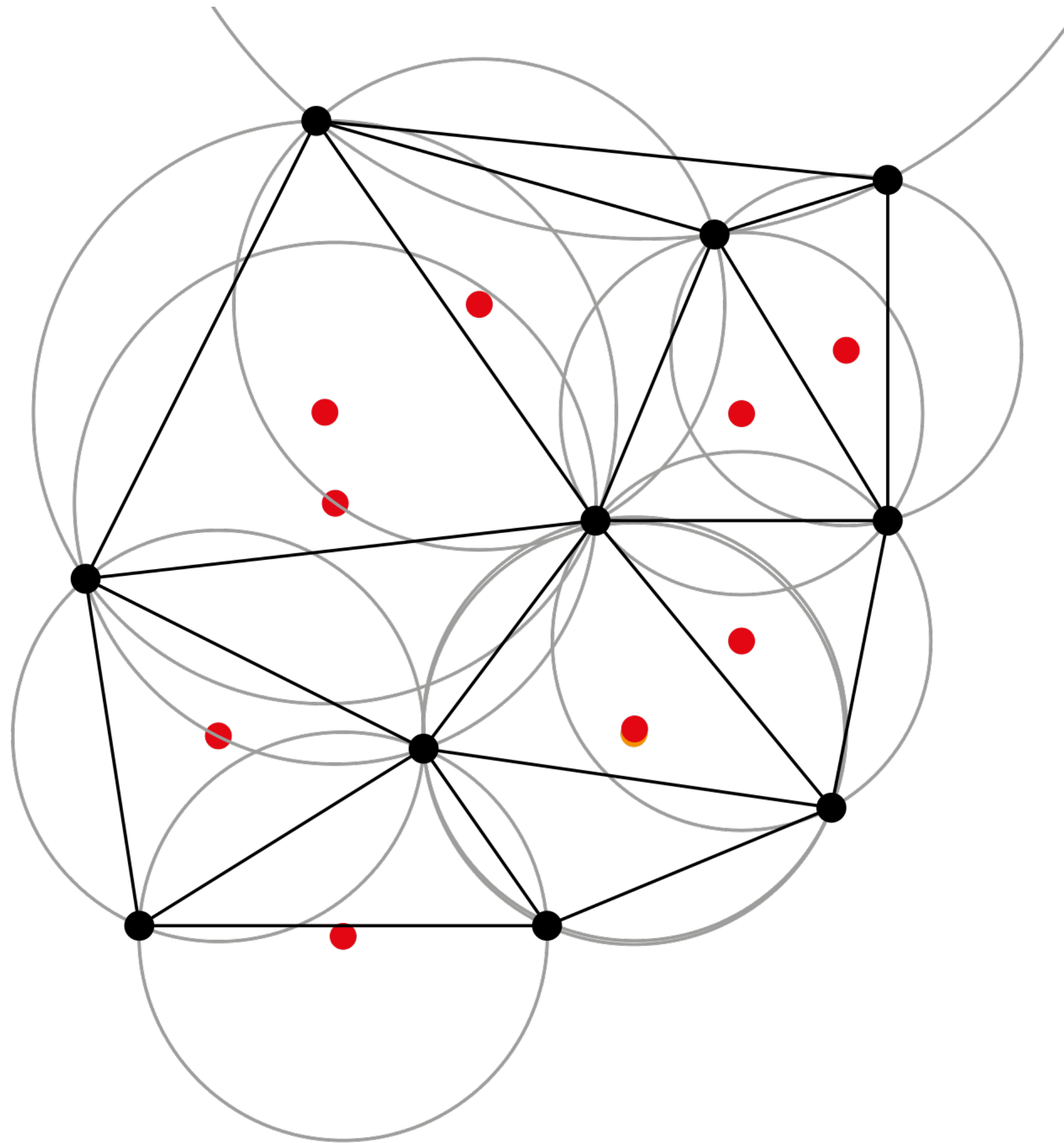


Voronoi tessellation (of the plane)

Partition of the plane into regions around seed points (“Voronoi cells”), for each of which all of their respective points are closer to the region’s seed than to any other seed

A **Delaunay triangulation** maximizes the minimum of all angles

It is the dual graph of the **Voronoi diagram**

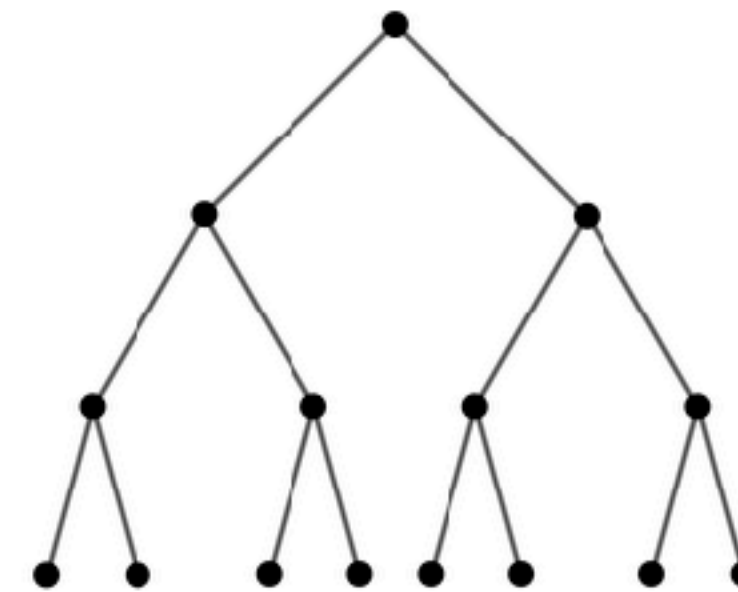
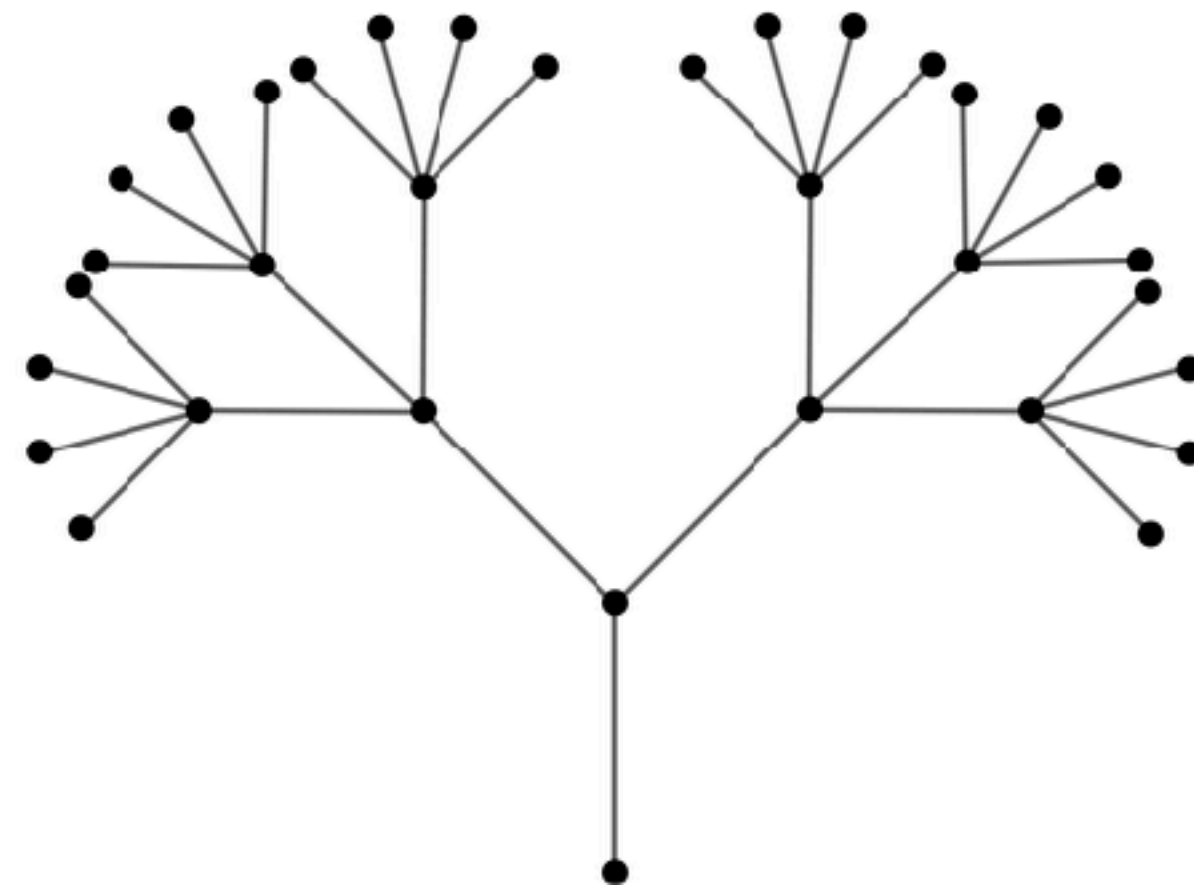
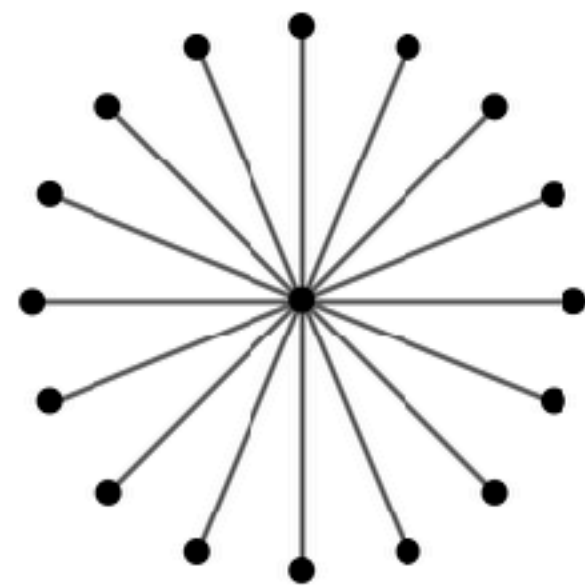


**The minimum spanning tree
is a subset of the Delaunay triangulation**

The minimum spanning tree is a subset of the Delaunay triangulation

tree (graph)

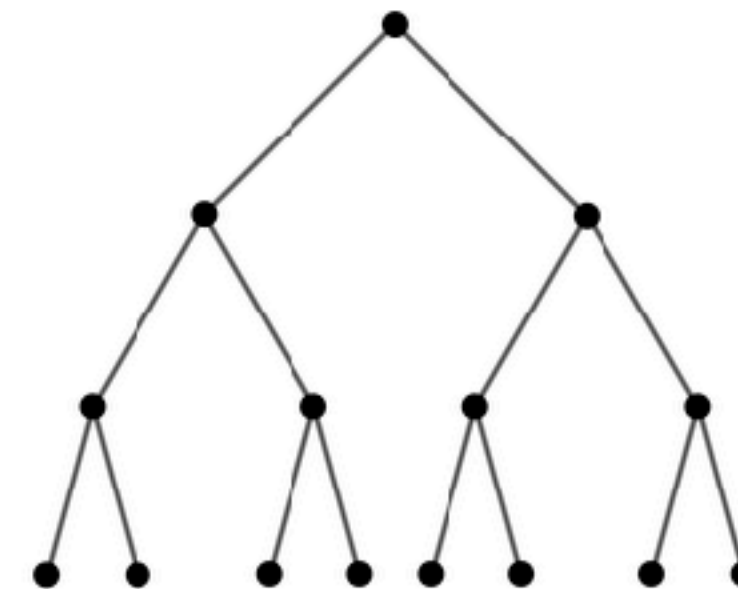
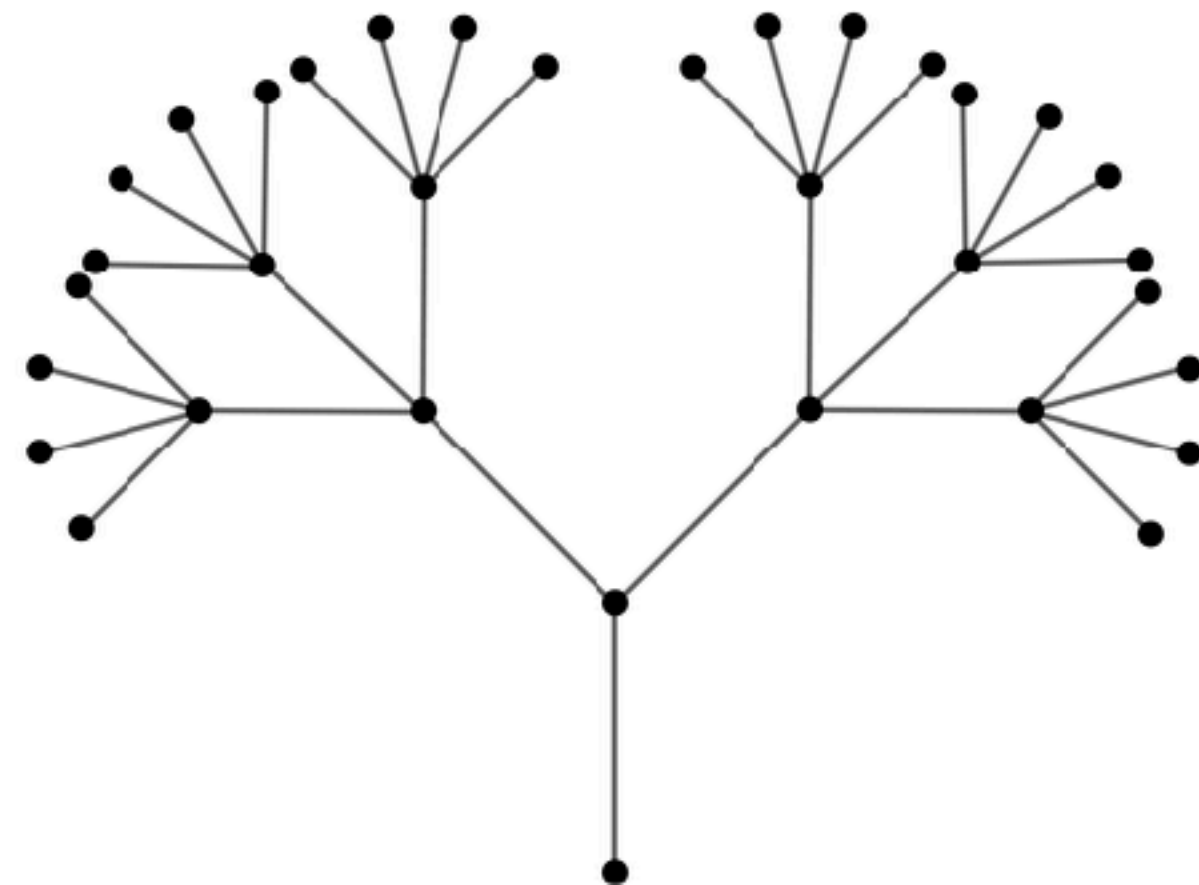
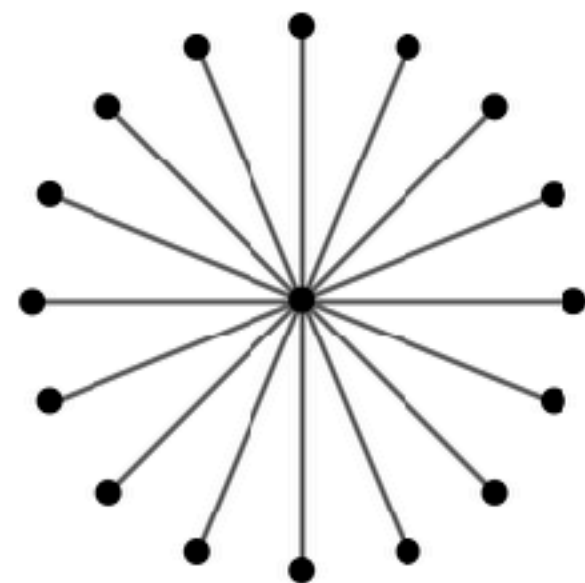
graph where every node pair is connected by exactly one path



The minimum **spanning tree** is a subset of the Delaunay triangulation

tree (graph)

graph where every node pair is connected by exactly one path

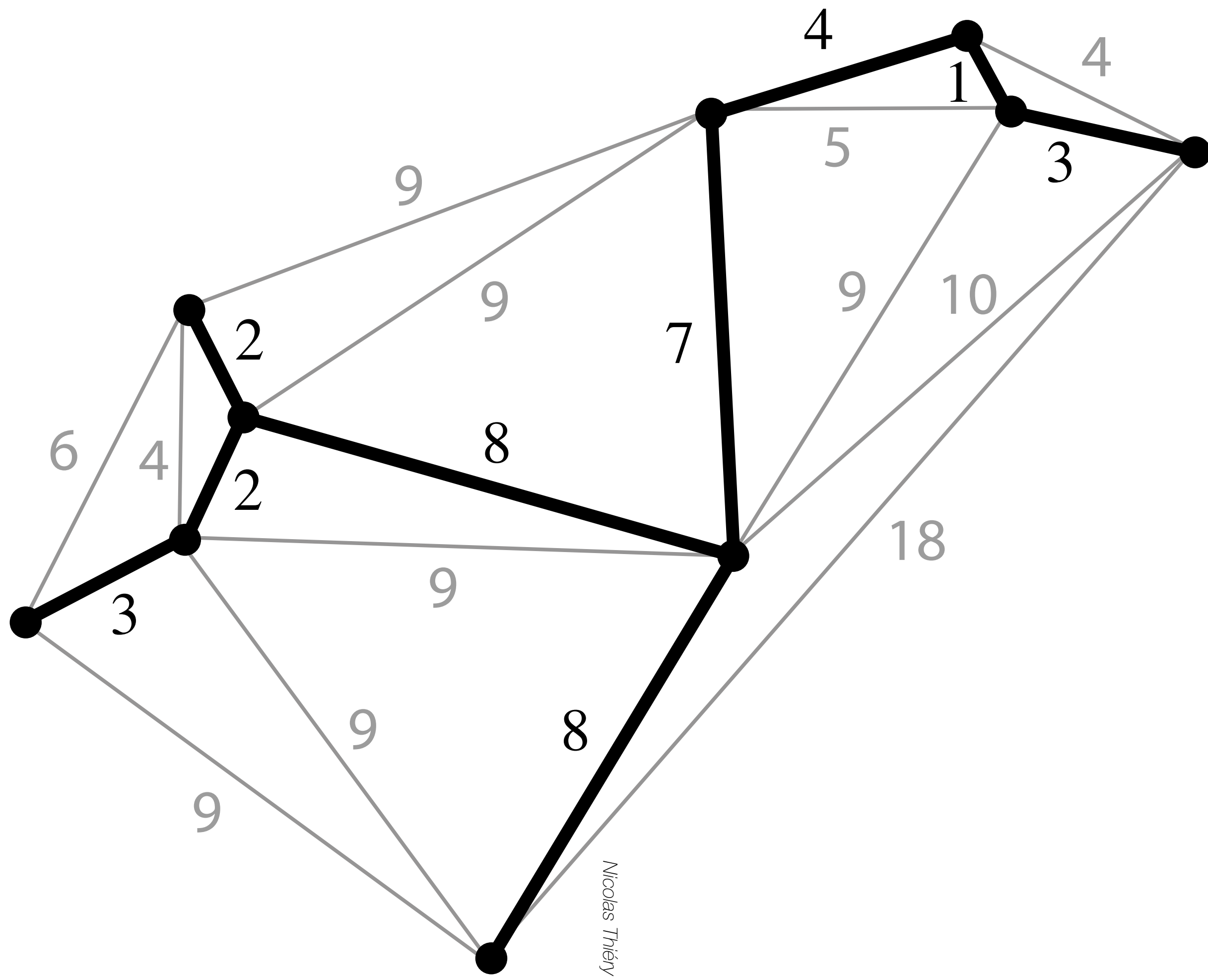


spanning tree (graph)

a subset of a graph that is a tree and connects all nodes

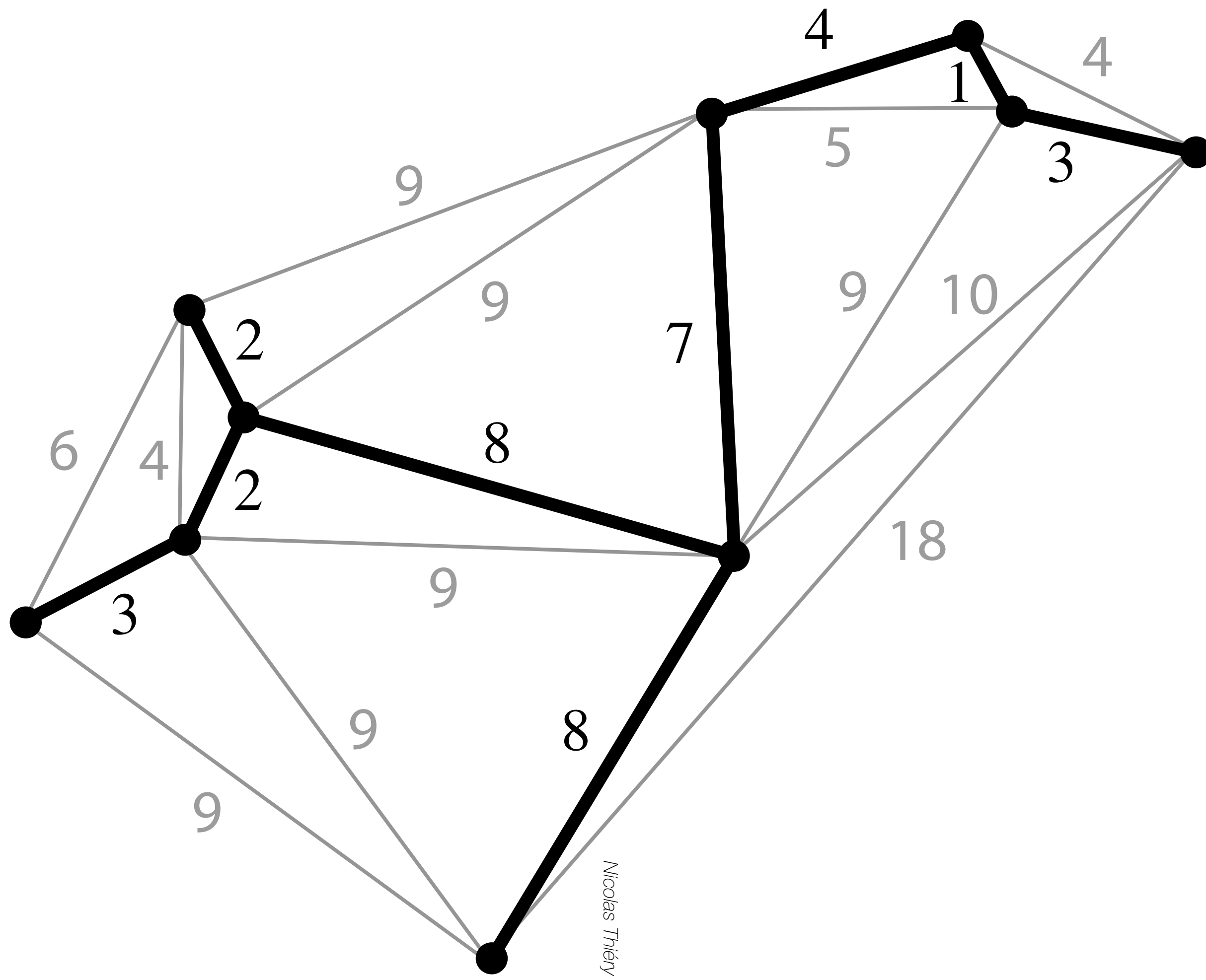
**The minimum spanning tree
is a subset of the Delaunay triangulation**

The **MST** is a subgraph that connects all nodes optimally



Given a connected, undirected, weighted graph, a **spanning tree** of the graph is a subgraph that is a tree and connects all the nodes.

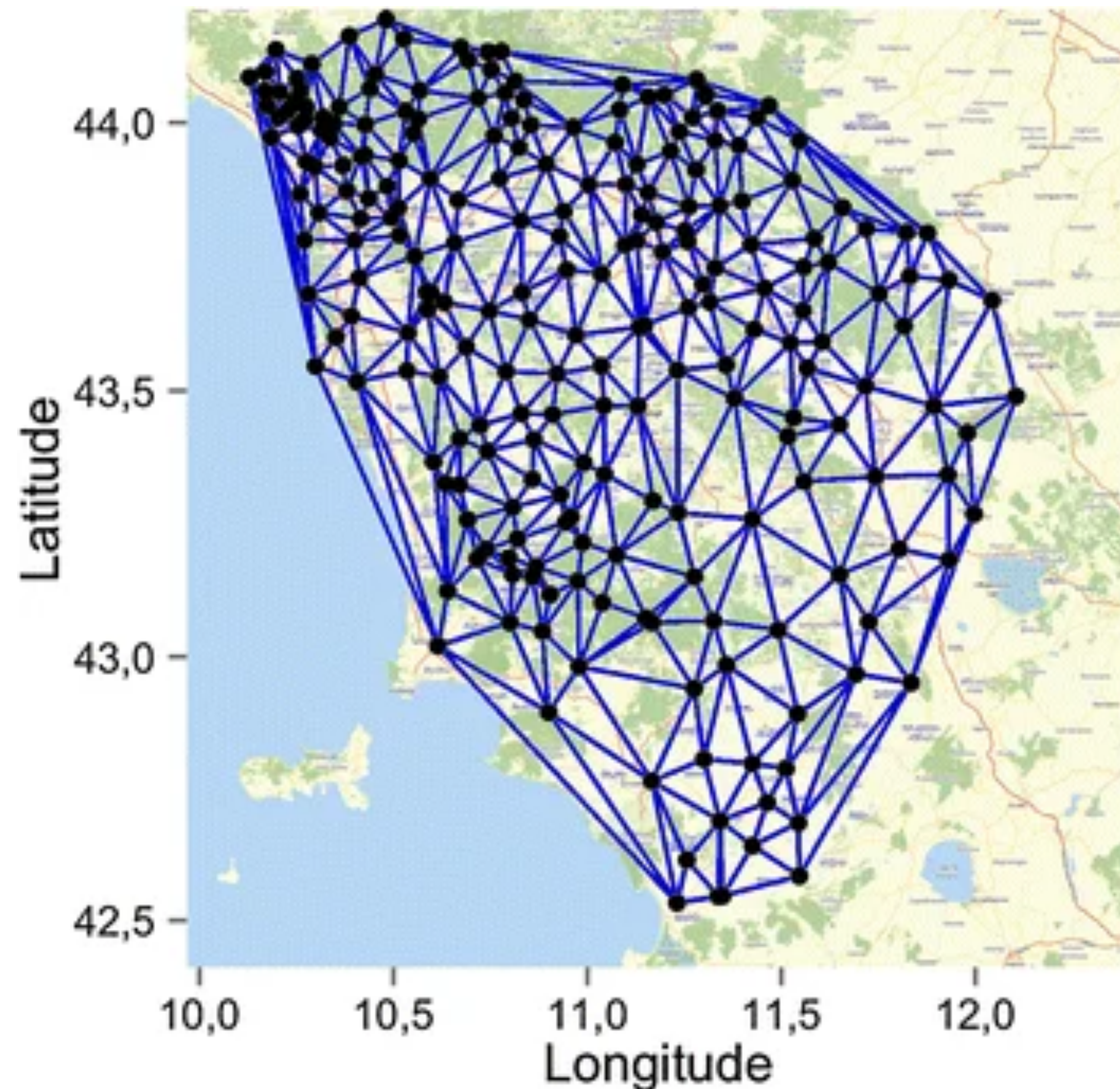
The **MST** is a subgraph that connects all nodes optimally



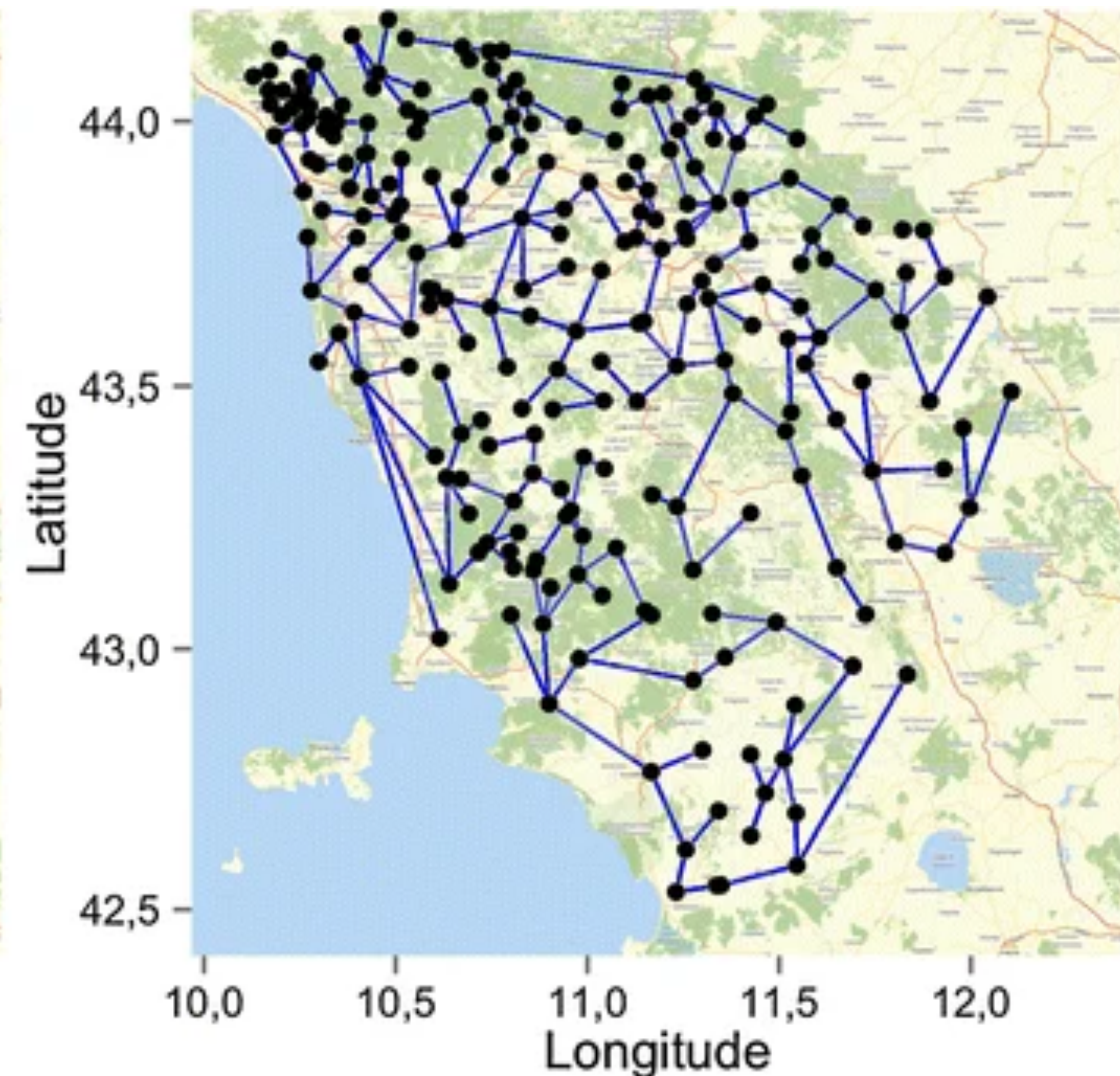
Given a connected, undirected, weighted graph, a **spanning tree** of the graph is a subgraph that is a tree and connects all the nodes.

A **minimum spanning tree (MST)** is a spanning tree such that the sum of its link lengths is smaller than or equal to the sum of the link lengths of any other spanning tree.

The MST is a subset of the Delaunay triangulation

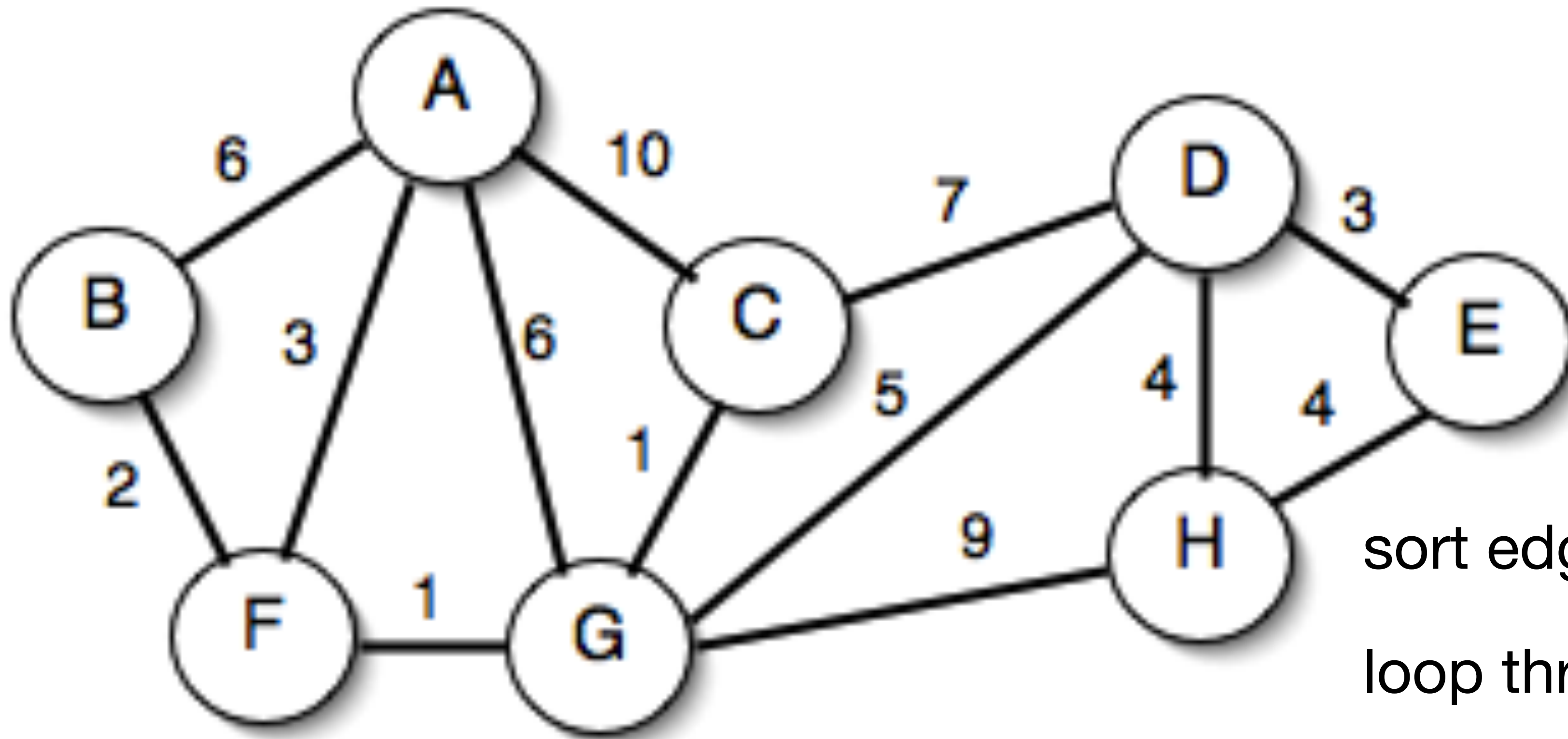


(a) Delaunay Triangle Network (D-TIN)



(b) Minimum Spanning Tree

Kruskal's algorithm for finding a minimum spanning tree



<https://tildesites.geneseo.edu/~baldwin/csci242/spring2013/0404mst.html>

sort edges by ascending weight
loop through edges

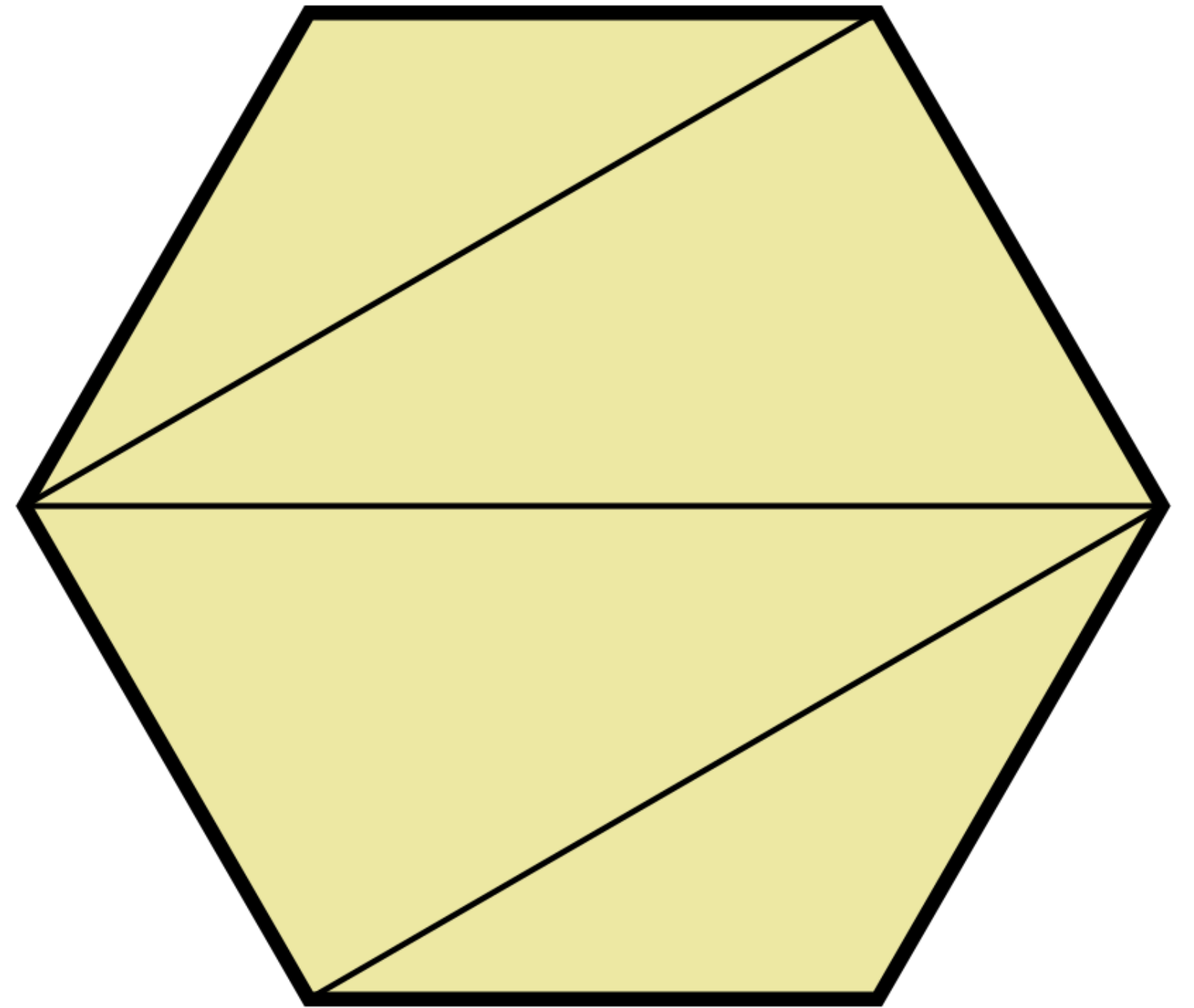
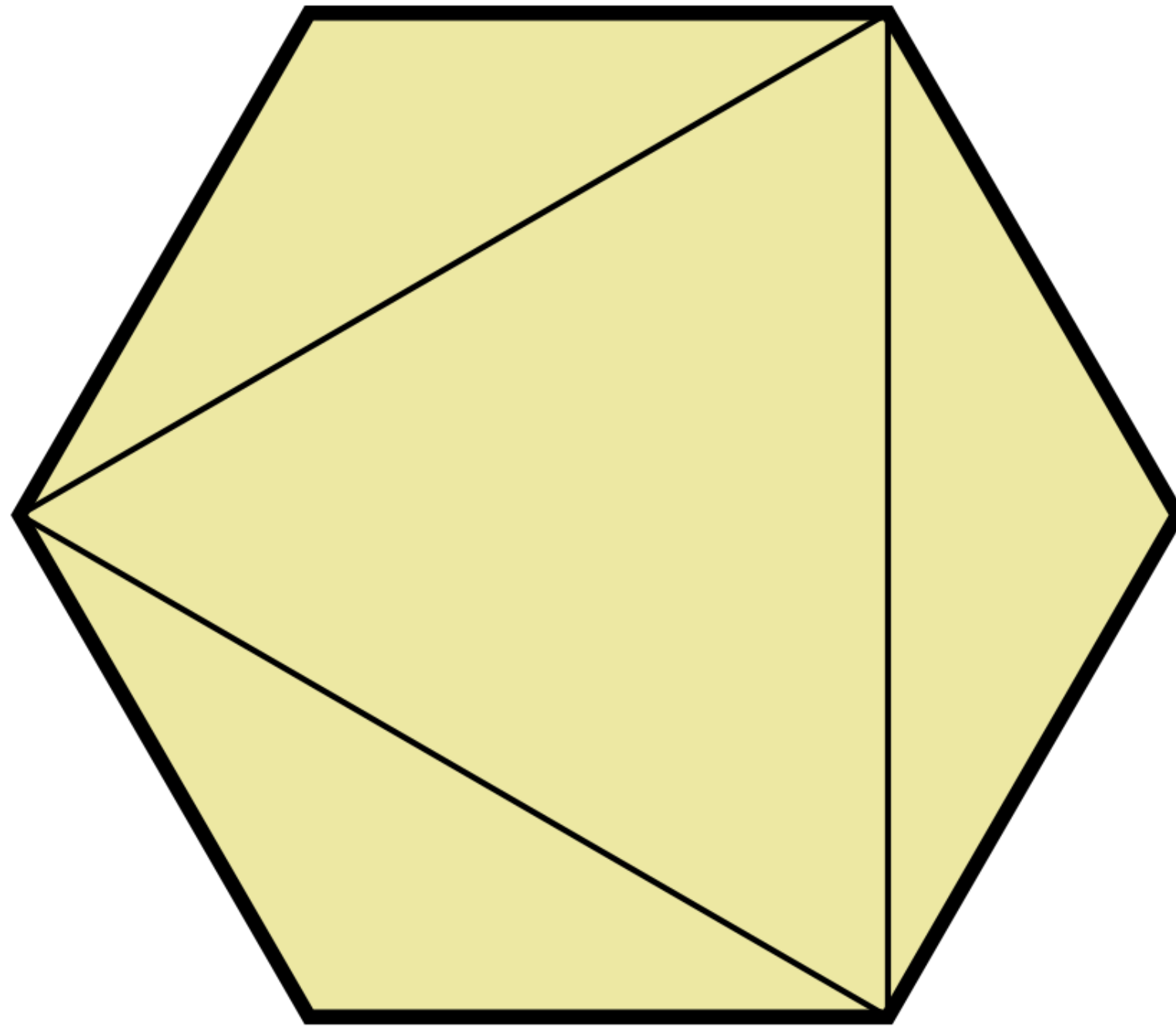
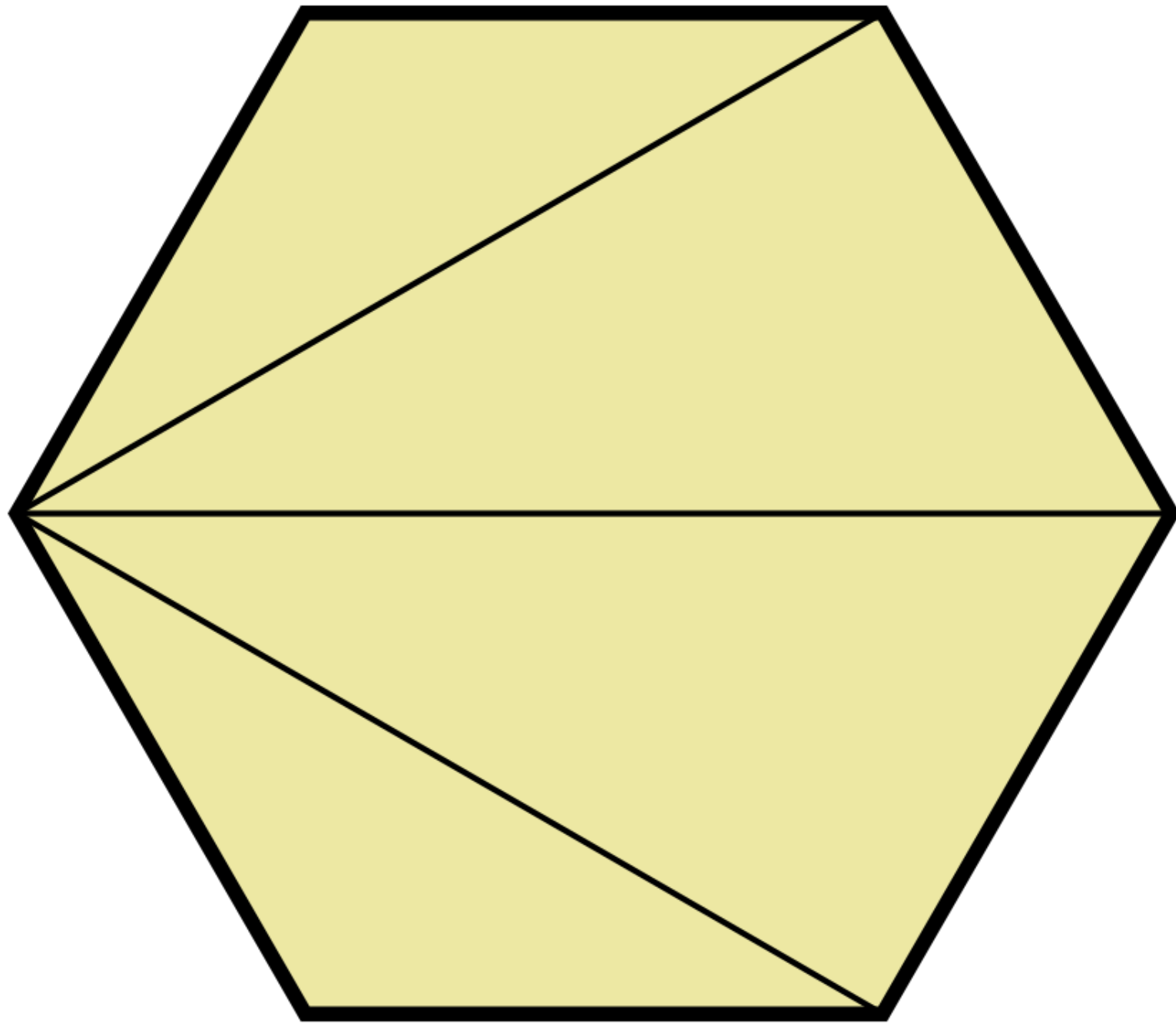
add edge,

UNLESS it creates cycle

stop when: all nodes are part of graph

& when 2 edges have the same weight,
select one of them at random.

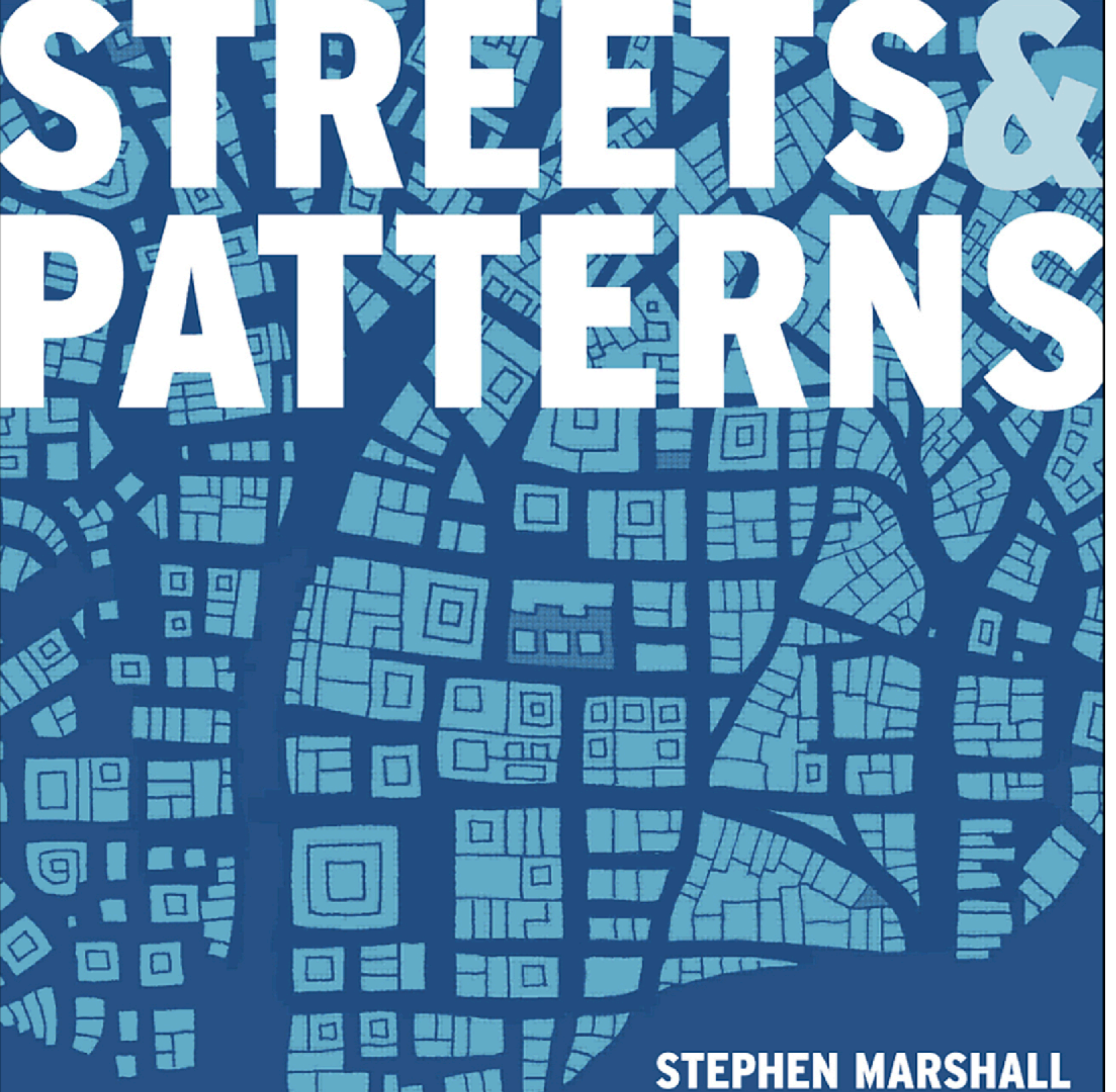
A **minimum weight triangulation** minimizes sum of edge lengths



Many other triangulations exist

		minimize	maximize
minimum	angle		$O(n \log n)$ (Delaunay triangulation)
maximum		$O(n^2 \log n)$ [8] [9]	
minimum	area	$O(n^2)$ [10]	$O(n^2 \log n)$ [11]
maximum		$O(n^2 \log n)$ [11]	
maximum	degree	NP-complete for degree of 7 [12]	
maximum	eccentricity	$O(n^3)$ [9]	
minimum	edge length	$O(n \log n)$ (Closest pair of points problem)	NP-complete [13]
maximum		$O(n^2)$ [14]	$O(n \log n)$ (using the Convex hull)
sum of		NP-hard (Minimum-weight triangulation)	
minimum	height		$O(n^2 \log n)$ [9]
maximum	slope	$O(n^3)$ [9]	

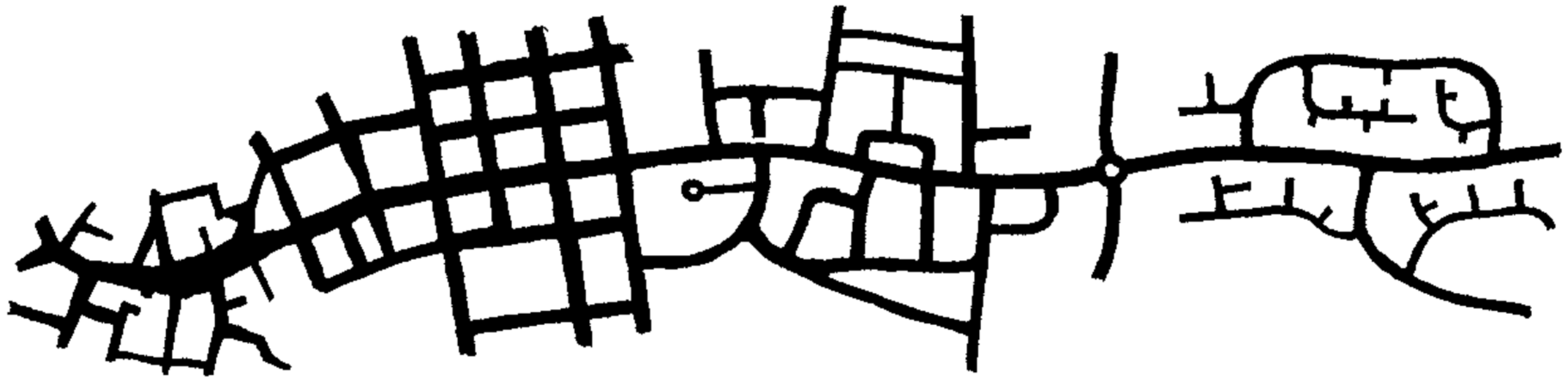
Street Patterns



STREETS & PATTERNS

STEPHEN MARSHALL

Stephen Marshall: Streets & Patterns (2005)

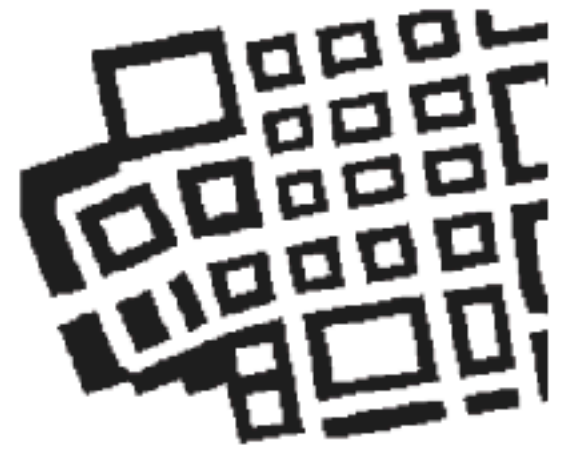


A

B

C

D

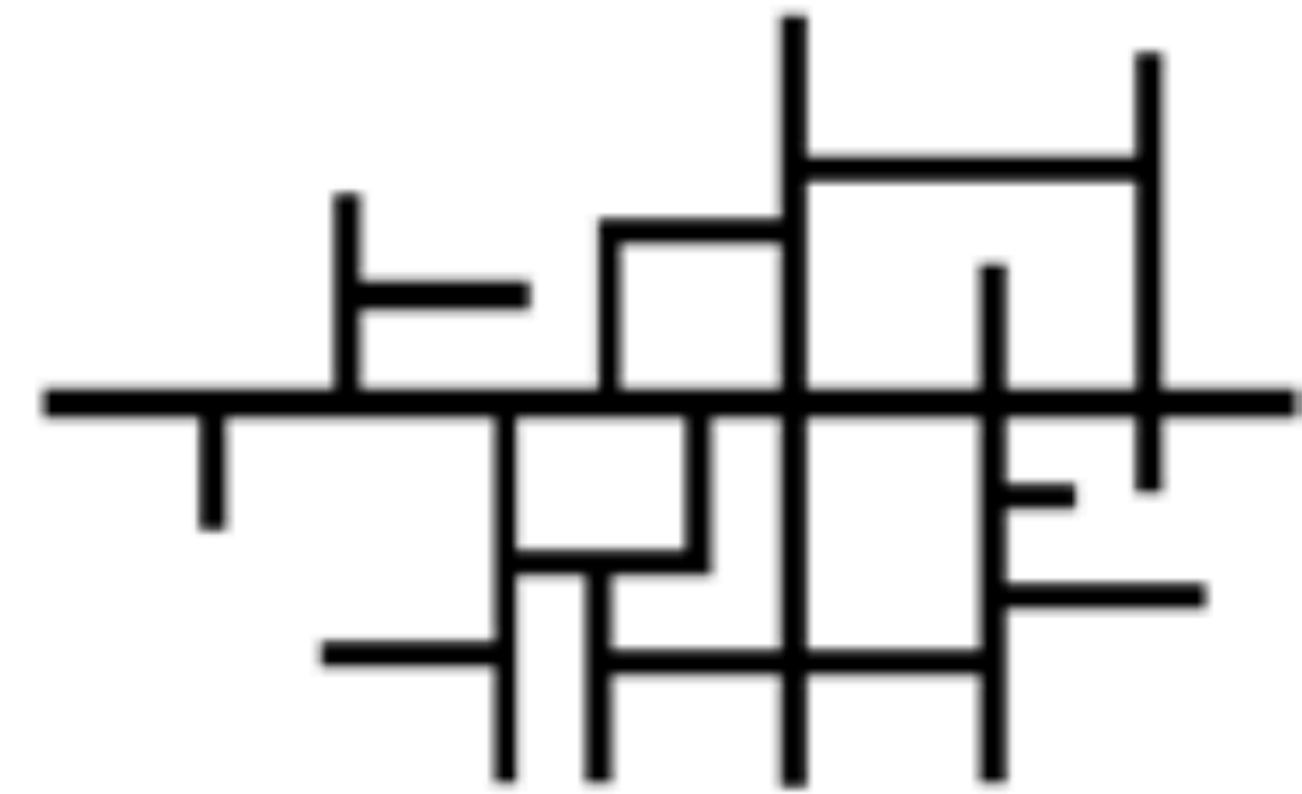
Type	Example pattern	Typical location	Frontages	Transport era
A-type <i>Altstadt</i>		Historic core	Built frontages	Era of pedestrian and horseback
B-type Bilateral		Gridiron (central, or extension, or citywide)	Built frontages	Era of horse and carriage
C-type Characteristic/ Conjoint		Anywhere; including individual villages or suburban extensions: often astride arterial routes	Built frontages or buildings set back in space ('pavilions')	Any Era of public transport; car
D-type Distributory		Peripheral development: off-line pods or superblock infill	Buildings set back in space, access only to minor roads	Era of the car

Composition



Irregular, fine scale angular, streets mostly short or crooked, varying in width, going in all directions.

Configuration



Mixture of configurational properties (T- and X-junctions, some culs-de-sac; moderate connectivity).

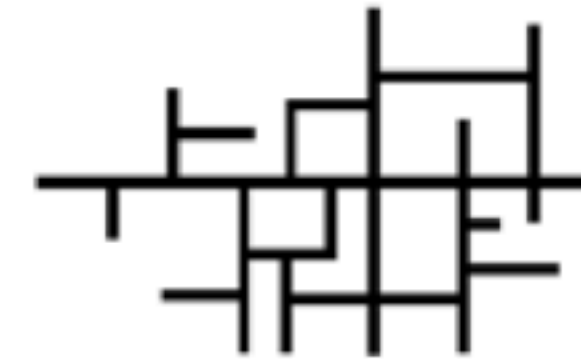
Composition

Configuration

A-type

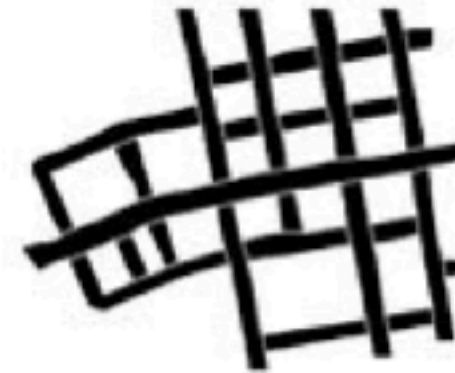


Irregular, fine scale angular, streets mostly short or crooked, varying in width, going in all directions.

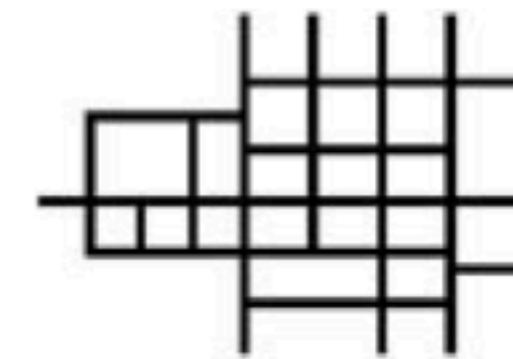


Mixture of configurational properties (T- and X-junctions, some culs-de-sac; moderate connectivity).

B-type



Regular, orthogonal, rectilinear, streets of consistent width, going in two directions.

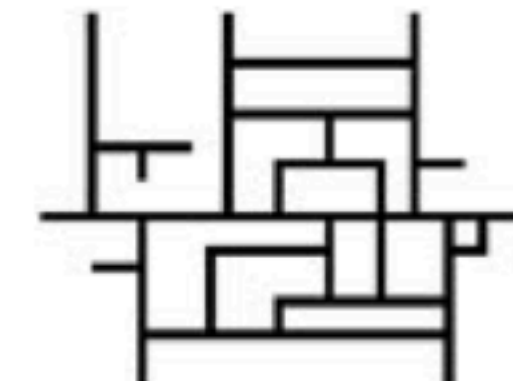


Mainly grid with crossroads (high connectivity). Continuity of cross routes.

C-type



Mixture of regularity and irregularity, streets typically of consistent width; curved or rectilinear formations, meeting at right angles.

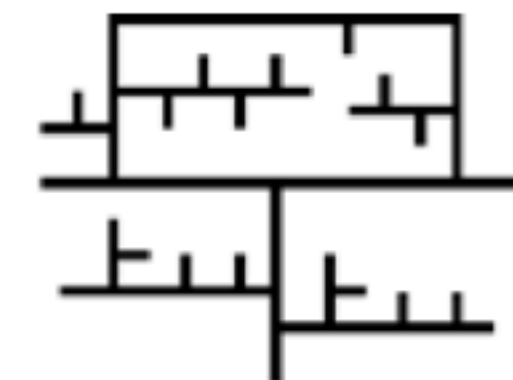


Mixture of configurational properties (T- and X-junctions, some culs-de-sac; moderate connectivity).

D-type



Based on consistent road geometry. Curvilinear or rectilinear formations, mostly meeting at right angles.



Loop roads with many branching routes in tree-like configurations (mainly T-junctions, many culs-de-sac; low connectivity).



Tokyo Grid



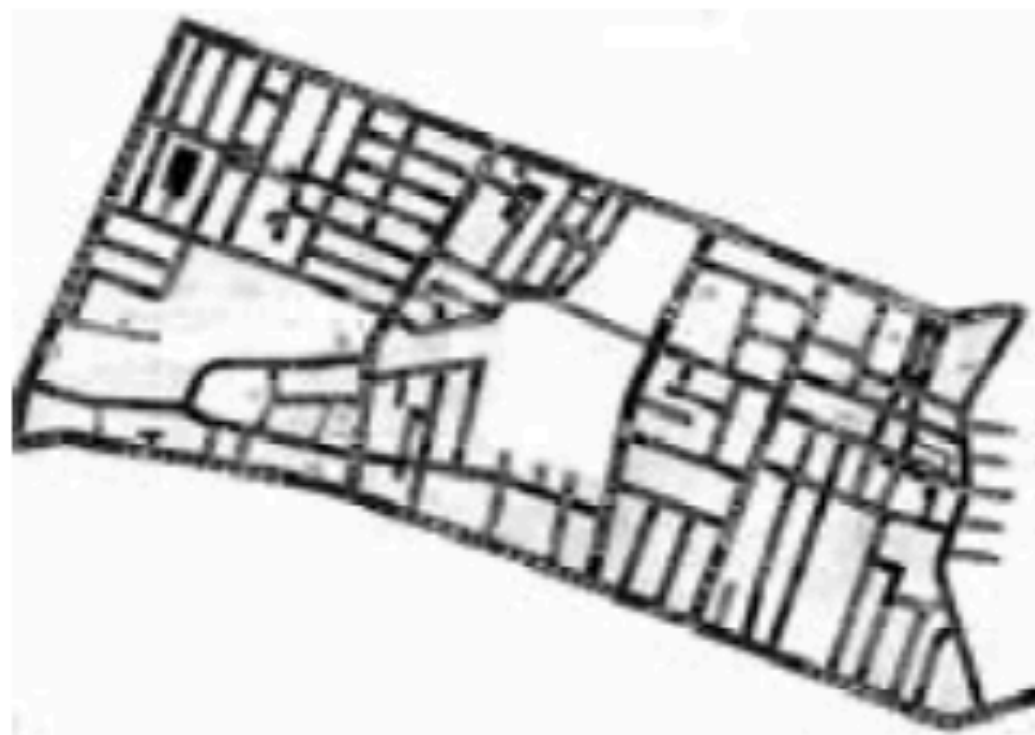
Glasgow Southside



Reykjavik Central



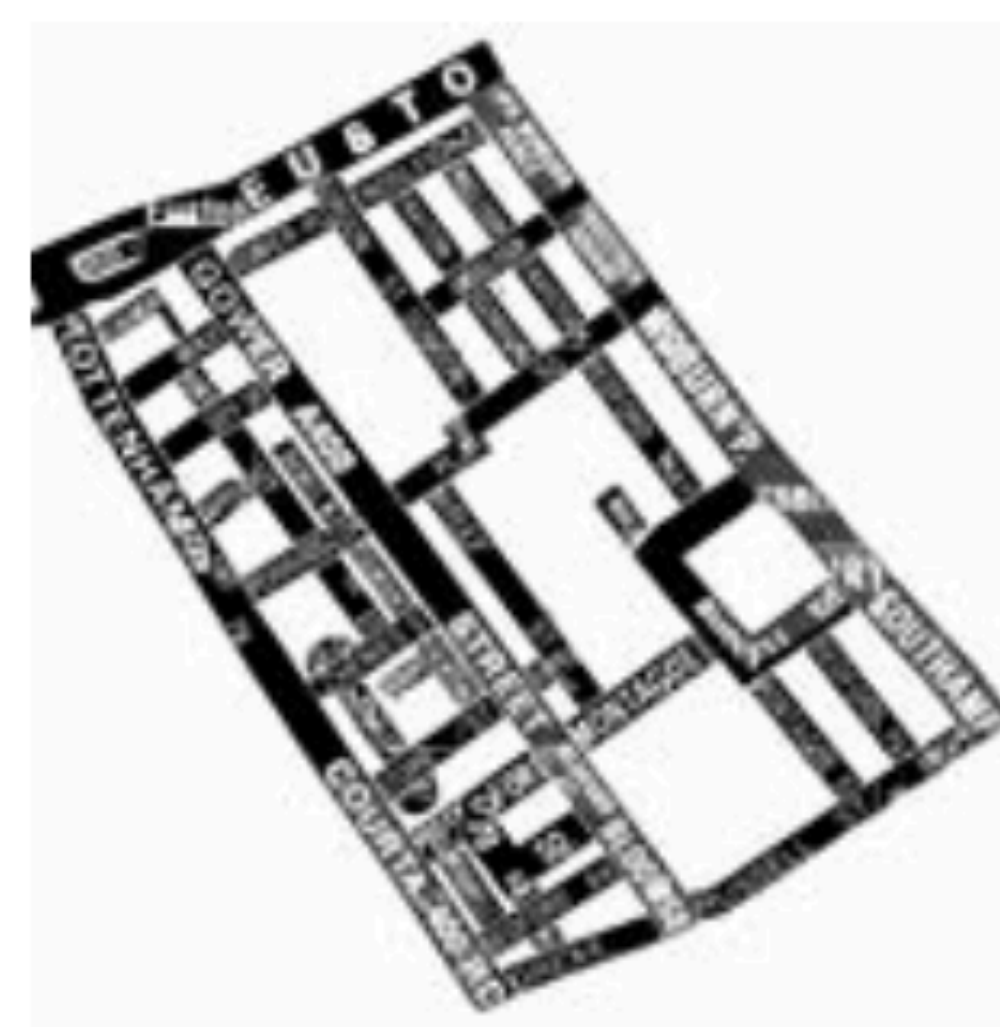
Glasgow Grid



Copenhagen Inner



Cornhill



Bloomsbury



Tunis Medina

Today's lecture in 3 sentences

A spatial network has nodes in metric space and is often (but not necessarily!) planar.

We can model "optimal" networks with different triangulation algorithms.

Street network degree distributions are boring, but street network patterns are interesting.